



CAPSTONE PROJECT REPORT

Report 3 – Software Requirement Specification

– Hanoi, Sep 2025 –

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Definitions and Acronyms

Acronym	Meaning
A	Added
M	Modified
D	Deleted
UC	Use Case
CRUD	Create, Read, Update, Delete
UI	User Interface
APIs	Application Programming Interfaces
QR system	Quick Response system
Genemi AI	Genemi Artificial Intelligence
RAG	Retrieval-Augmented Generation

I. Record of Changes

Date	A* M, D	In charge	Change Description
12/09/2025	A	QuatBT	Initial creation of SRS document.
20/09/2025	M	Team G44	Update Software Requirement Specification
05/10/2025	M	Team G44	Update Software Requirement Specification
18/10/2025	M	Team G44	Update Software Requirement Specification
02/11/2025	M	Team G44	Update Software Requirement Specification
15/11/2025	M	Team G44	Update Software Requirement Specification
28/11/2025	M	Team G44	Update Software Requirement Specification
05/12/2025	M	Team G44	Final review and formatting.
28/12/2025	M	Team G44	Update Software Requirement Specification

*A - Added M - Modified D - Deleted

II. Software Requirement Specification

1. Overall Requirements

1.1 Context Diagram

The SkillUp Online Learning Platform is a comprehensive web-based educational system that modernizes and streamlines the online learning experience for students, lecturers, and educational institutions. This system replaces traditional fragmented learning management approaches by providing an integrated platform that supports course creation and management, interactive learning, real-time collaboration, secure payment processing, and intelligent content moderation. The context diagram below illustrates the external entities and system interfaces for release 1.0.

The primary external entities interacting with the SkillUp system include:

- **Guest:** A Guest is an unauthenticated user who can access publicly available features of the SkillUp platform, such as browsing courses, searching content, and viewing public news or forum posts. Guests may register an account to gain full access to learning and interactive features.
- **Students:** Learners who browse, purchase, and consume educational content, participate in discussions, complete quizzes, and provide course ratings and feedback.
- **Lecturers:** Content creators who develop courses, upload learning materials, manage course structure, interact with students, and track their teaching performance.
- **Content Moderators:** Staff members responsible for reviewing and moderating user-generated content, managing forum posts, handling content reports, and ensuring community guidelines compliance.
- **System Moderators:** Operational staff responsible for managing lecturer applications, handling support tickets, monitoring system activities, and overseeing daily platform operations.
- **Administrators:** High-level administrators who manage platform configuration, system settings, user roles, and moderator accounts.
- **Payment Gateway (PayOS):** External payment processing service that handles secure transaction authorization, payment verification, and refund processing for course purchases.
- **Cloud Storage Service (Cloudinary):** Third-party media storage and delivery platform for hosting course videos, images, and other multimedia content with CDN support.
- **Email Service Provider:** External SMTP service for sending transactional emails including registration confirmations, password resets, notifications, and system alerts.
- **AI Services (Gemini/Ollama):** External artificial intelligence APIs for subtitle generation, content analysis, and RAG (Retrieval-Augmented Generation) capabilities for enhanced learning support.
- **Vector Database (Qdrant):** External vector search service for implementing semantic search and intelligent content recommendation features.

The system is expected to evolve over several releases, with future enhancements including integration with additional payment methods (credit/debit card processors, digital wallets), third-party authentication providers (Google, Facebook OAuth), video conferencing platforms for live

classes, mobile applications (iOS and Android), advanced analytics and reporting dashboards, AI-powered personalized learning paths, and integration with external learning content providers and certification authorities.

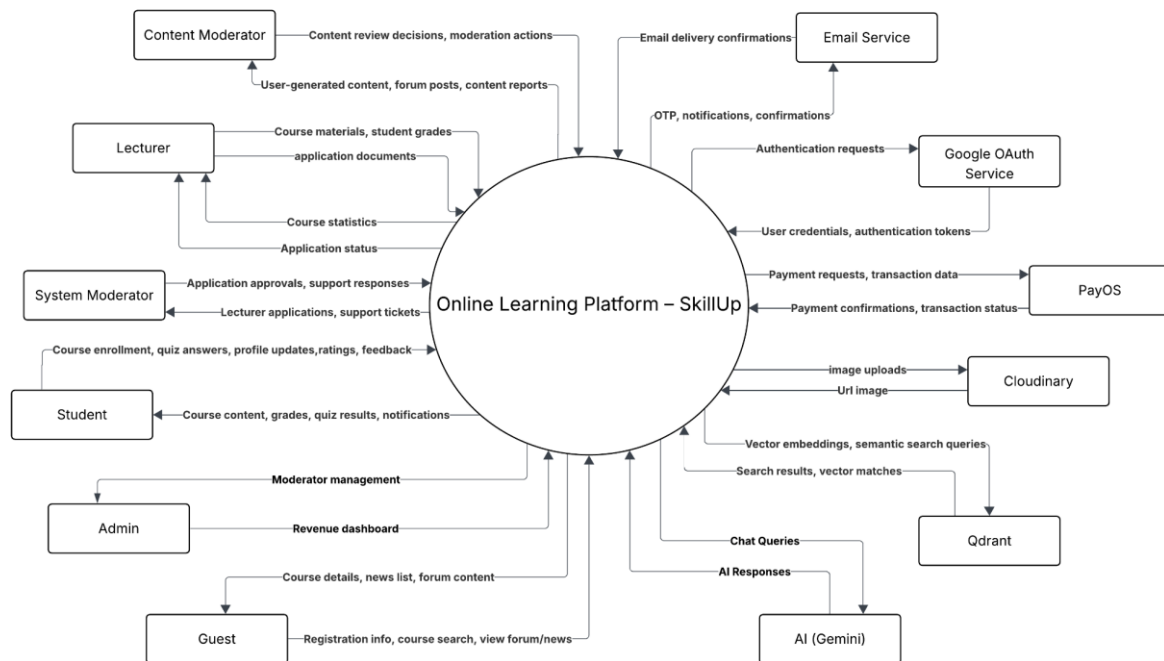


Figure 1.1 - Context diagram of OLP-SkillUp

1.2 Main Workflows

1.2.1 Register to become a lecturer

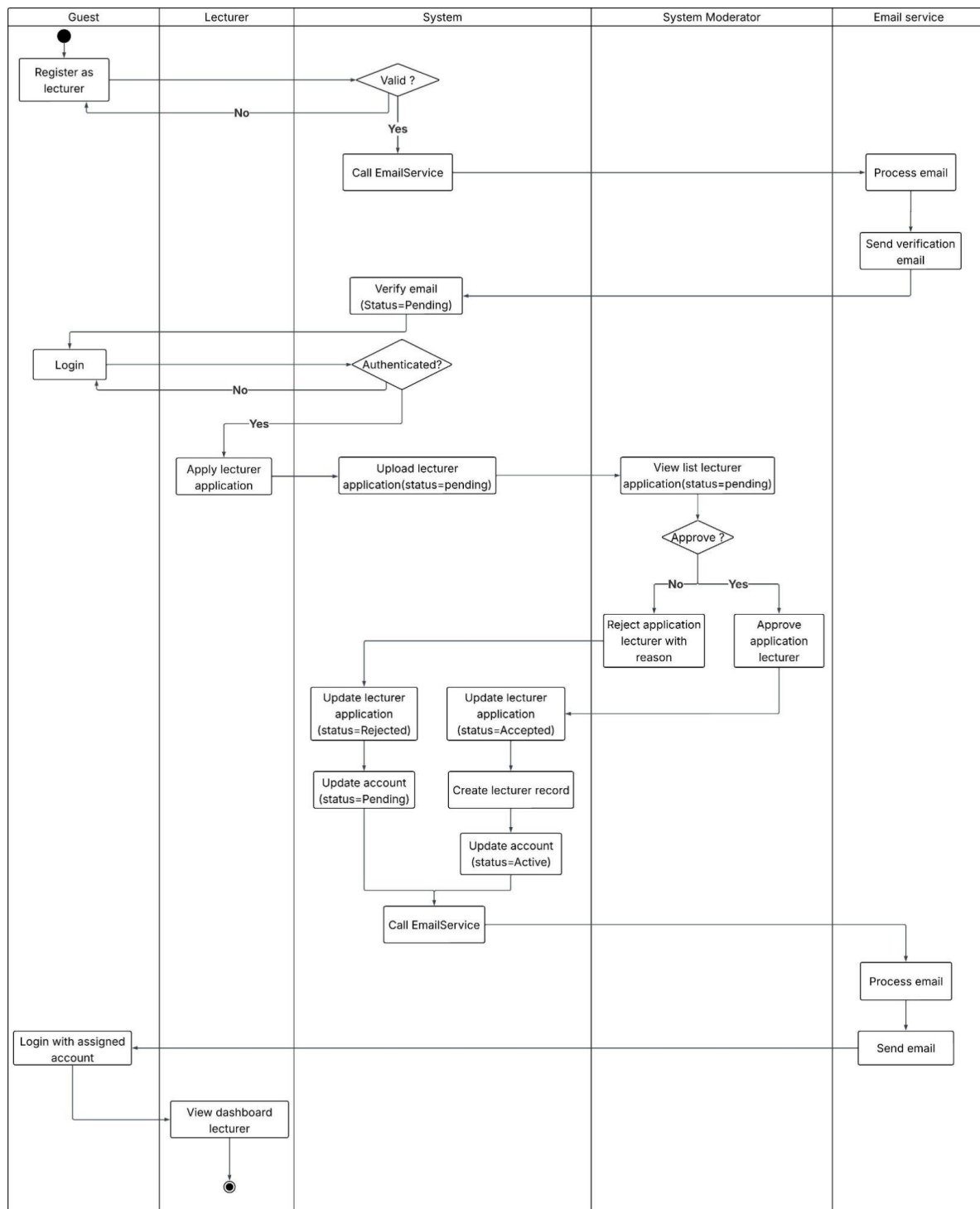


Figure 1.2.1 - Main workflow register to become a lecturer

1.2.2 Create and update course

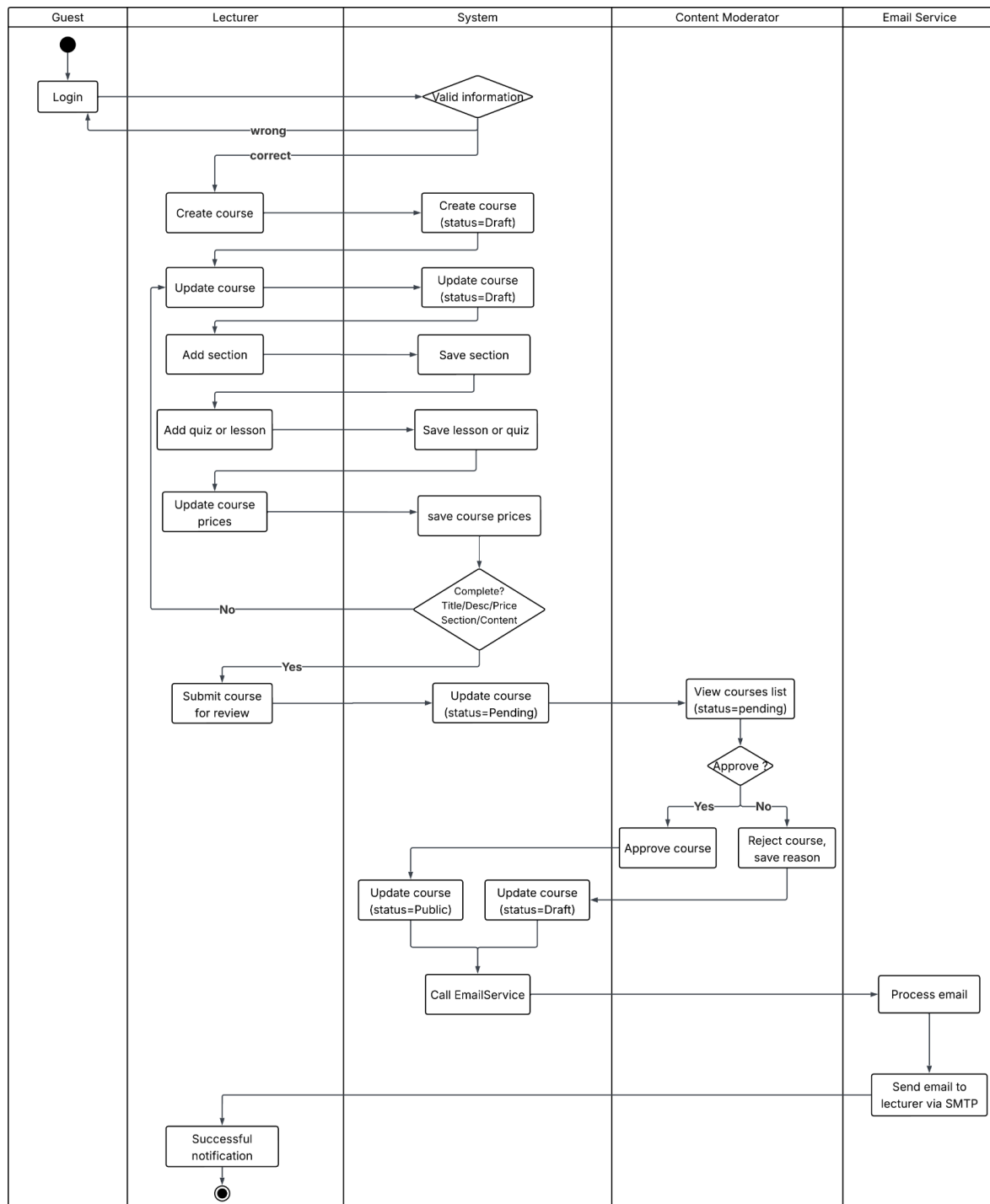


Figure 1.2.2 - Main workflow create course

1.2.3 Buy course and learning materials

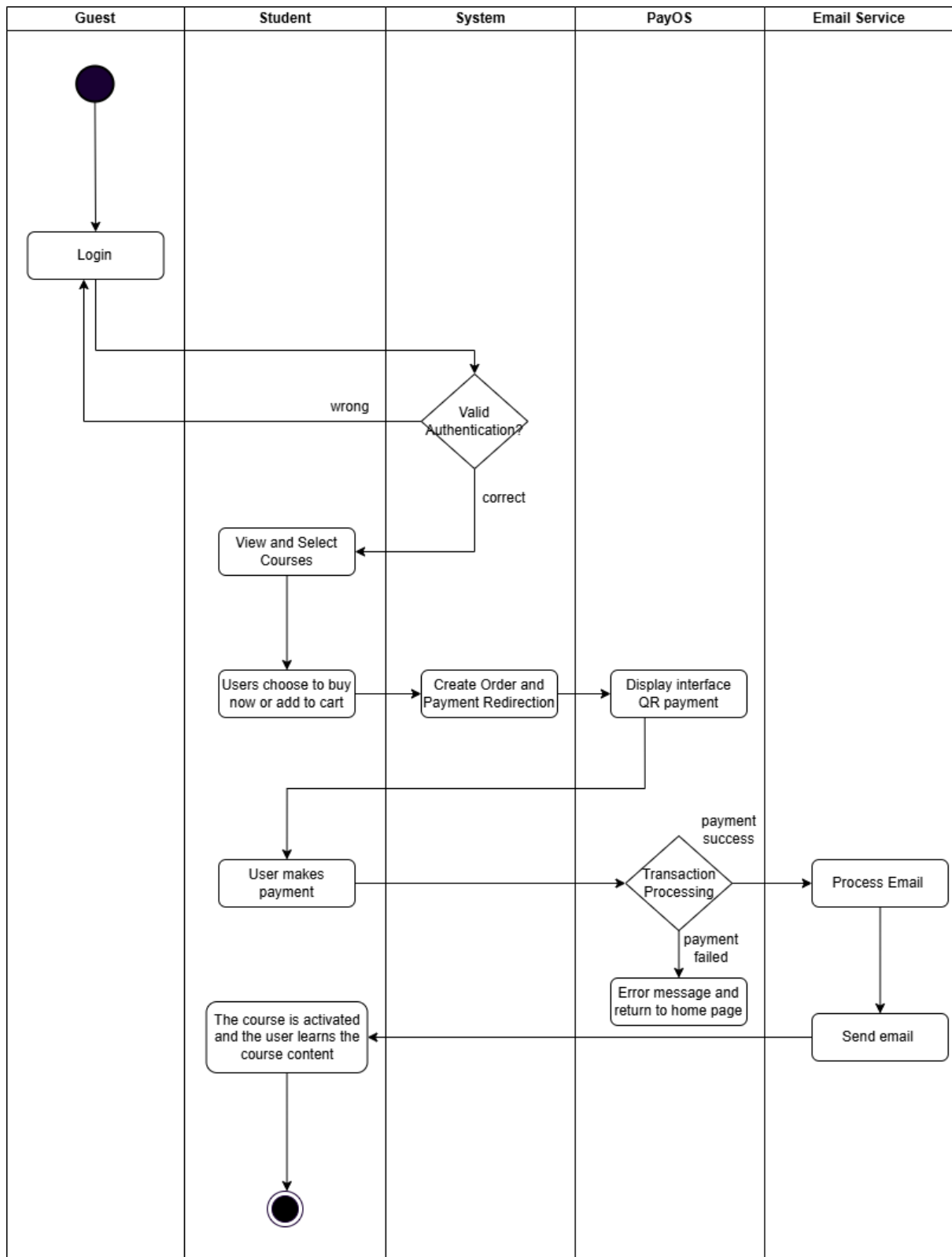


Figure 1.2.3 - Main workflow buy course and learning materials

1.2.4 Create quiz

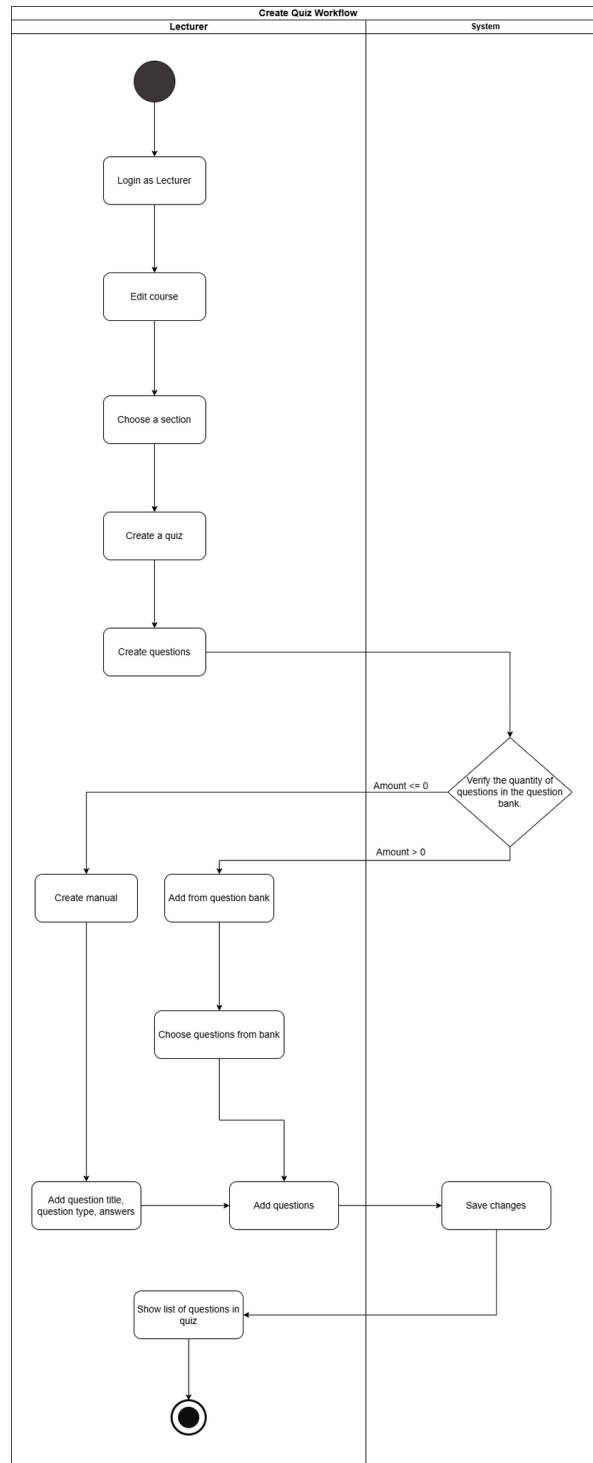


Figure 1.2.4 - Main workflow create quiz

1.2.5 AI Lesson Q&A

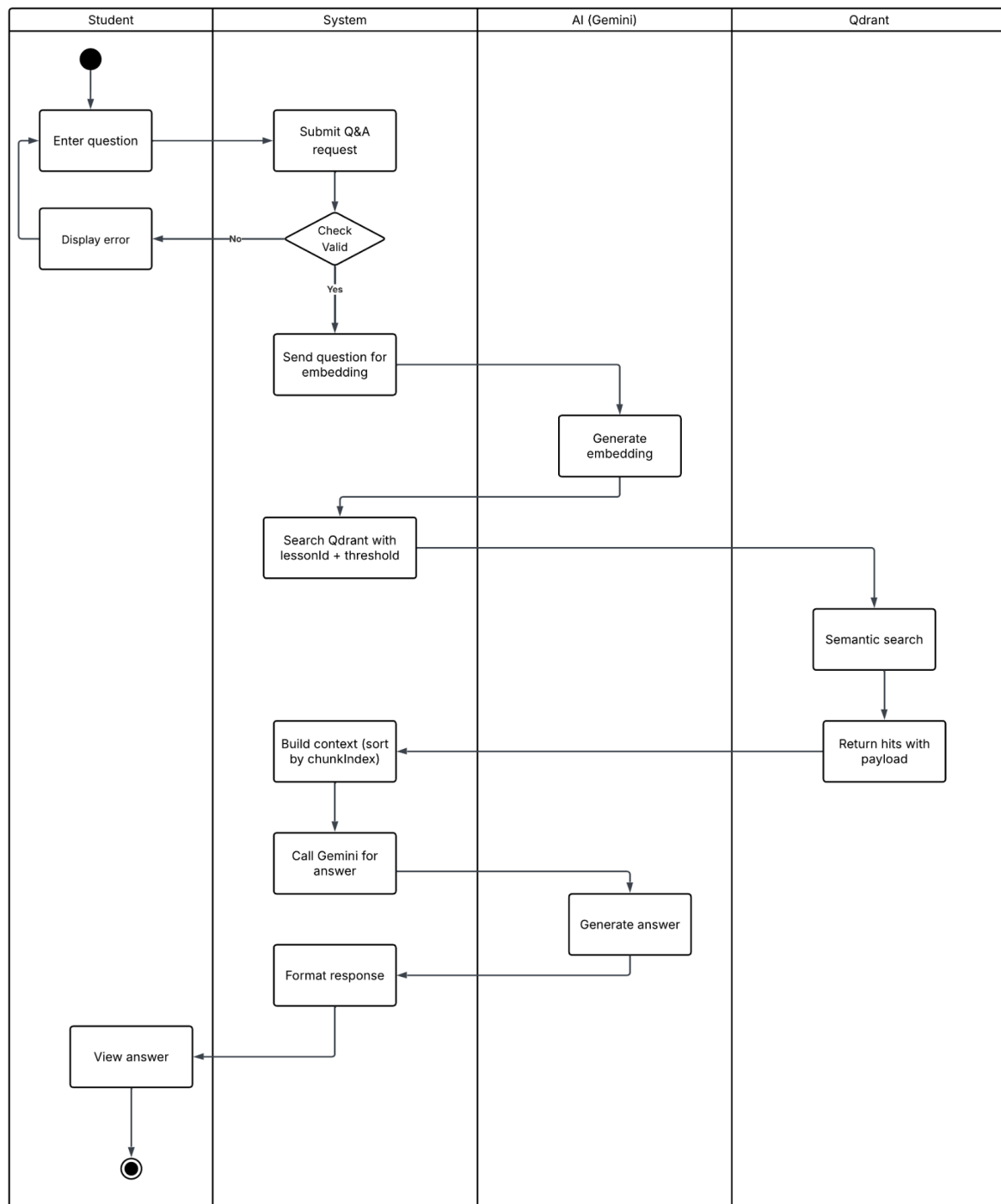


Figure 1.2.5 - Main workflow AI lesson Q&A

1.3 User Requirements

1.3.1 Actors

#	Actor	Description
1	Administrator	An administrator who has access to the system dashboard, with the ability to monitor overall activities, view key metrics, and manage Moderators by adding, updating, or removing their roles and permissions.
2	Guest	A non-registered visitor who can browse the homepage, view available courses, add courses to the shopping cart, and read news updates, but cannot enroll or access premium features without creating an account.
3	Student	A registered user who can enroll in courses, manage their profile, access customer support, recharge their wallet, provide feedback, and navigate their personalized student dashboard.
4	System Moderator	Moderators have content monitoring and moderation rights. They can manage student and lecturer accounts, manage and approve user's tickets, assess lecturer applications, manage their own profile, and view statistics in the System Moderator Dashboard.
5	Content Moderator	A Content Moderator oversees posts, comments, news, categories, and promotional images to ensure all content follows guidelines and keeps the platform safe and organized.
6	Lecturer	A lecturer is a person who creates and manages course content. Can create/edit/delete courses, add lectures (video, text, quiz, slide), manage quiz, view student list, receive revenue payments, and track sales performance via Lecturer Dashboard.

1.3.2 Use Cases (UC)

ID	Use Case	Feature	Use Case Description
01	Login	Common Feature	Login allows users to easily access the course library, track their learning progress, and take advantage of features that support skill development on SkillUp.
02	Register	Common Feature	Register allows new users to create an account on SkillUp by providing personal information. Upon successful registration, users receive a verification email to activate their account and gain access to platform features.
02	Forgot Password	Common Feature	Forgot Password allows users who cannot remember their login credentials to initiate a password recovery process. Users enter their registered email address, and the system sends a password reset link to verify their identity.
03	Reset Password	Common Feature	Reset Password allows users to create a new password after clicking the verification link received

			via email. Users enter and confirm a new password that meets security requirements, updating their account credentials.
04	Change Password	Common Feature	Change Password allows authenticated users to update their current password for security purposes. Users must provide their current password for verification, then enter and confirm a new password meeting system requirement.
05	View Lecturer's Courses	Course Feature	The Lecturer views a comprehensive list of all courses they have created, along with their current status, recent activity, and quick actions.
06	Create New Course	Course Feature	The Lecturer initializes a new course shell by providing basic metadata. This creates a "Draft" course in the system.
07	Update Course & Manage General Settings	Course Feature	The Lecturer updates the core metadata, image, price, and coupon information for an existing course. This Use Case covers the actions within the Course Editor sidebar tabs.
08	Manage Sections	Course Feature	The Lecturer structures the course content by creating, modifying, and deleting sections (chapters) within the "Khung chương trình" tab.
09	Manage Lessons	Course Feature	The Lecturer populates sections with learning units, choosing between a standard Bài học (Video/Text) or a Quiz format, and managing supplementary material.
10	Submit Course for Publish	Course Feature	The Lecturer submits the finalized course for review by a Content Moderator once all core prerequisites are met.
11	Ban/unban course as Moderator Content	Course Feature	Content Moderator manages courses in the system, including viewing course list, approved/rejected pending courses, and ban/unban published courses.
12	Approve/Rejected Course as Moderator Content	Course Feature	Content Moderator manages courses in the system, including viewing course list, approved/rejected pending courses, and ban/unban published courses.
13	View section	Section Feature	The "View Section Detail" use case allows an enrolled Student to view the contents of a specific chapter (section) inside the learning interface. It displays the list of lessons, current progress within that chapter, and allows the student to navigate to specific lessons.
14	View lesson	Lesson Feature	The "View Lesson Content" use case allows an enrolled Student to view the specific content of a lesson (Video, Article, or Quiz). The student can read

			or watch the material, mark the lesson as completed, and navigate to the next or previous lesson.
15	Take quiz	Quiz Feature	The "Take Quiz" use case allows an enrolled Student to attempt a quiz assigned in a course. The student can view questions, select answers, navigate between questions, save progress, and submit the quiz for grading.
16	View detail result	Quiz Feature	The "View Quiz Result Detail" use case allows a Student to review their performance after submitting a quiz. The system displays an overall summary (score, pass/fail status) and a detailed breakdown of each question, showing the user's selected answers versus the correct answers with specific feedback.
17	Add cart item	Cart Feature	The "Add Course to Cart" use case allows a logged-in Student to select a specific course and add it to their shopping cart database. This allows the student to collect multiple courses before proceeding to a single checkout process.
18	Remove cart item	Cart Feature	The "Remove Item from Cart" use case allows a logged-in Student to remove a specific course from their shopping cart if they decide not to purchase it. The system updates the cart total immediately after removal.
19	Apply voucher	Cart Feature	The "Apply Voucher" use case allows a Student to enter a promotional code during the shopping cart review process. If the code is valid, the system calculates the discount amount and deducts it from the total price.
20	Create Lecturer Application	Lecturer Application	The "Submit Lecturer Application" use case allows a newly registered Lecturer to complete their profile by submitting professional details (experience, CV, degrees). This step is mandatory to activate the teaching privileges (create courses).
21	View detail Lecturer Application	Lecturer Application	The "View My Application Detail" use case allows a Lecturer (who has submitted a registration form) to review the details of their application. They can check the current approval status (Pending, Approved, Rejected), review the submitted experience text and credentials, and access the "Edit" function if the application is still pending or was rejected.
22	Manage Lecturer Application	Lecturer Application	The "Manage Lecturer Application" use case allows a System Moderator to review the details of lecturer applications. They can check the current approval

			status (Pending, Approved, Rejected), review the submitted experience text and credentials, and choose to reject or accept the application.
23	Buy course	Shopping cart & payment feature	The "Buy Course" use case allows a logged-in Student to purchase courses. There are two entry points: Direct Purchase (Mua ngay): Buying a single course directly from the detail page. Cart Checkout (Thanh toán giỏ hàng): Buying multiple courses saved in the cart. Upon success, the system enrolls the student in the course(s) and records the transaction.
24	View history transaction	Shopping cart & payment feature	The "View Transaction History" use case allows a logged-in Student to view a chronological list of their financial activities within the platform. This includes wallet top-ups, course purchases, and refunds (if any). It helps users track their spending and verify payment statuses.
25	Feedback course	Shopping cart & payment feature	The "View Transaction History" use case allows a logged-in Student to view a chronological list of their financial activities within the platform. This includes wallet top-ups, course purchases, and refunds (if any). It helps users track their spending and verify payment statuses.
26	View Profile	Profile Management Feature	The "View Lecturer Profile" use case allows a logged-in Lecturer to view their account details. This includes Public Information (visible to students like Title, Profession, Bio) and Private Information (Financial details for payouts like Bank Account).
27	Update Profile	Profile Management Feature	The "Update Lecturer Profile" use case allows a logged-in Lecturer to modify their personal, professional, and financial details. Note that profile picture updates are not supported in this function; users can only update text-based information.
28	Manage Quiz	Quiz Feature	Allows Lecturers to create new quizzes, update existing test information (title , quiz time , passing score) , or delete quizzes in a specific section to assess student knowledge.
29	Add question to quiz	Quiz Feature	Allows Lecturers to add questions to a specific quiz to build the content. Lecturers can either create new

			questions manually or select existing questions from the Question Bank
30	Manage Question	Question Feature	Allows Lecturer to view a list of existing questions in a Quiz, make edits to question content, or delete questions from the Quiz.
31	Manage Question Bank	Question Feature	Allows Lecturers to manage a shared question bank for each Section of the course. Functions include viewing the list, creating new questions, editing, deleting, and importing in bulk from Excel files.
32	Manage Answer	Answer Feature	Allows Lecturers to manage answer options for a specific question. Includes adding answers, editing content, deleting answers, uploading answer illustrations, and configuring correct answers.
33	View Enrolled Students	Answer Feature	Allows Lecturers to view a list of all students enrolled in their courses. Lecturers can search and filter the list by specific course to track learning progress.
34	Manage Profile	User Feature	The “View Profile” use case allows a student & lecturer to view their personal information and account details on the SkillUp platform
35	Manage Ticket for Student and Lecturer	Ticket Feature	The “Manage Ticket” feature allows a student or lecturer to create, view and update their support ticket, which will be sent to a Moderator for consideration.
36	Manage Ticket for System Moderator	Ticket Feature	The "Manage Ticket" feature allows a System Moderator to view and handle all Tickets submitted by the Students and Lecturers.
37	Manage News for Content Moderator	News Feature	Allows a moderator to create, edit, publish, and delete news articles that will be displayed to students and other users on the platform.
38	View News	News Feature	Allows students and guests to view a list of published news articles and click on one to read its full content.
39	View List Post	Post Feature	Content Moderator is authenticated and inside “Quản lý bài đăng”.
40	Manage voucher	Voucher Feature	The Lecturer is authenticated and is on the course update screen, at the "Phiếu giảm giá" step. The Lecturer can view the list of vouchers for the course, create new vouchers, edit existing ones, or delete them.
41	Create Post	Forum Feature	Allows authenticated users to create new discussion posts in forum categories

42	Edit/Delete Post	Forum Feature	Allows post authors to edit or delete their own posts.
43	View Post Detail	Forum Feature	Allows users to view the full content of a forum post, including all comments, likes, and replies in a threaded conversation format.
44	Manage Forum Categories	Forum Feature	Allows Content Moderators to create, edit, or delete forum categories to organize discussions
45	Report Post	Report Feature	Allows users to report inappropriate forum posts that violate community guidelines.
46	Manage Post Reports	Report Feature	Allows Content Moderators to view, investigate, and take action on reported posts.
47	Comment on Lesson	Comment Feature	Allows enrolled Students to post questions or comments on specific lessons within a course.
48	Comment on Post	Comment Feature	Allows authenticated users to reply to forum posts, creating threaded discussions.
49	Edit/Delete Comment	Comment Feature	Allows comment authors to edit or delete their own comments on lessons or posts.
50	Like Comment on Lesson	Comment Feature	Allows enrolled Students to like helpful comments on lessons
51	Like Comment on Post	Comment Feature	Allows users to like comments on forum posts, indicating agreement or appreciation.
52	Manage Comment Reports	Comment Feature	Allows Content Moderators to review reported comments, view context, and take action
53	View Courses by SubCategory	Course Feature	Allows users to filter and view courses within a specific subcategory
54	View Course Detail	Course Feature	Displays comprehensive course information including description, curriculum outline, instructor profile, ratings, reviews, preview videos, and enrollment button.
56	Report Course	Report Feature	Allows enrolled students to report issues with courses such as outdated content, misleading information, copyright violations, or poor quality.
57	Manage Course Reports	Report Feature	Allows Content Moderators to review course reports, investigate claims, contact instructors for corrections, and take action
58	View Notifications	Notification Feature	Allows authenticated users to view real-time notifications about course updates, comment replies,

			ticket responses, application status, purchases, and system announcements.
59	Comment in Lesson	Comment Feature	Allows enrolled students to use real-time chat during lesson viewing to ask questions and interact with other students currently watching the same lesson.
60	Auto-Generate Subtitles	AI Feature	Allows Lecturers to automatically generate subtitles for uploaded videos using AI services.

1.3.3 Use Case Diagrams

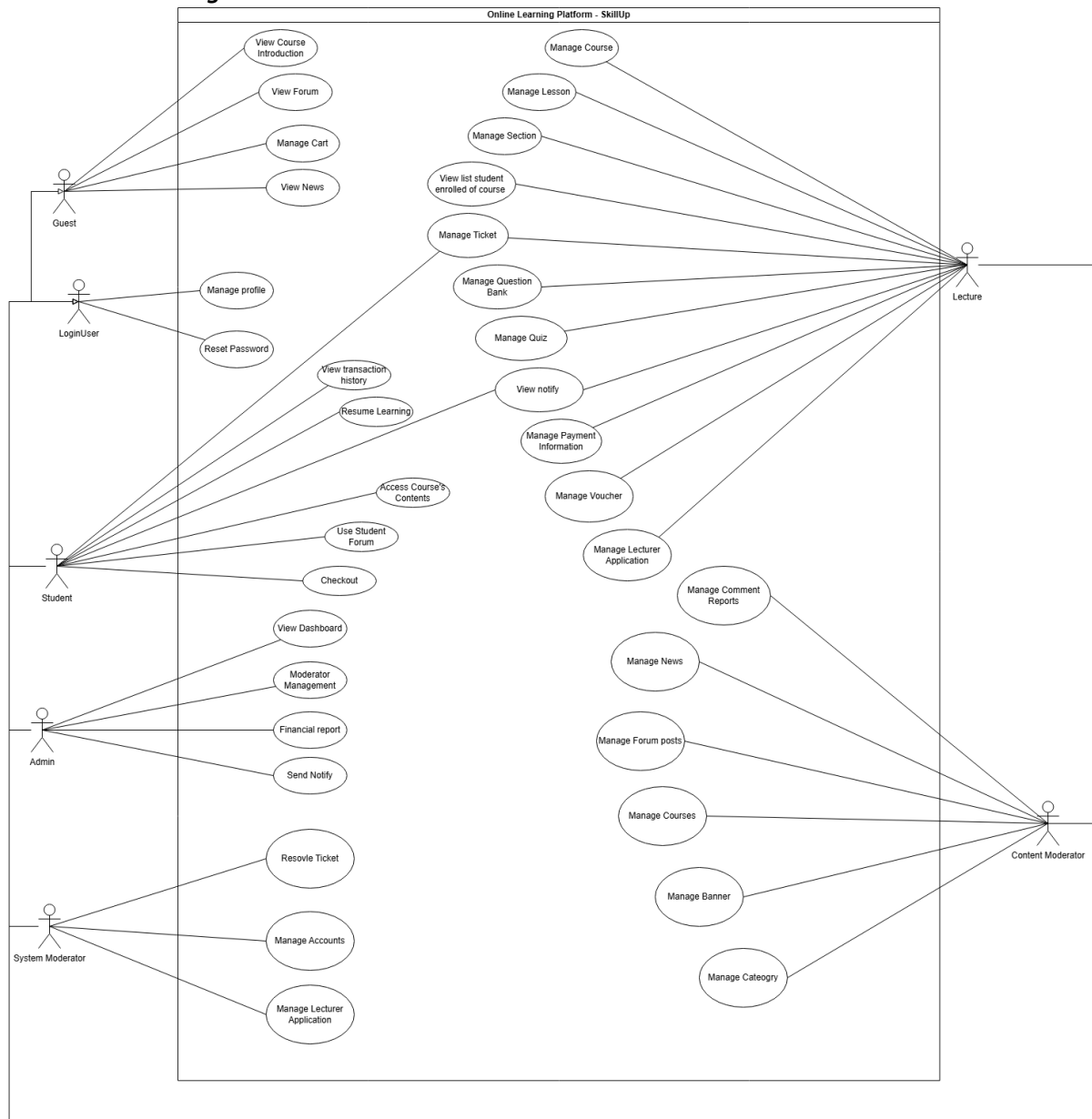


Figure 1.3.3 - Overview use case diagram

1.3.3.1 UCs for Logged In User

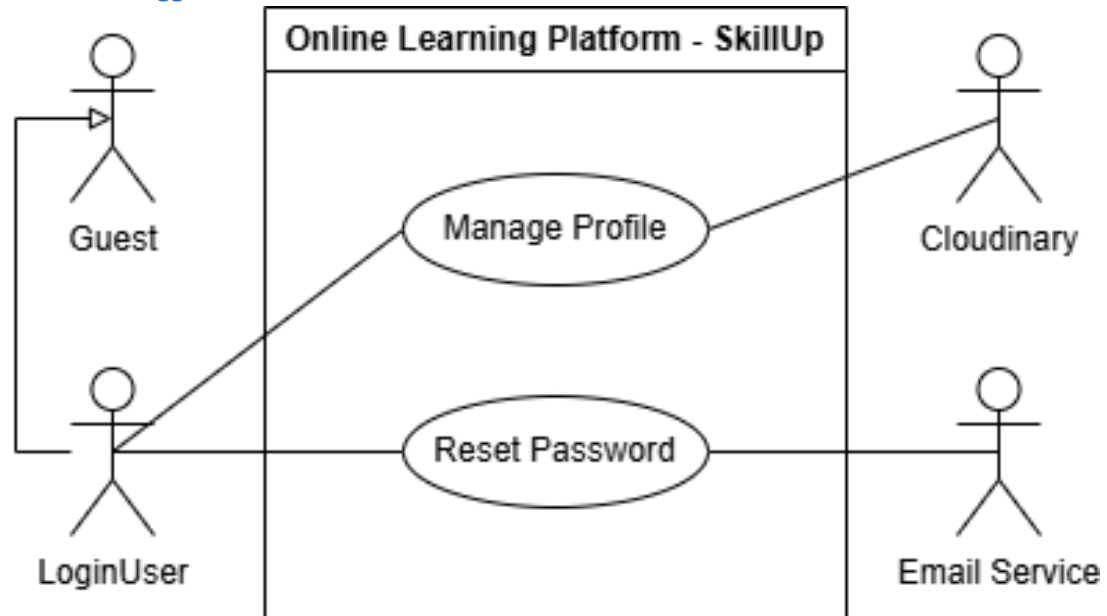


Figure 1.3.3.1 - use case logged in user diagram

1.3.3.2 UCs for Guest

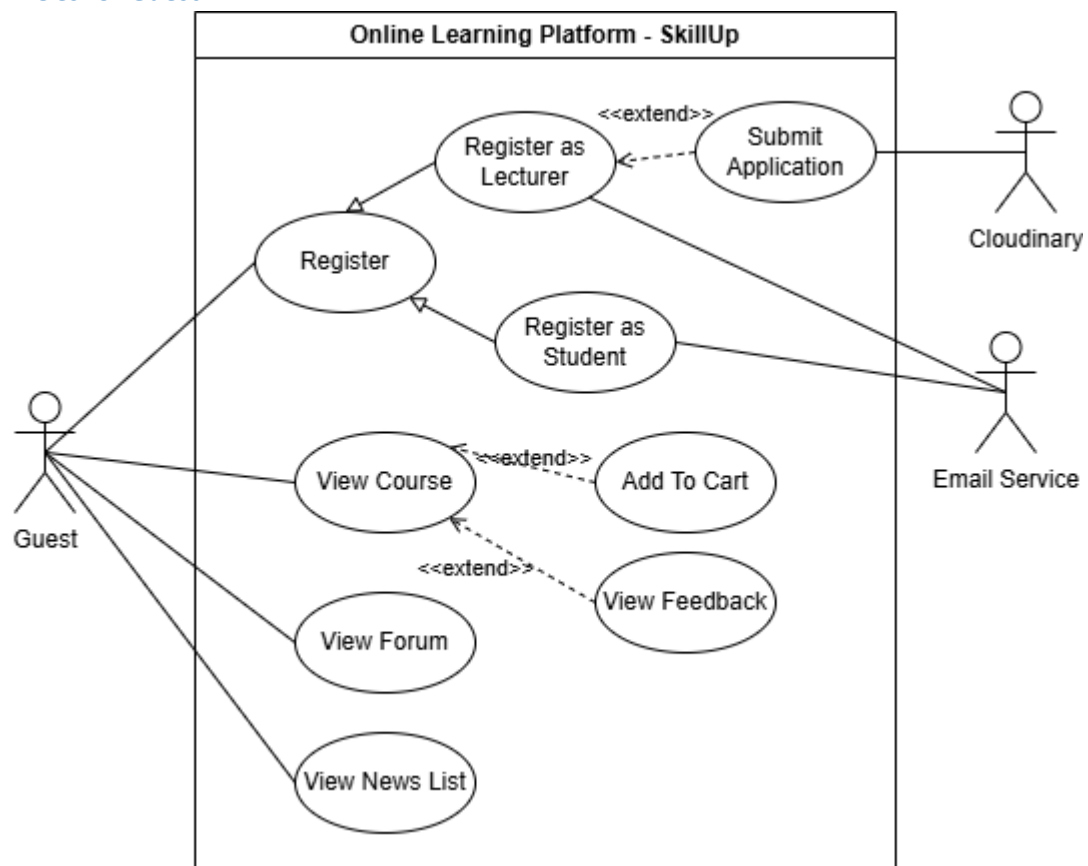


Figure 1.3.3.2 - use case guest diagram

1.3.3.3 UCs for Student

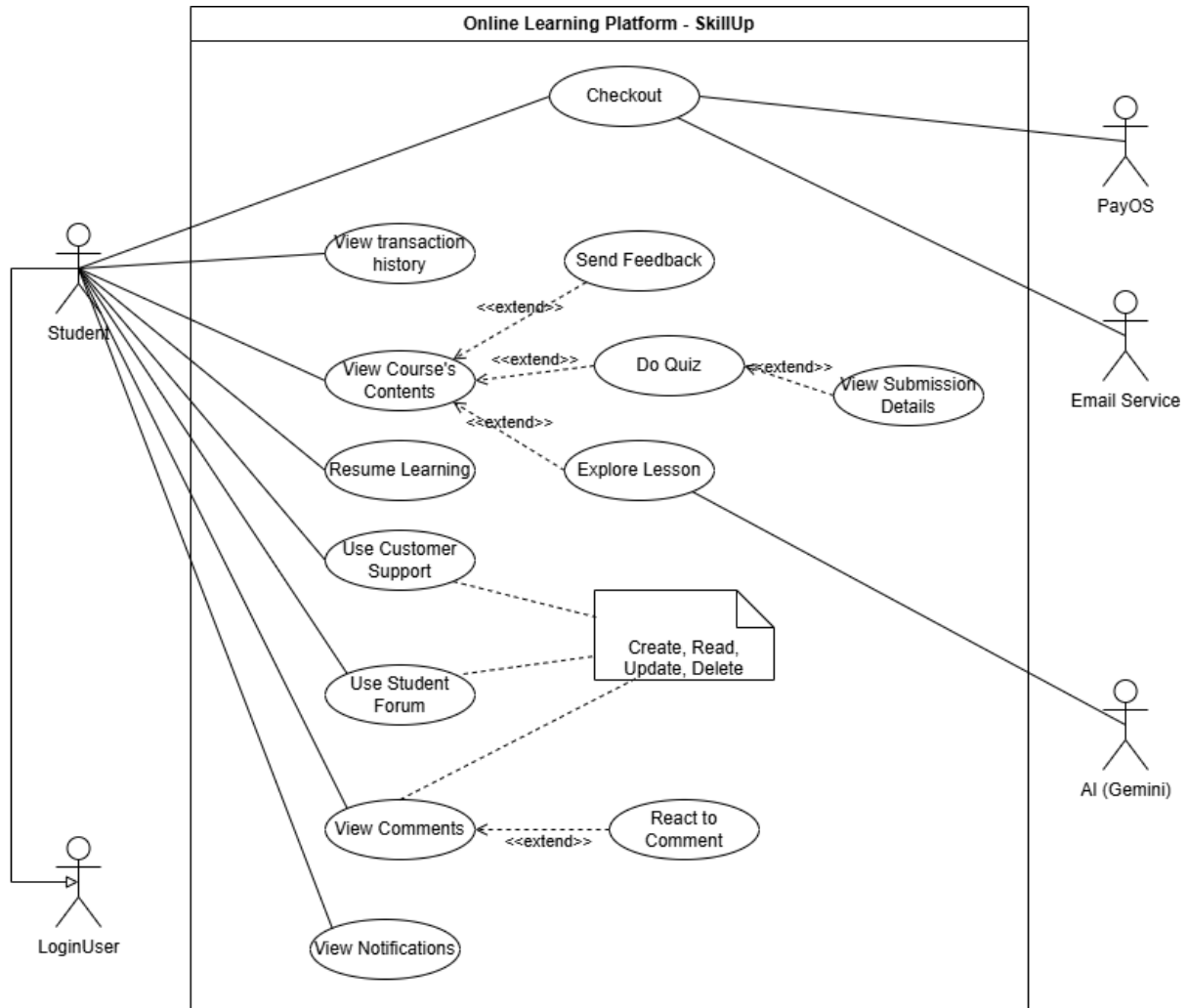


Figure 1.3.3.3 - use case student diagram

1.3.3.4 UCs for Lecturer

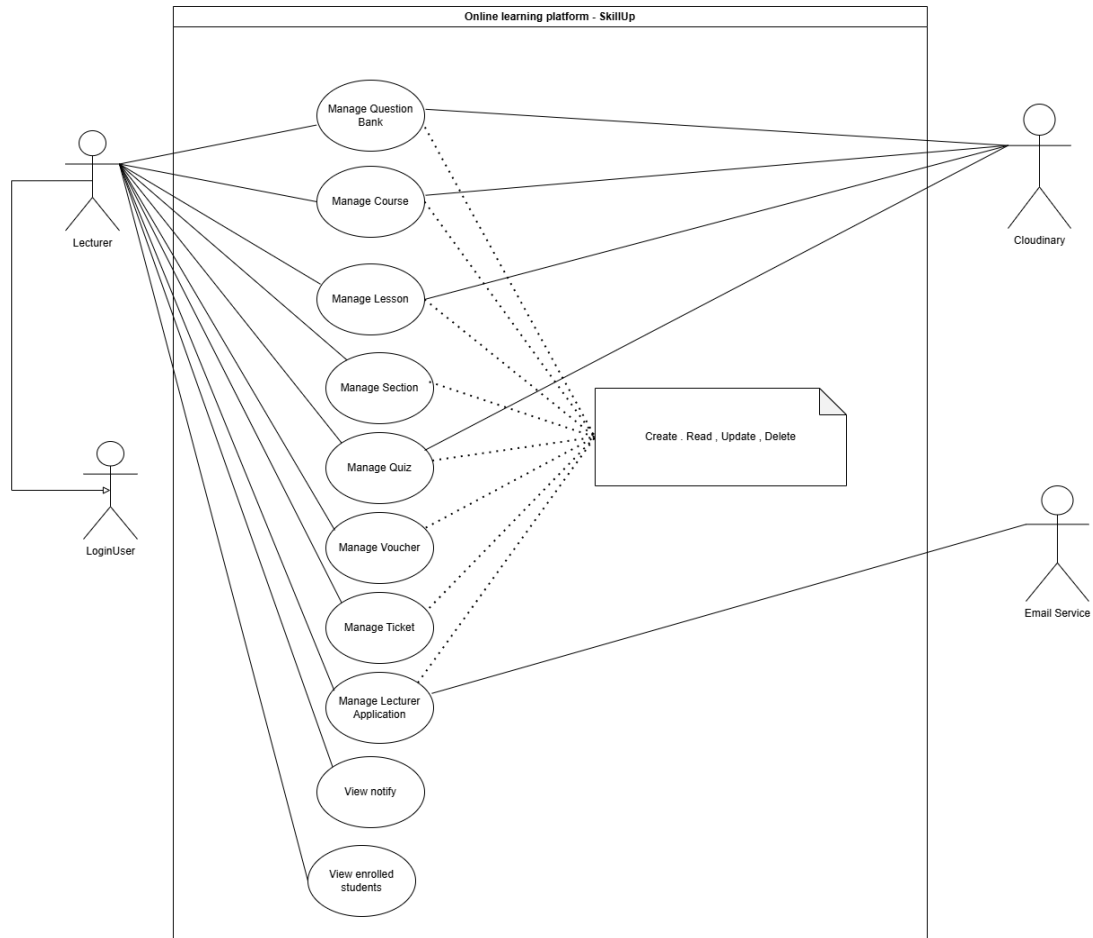


Figure 1.3.3.4 - use case lecturer diagram

1.3.3.5 UCs for Content Moderator

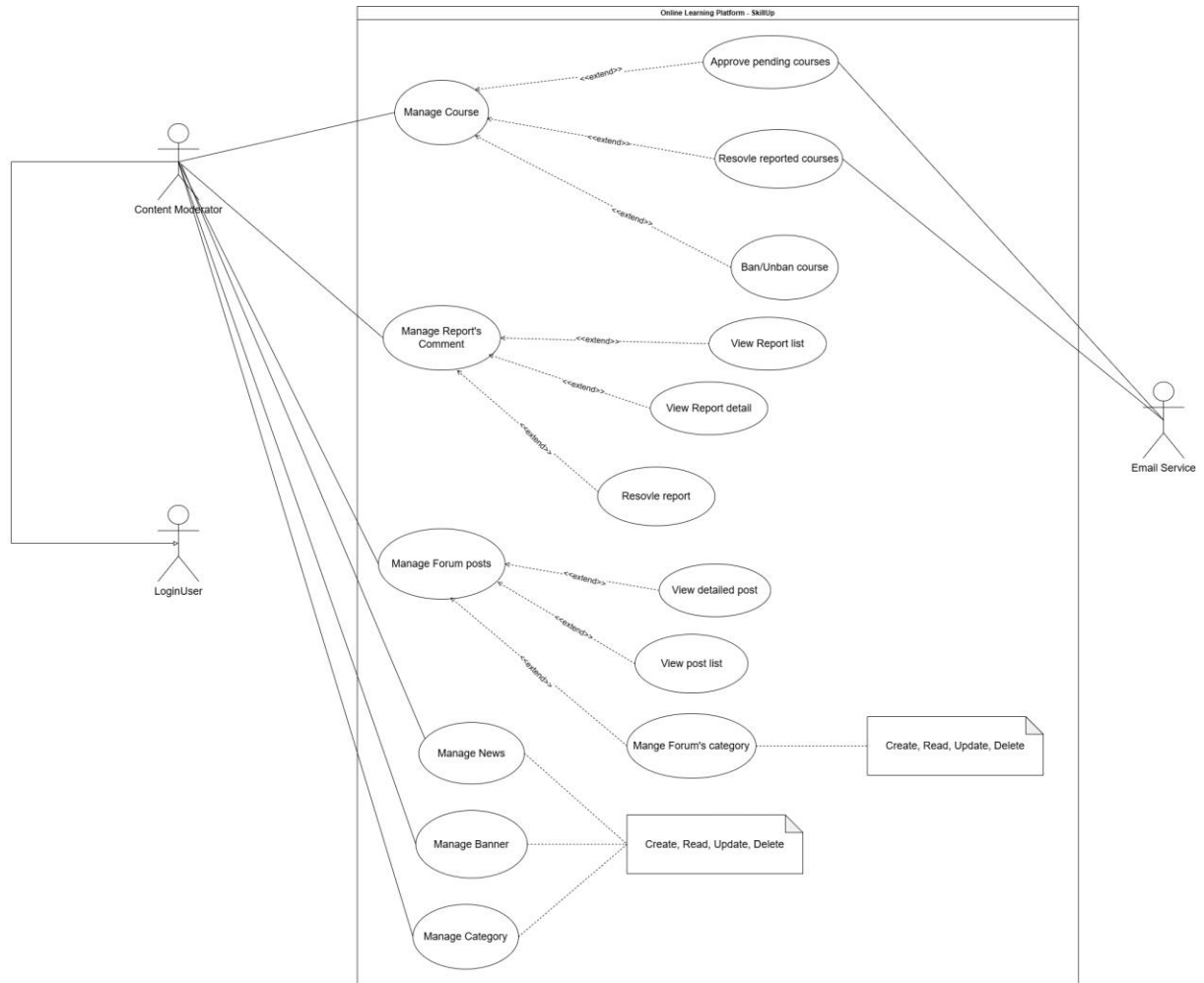


Figure 1.3.3.5 - use case content moderator diagram

1.3.3.6 UCs for System Moderator



Figure 1.3.3.6 - use case system moderator diagram

1.3.3.7 UCs for Admin

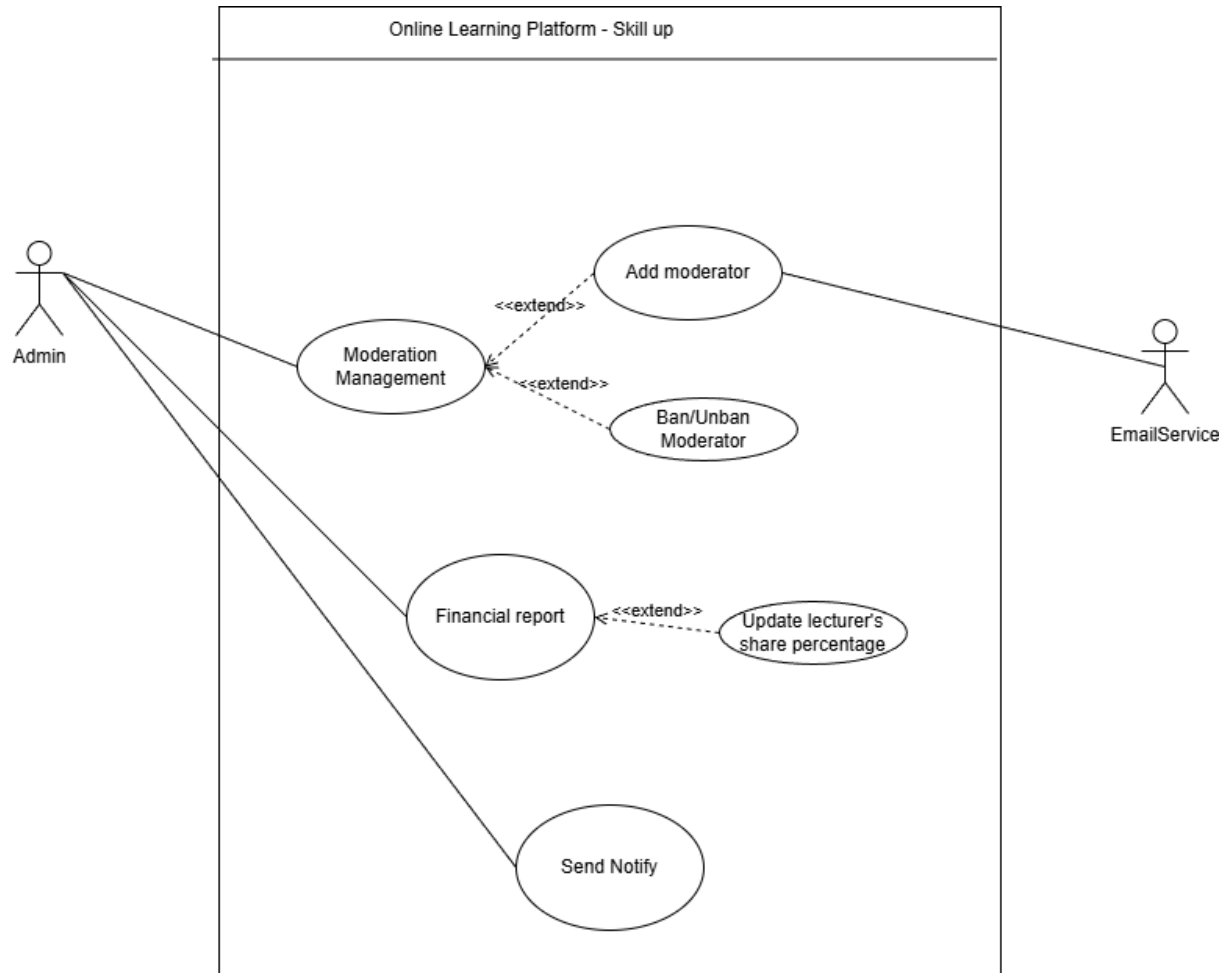


Figure 1.3.3.7 - use case admin diagram

1.4 System Functionalities

1.4.1 Screens Flow

1.4.1.1 Guest Screens Flow

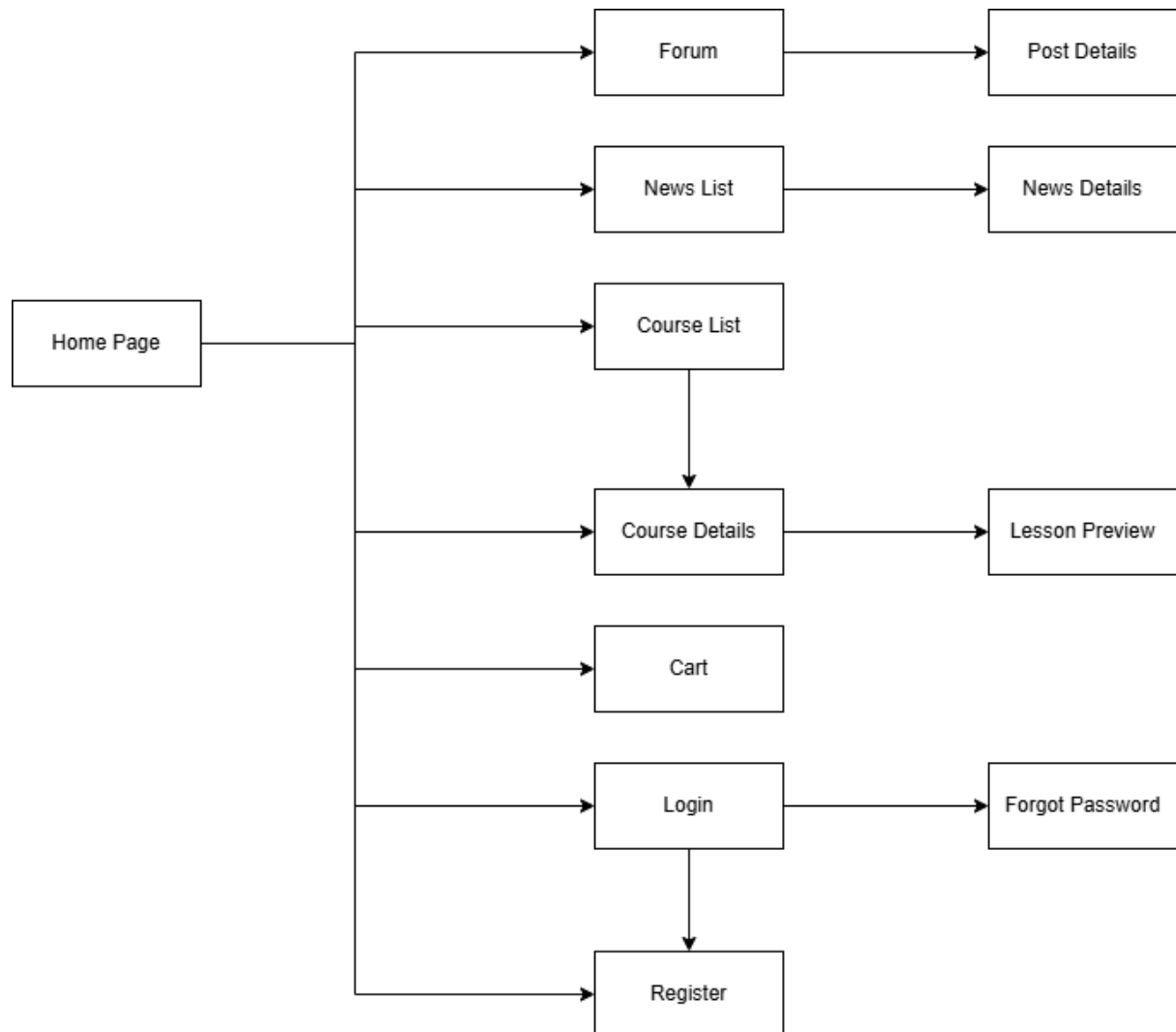


Figure 1.4.1.1 - Guest Screens Flow

1.4.1.2 Content Moderator Screens Flow

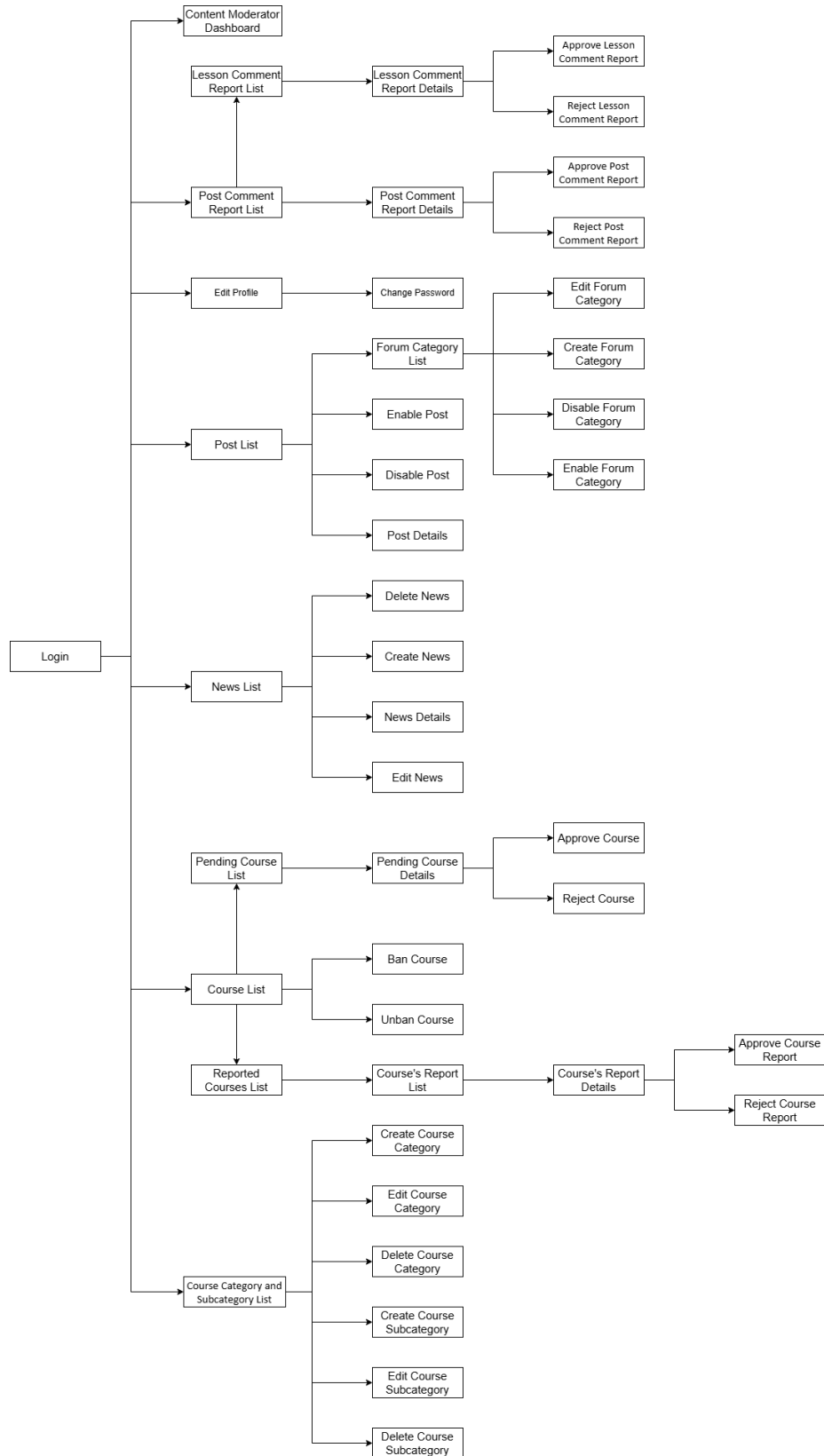


Figure 1.4.1.2 - Content moderator Screens Flow

1.4.1.3 Lecturer Screens Flow

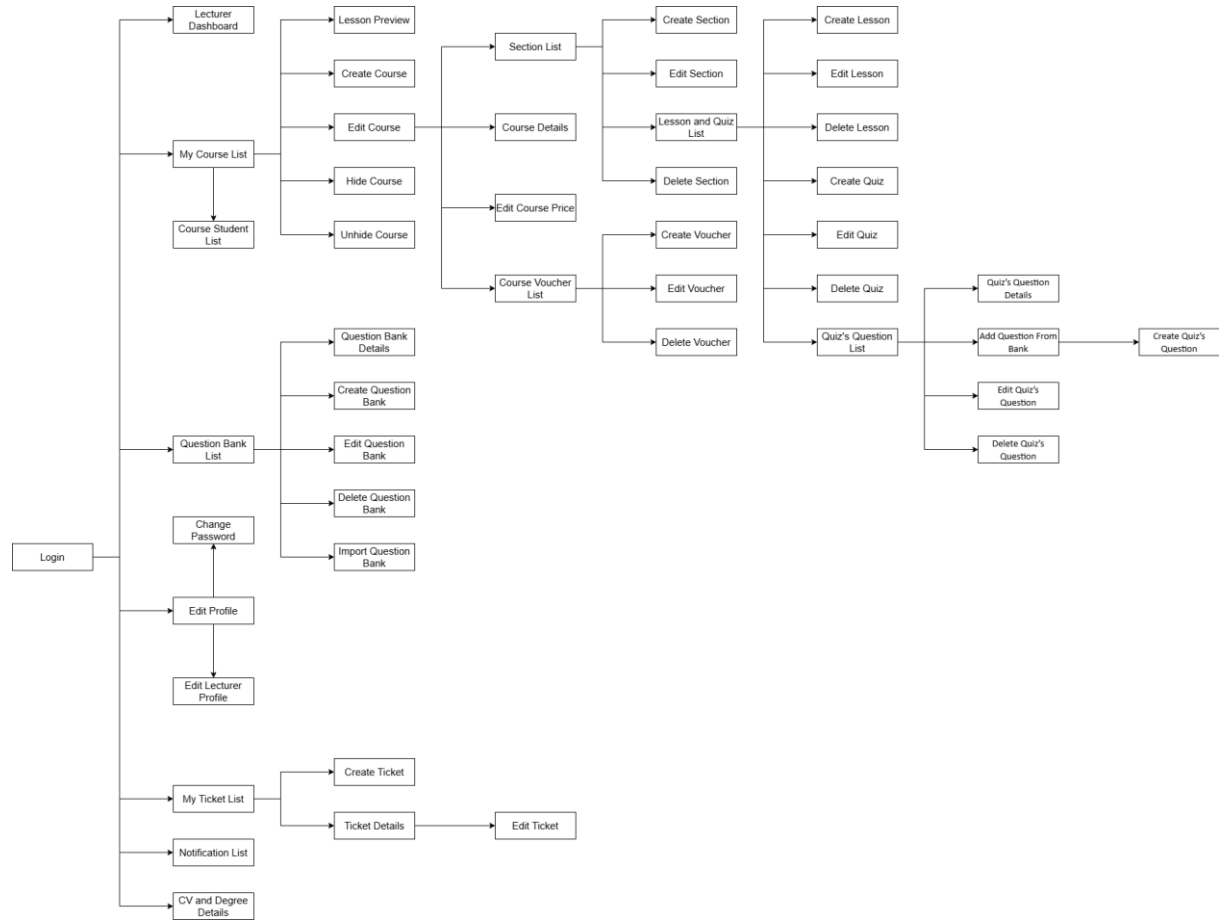


Figure 1.4.1.3 - Lecturer Screens Flow

1.4.1.4 Admin Screens Flow

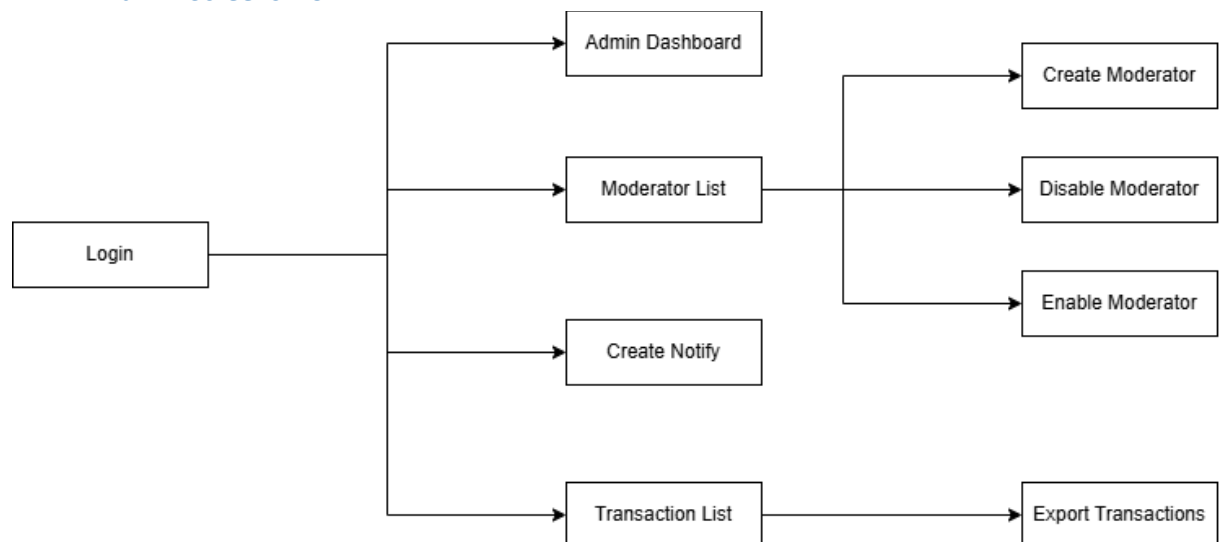


Figure 1.4.1.4 - Admin Screens Flow

1.4.1.5 System Moderator Screens Flow

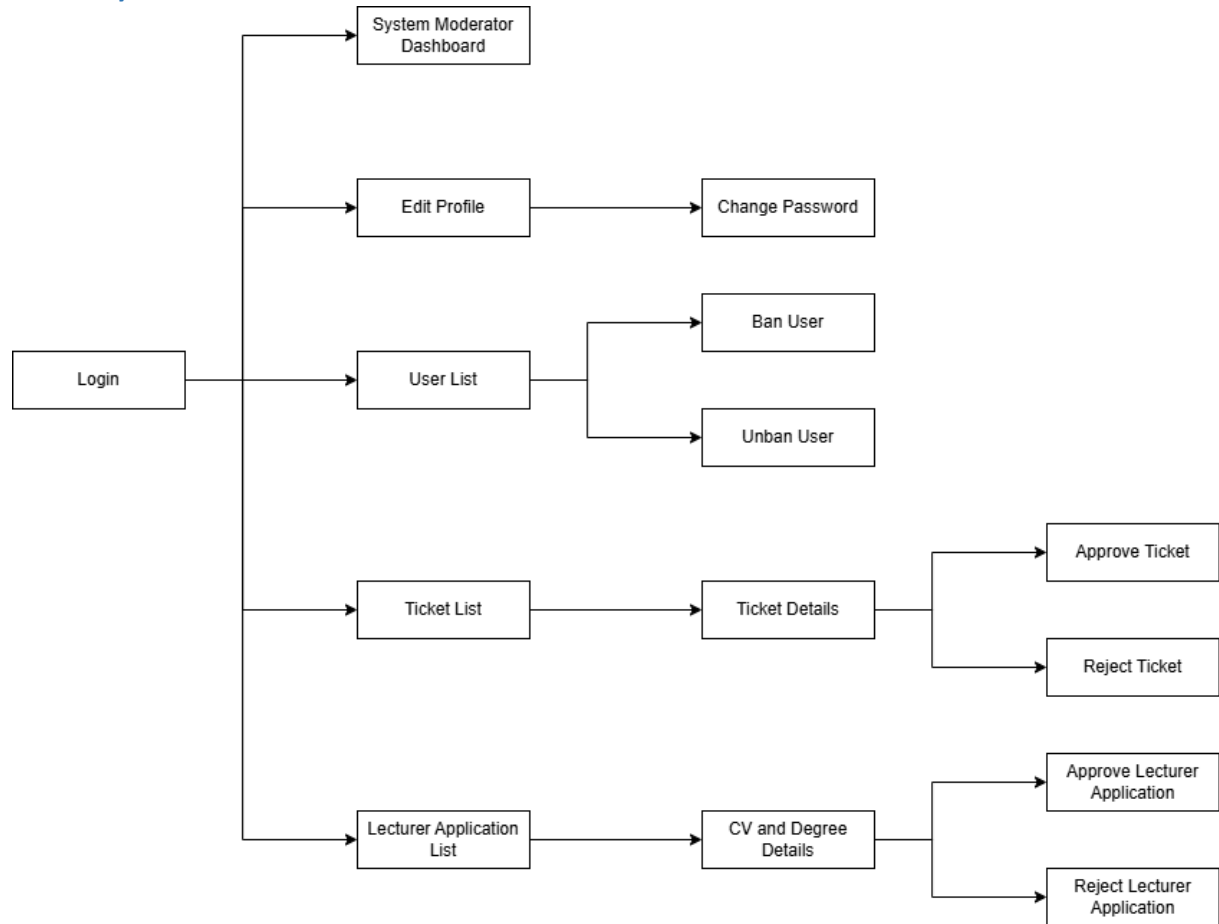


Figure 1.4.1.5 - System moderator Screens Flow

1.4.1.6 Student Screens Flow

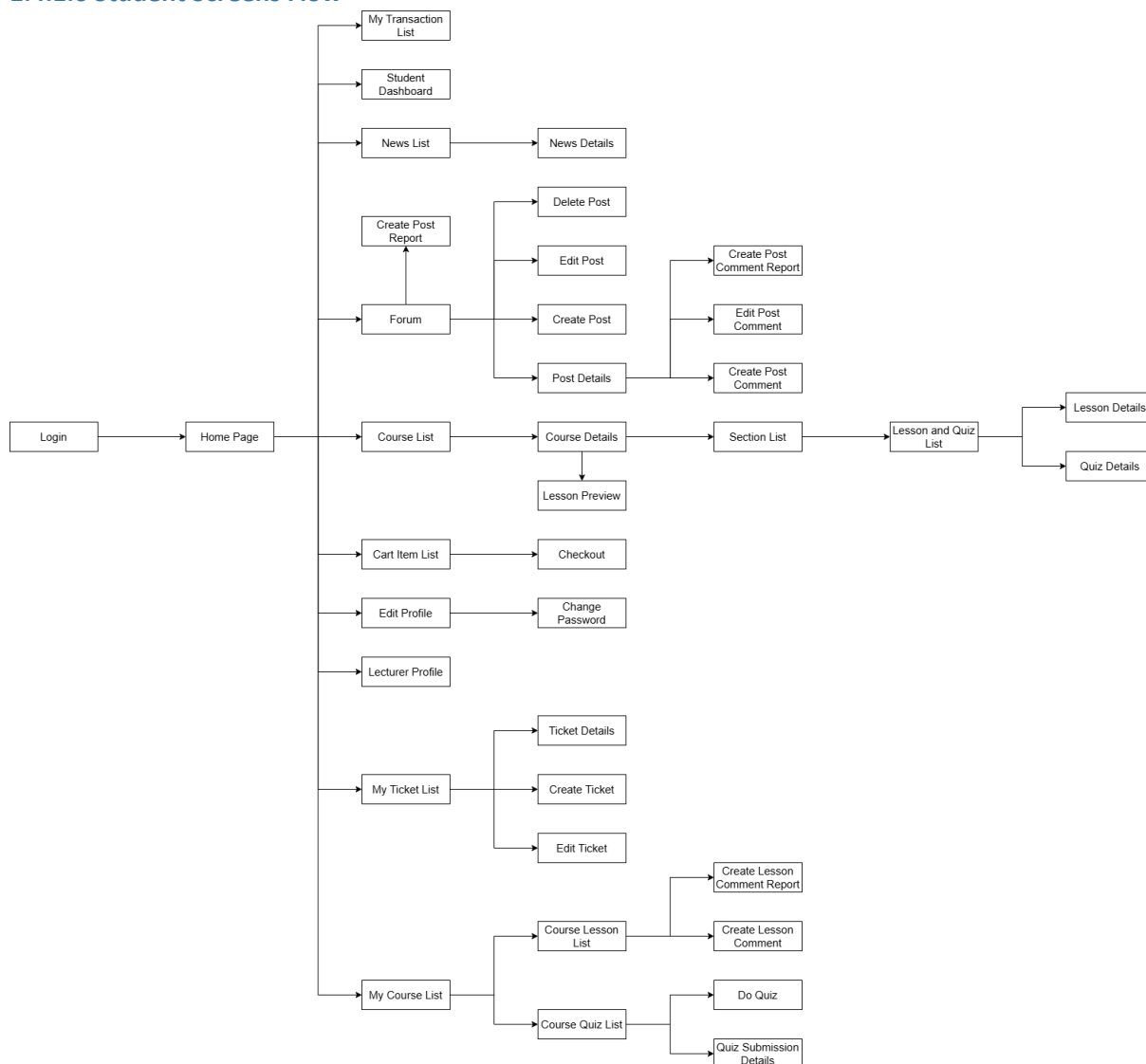


Figure 1.4.1.6 - Student Screens Flow

1.4.2 Screen Authorization

The following table illustrates the authorization structure for the system's screens. Each screen may be accessible by one or more user roles, with an "x" marking where access is permitted. This helps clarify role restrictions and ensures that users only interact with functions relevant to their assigned permissions.

Screen	Guest	Admin	System Moderator	Content Moderator	Lecturer	Student
Home Page	x					x
Login	x	x	x	x	x	x
Forgot Password	x					
Register	x					
Forum	x					x
Post Details	x			x		x
News List	x			x		x

News Details	x			x		x
Course List	x			x		x
Ban Course				x		
Unban Course				x		
Pending Course List				x		
Pending Course Details				x		
Course Details	x				x	x
Lesson Preview	x				x	x
Cart Item List	x					x
Content Moderator Dashboard				x		
Create News				x		
Edit News				x		
Delete News				x		
Approve Course				x		
Reject Course				x		
Reported Courses List				x		
Reported Courses Details				x		
Course's Report List				x		
Course's Report Details				x		
Approve Course Report				x		
Reject Course Report				x		
Post List				x		
Disable Post				x		
Enable Post				x		
Forum Category List				x		
Create Forum Category				x		
Edit Forum Category				x		
Disable Forum Category				x		
Enable Forum Category				x		
Post Comment Report List				x		
Post Comment Report Details				x		
Approve Post Comment Report				x		
Reject Post Comment Report				x		
Lesson Comment Report List				x		

Lesson Comment Report Details				x		
Approve Lesson Comment Report				x		
Reject Lesson Comment Report				x		
Course Category and Subcategory List				x		
Create Course Category				x		
Edit Course Category				x		
Delete Course Category				x		
Create Course Subcategory				x		
Edit Course Subcategory				x		
Delete Course Subcategory				x		
Lecturer Dashboard					x	
Hide Course					x	
Unhide Course					x	
Create Course					x	
Edit Course					x	
Create Section					x	
Section List					x	x
Edit Section					x	
Delete Section					x	
Lesson and Quiz list					x	x
Lesson Details						x
Create Lesson					x	
Edit Lesson					x	
Delete Lesson					x	
Quiz Details						x
Create Quiz					x	
Edit Quiz					x	
Delete Quiz					x	
Course Voucher List					x	
Edit Voucher					x	
Create Voucher					x	
Delete Voucher					x	
Edit Course Price					x	
Question Bank List					x	
Question Bank Details					x	
Create Question Bank					x	

Edit Question Bank					x	
Delete Question Bank					x	
Import Question Bank					x	
Quiz's Question List					x	
Quiz's Question Details					x	
Create Quiz's Question					x	
Edit Quiz's Question					x	
Delete Quiz's Question					x	
Add Question From Bank					x	
Edit Profile			x	x	x	x
Lecturer Profile						x
Edit Lecturer Profile					x	
Change Password			x	x	x	x
System Moderator Dashboard			x			
Course Student List					x	
User List			x			
Ban User			x			
Unban User			x			
Ticket List			x			
Create Ticket					x	x
My Ticket List					x	x
Ticket Details			x		x	x
Edit Ticket					x	x
Approve Ticket			x			
Reject Ticket			x			
Lecturer Application List			x			
CV and Degree Details			x		x	
Approve Lecturer Application			x			
Reject Lecturer Application			x			
Admin Dashboard		x				
Moderator List		x				
Create Moderator		x				
Disable Moderator		x				
Enable Moderator		x				
Create Notify		x				
Student Dashboard						x
Create Post						x
Edit Post						x
Delete Post						x
Create Post Report						x

Create Post Comment						x
Edit Post Comment						x
Create Post Comment Report						x
My Course List					x	x
Create Lesson Comment						x
Delete Lesson Comment						x
Like Lesson Comment						x
Create Lesson Comment Report						x
Create Quiz Submission						x
Quiz Submission Details						x
Transaction List		x				
Export Transactions		x				
My Transaction List						x
Notification List					x	x

1.4.3 Non-UI Functions

#	Feature	System Function	Description
1	Email	Send Email	Send email with contents that are defined within workflow
2	Authentication	Authentication	Used to authentication users within system
3	AI (Gemini)	AI system	Supporting user experience(e.g recommend course, giving hint quiz,..)

1.5 Entity Relationship Diagram

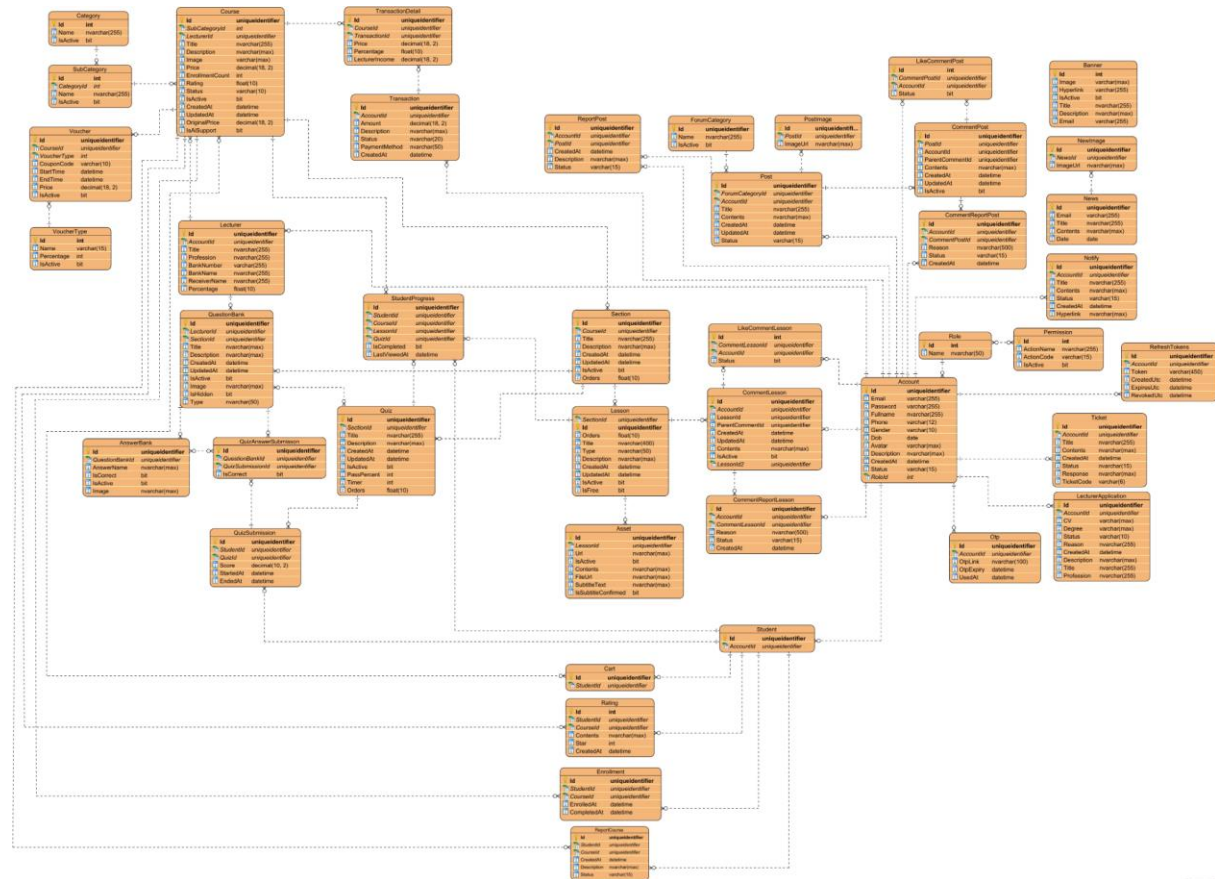


Figure 1.5.1 - Overview entity relationship diagram

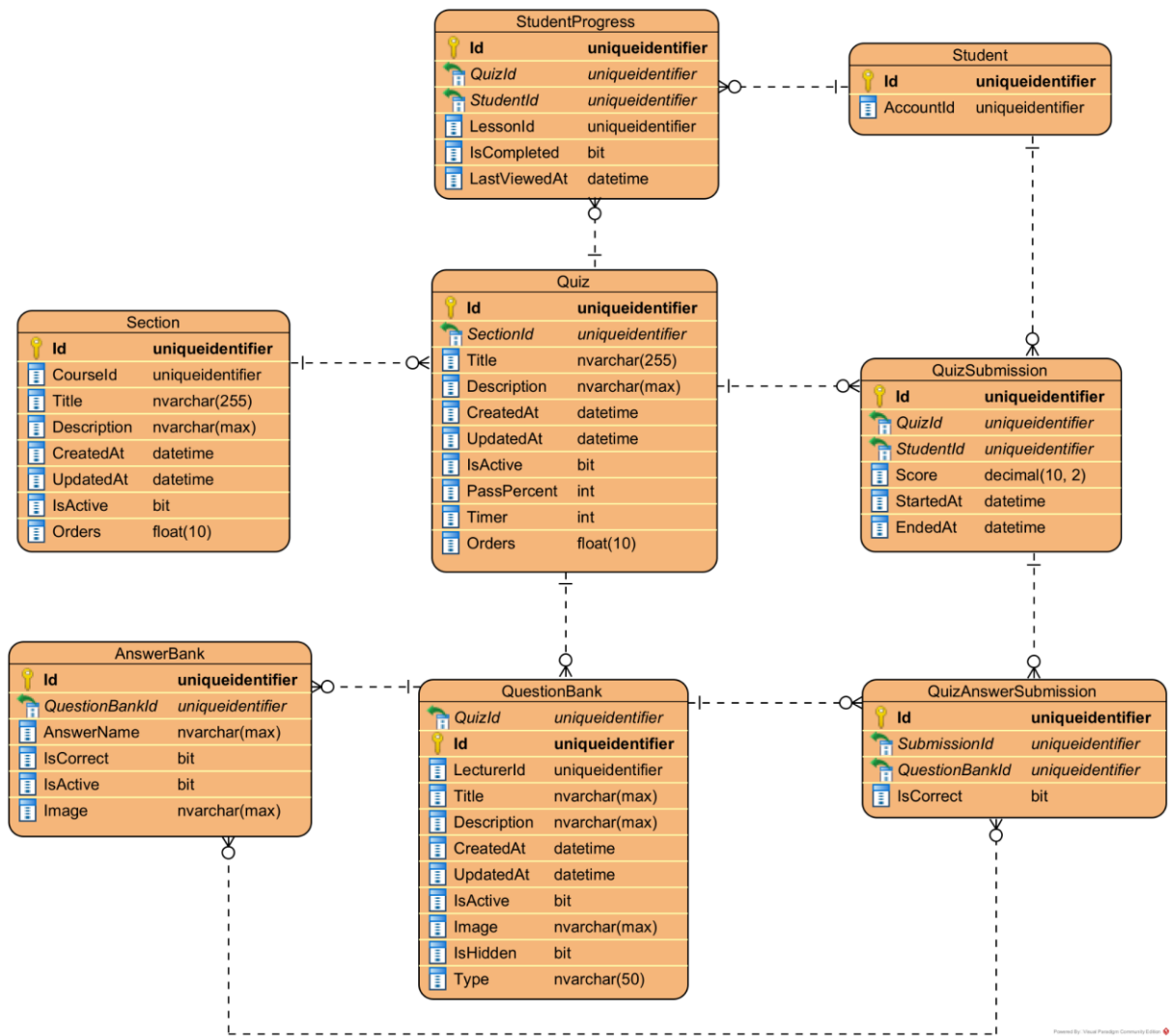


Figure 1.5.2 - Quiz entity relationship diagram

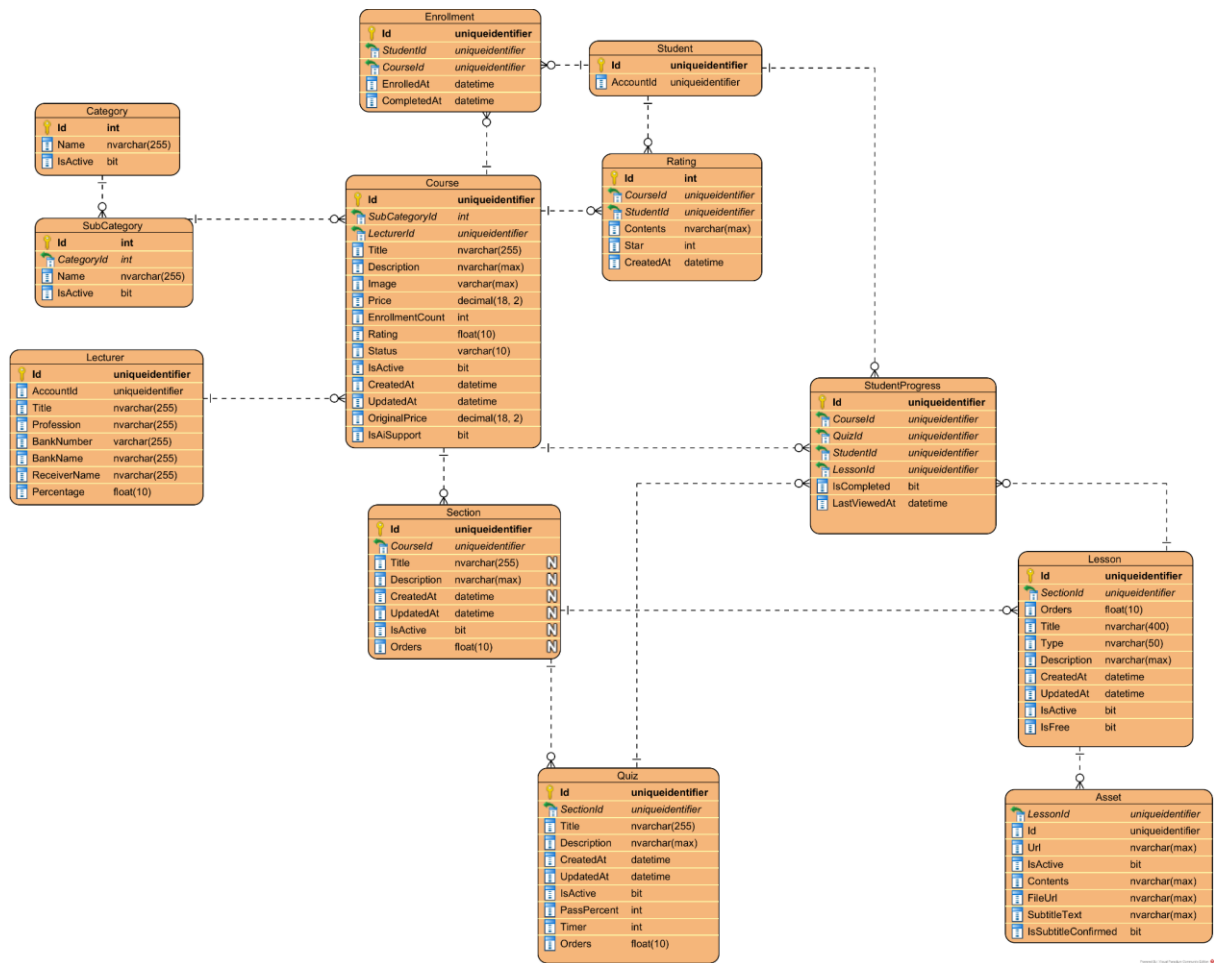


Figure 1.5.3 - Course relationship diagram



Figure 1.5.4 - LecturerApplication relationship diagram

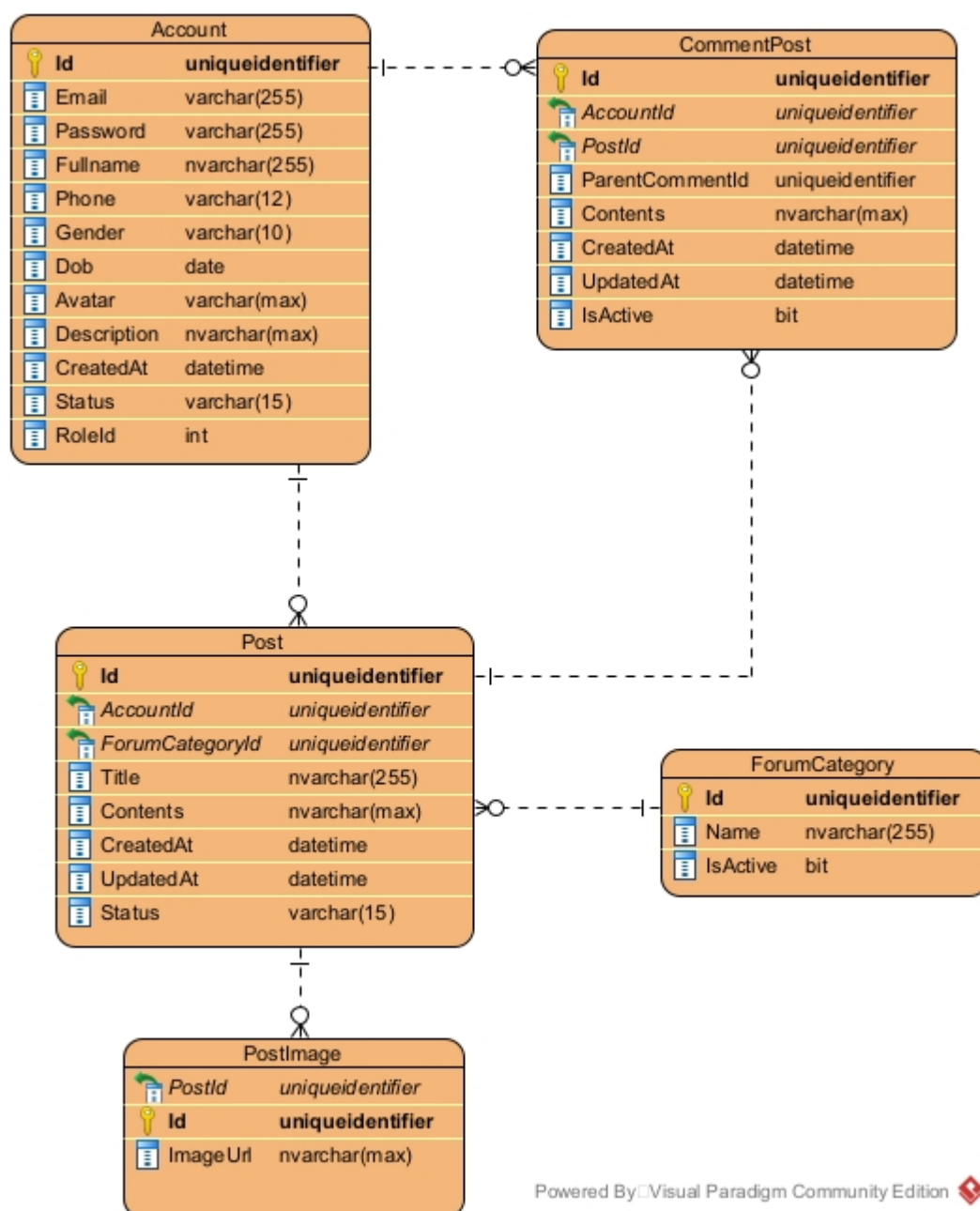


Figure 1.5.5 - Forum relationship diagram

Entities Description

#	Entity	Description
1	Category	Defines high-level classifications for organizing courses and posts.
2	SubCategory	Represents sub-classifications belonging to a specific category.
3	Voucher	Represents discount vouchers that can be applied to course purchases
4	VoucherType	Defines types of vouchers, including discount rules and conditions.
5	Course	Represents an online course with details such as title, price, and status.

#	Entity	Description
6	TransactionDetail	Stores detailed information of each course within a transaction.
7	Transaction	Records payment transactions made by users for course enrollments.
8	Lecturer	Represents instructors who create and manage courses.
9	QuestionBank	Stores questions used for quizzes and assessments.
10	StudentProgress	Tracks a student's learning progress within enrolled courses.
11	AnswerBank	Stores possible answers for questions in the question bank.
12	QuizAnswerSubmission	Records answers submitted by students for quiz questions.
13	Quiz	Represents a quiz associated with a course or lesson...
14	QuizSubmission	Stores quiz submission results of students, including scores and status.
15	ReportPost	Records reports submitted by users for inappropriate posts.
16	ForumCategory	Defines categories used to organize discussion forum posts.
17	PostImage	Stores images associated with forum posts.
18	Post	Represents discussion posts created by users in forums.
19	Section	Divides a course into structured sections or modules.
20	LikeCommentLesson	Stores like on comments related to lessons.
21	Lesson	Represents individual lessons within a course section.
22	CommentLesson	Stores user comments on lessons.
23	Asset	Stores learning materials such as videos, documents, or files.
24	CommentReportLesson	Records reports submitted for inappropriate lesson comments.
25	Cart	Stores courses added to a user's cart before checkout.
26	Rating	Stores user ratings and reviews for courses.
27	Enrollment	Records student enrollments in courses.
28	ReportCourse	Stores reports submitted by users for inappropriate courses.
29	Student	Represents learners enrolled in courses on the platform.
30	LikeCommentPost	Stores like on comments related to forum posts.
31	CommentPost	Stores user comments on forum posts.
32	CommentReportPost	Records reports submitted for inappropriate post comments.
33	Banner	Stores banner images displayed on the platform.
34	NewImage	Stores images associated with news or announcements.
35	News	Represents news or announcements published on the platform.
36	Notify	Stores system notifications sent to users.
37	Role	Defines user roles such as Admin, Lecturer, or Student.
38	Permission	Defines access permissions assigned to roles.
39	RefreshTokens	Stores refresh tokens used for authentication.
40	Account	Stores user account information and authentication data.
41	Ticket	Represents support tickets submitted by users.
42	Otp	Stores one-time passwords used for verification processes.
43	LecturerApplication	Stores applications submitted by users to become lecturers.

2. Use Case Specifications

2.1. Common Features

2.1.1 Login

Primary Actors	User	Secondary Actors	Google OAuth Service
Description	Login allows users to easily access the course library, track their learning progress, and take advantage of features that support skill development on SkillUp.		
Preconditions	• The user has a valid account in the system • The system is operating normally • A stable internet connection is available		
Postconditions	• The user logs in successfully • The user is redirected to the main page		
Normal Sequence/Flow	Login: 1. User clicks Login button from the page header or accesses an authenticated feature. 2. System shows the User Login screen. 3. User types in the login details ("Email", "Mật khẩu"). 4. User clicks the "Đăng nhập" button. 5. System validates required fields and email format. 6. System verifies the credentials against the database. 7. System checks account status (Active, InActive, Pending, Banned). 8. System issues access/refresh tokens and grants access. 9. System logs the login event and merges any guest cart items. 10. System the redirect path based on the user role and shows a success message (MSG01). 11. User is redirected to the appropriate page (Home or role dashboard).		
Alternative Sequences/Flows	Step 2.1_Google Login 1. User clicks the "Tiếp tục sử dụng dịch vụ bằng Google" button. 2. System redirects to the Google OAuth consent screen. 3. User types in the Google account details and chooses to login 4. Google validates user's login information successfully and redirect him/her back to the system 5. Return to step 10 of normal flow (MSG02) Step 6_Invalid credentials: 1. Email or password is incorrect. 2. System shows "Đăng nhập thất bại. Vui lòng thử lại!" (MSG03) 3. Return to step 3 of the normal flow. Step 7.1_Email not verified: 1. Credentials are correct, but email is not verified.		

	2. System shows "Tài khoản chưa được kích hoạt" (MSG04) 3. Return to step 3 of the normal flow. Step 7.2_Account banned: 1. Account is banned. 2. System shows "Tài khoản chưa được kích hoạt" (MSG04) 3. Return to step 3 of the normal flow.
Exception	E1_System/Database Error ● Trigger: Occurs at Normal Flow Step 6, 7, 8, or 9 (during validation, verification, or token issuance). ● Sequence: 1. The system encounters an internal error (e.g., database timeout, server 500 error, connection failure). 2. System logs the error details for administrators. 3. System displays a generic error message: "Đã có lỗi xảy ra từ hệ thống. Vui lòng thử lại sau." (MSG-SYS-01). 4. System clears the password field. 5. Postcondition: User remains on the Login screen; no session is created. E2_Google OAuth Service Failure ● Trigger: Occurs at Alternative Flow Step 2.1 (Step 2 or 4). ● Sequence: 1. The system cannot establish a connection with Google servers, or Google returns a server-side error (500). 2. System catches the timeout or error exception. 3. System redirects the user back to the Login screen. 4. System displays message: "Không thể kết nối với Google lúc này. Vui lòng sử dụng email và mật khẩu." (MSG-SYS-02). 5. Postcondition: User remains on the Login screen. E3_Google Login Cancelled/Denied ● Trigger: Occurs at Alternative Flow Step 2.1 (Step 3). ● Sequence: 1. User clicks "Cancel" or denies permission on the Google Consent screen. 2. Google redirects the user back to the system with a "Access Denied" or "User Cancelled" flag. 3. System displays message: "Đăng nhập bằng Google đã bị hủy." (MSG-INFO-01).

	4. Postcondition: User remains on the Login screen.
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2.1.2 Logout

Primary Actors	Student, Lecturer, Moderator, Admin	Secondary Actors	None
Description	User logs out from the SkillUp system		
Preconditions	User is logged into the system		
Postconditions	• User successfully logged out • User is redirected to home page		
Normal Sequence/Flow	Logout: 1. The user clicks Logout on the navbar. 2. The system displays a logout confirmation dialog. 3. The user confirms the logout. 4. The system deletes the JWT token from the client 5. The system redirects to the home page.		
Alternative Sequences/Flows	Step 3.1_User cancels logout 1. The system returns to the current page. Step 1_Auto logout (JWT expiration): 1. The JWT token expires automatically 2. The system redirects the user to the home page		
Exception	E1_Network Failure • Trigger: Occurs at Normal Flow Step 3 (User confirms logout). • Sequence: 1. User confirms the logout. 2. System attempts to send a request to the server to blacklist/invalidate the token (if using server-side invalidation). 3. Connection fails or times out. 4. System forces the deletion of the token from local storage/cookies (Client-side logout) to ensure security. 5. System redirects to the Home page. 6. System displays a message: "Đăng xuất thành công (Offline mode)." 7. Postcondition: User is logged out locally; the session may technically remain valid on the server until expiration, but the user cannot access the system from that browser. E2_Token Already Invalid (Race Condition) • Trigger: Occurs at Normal Flow Step 1. • Sequence:		

	1. User clicks "Logout". 2. System checks the token status and realizes it has already expired or been deleted (e.g., user logged out in a different tab). 3. System skips the confirmation dialog (Step 2). 4. System performs any necessary cleanup. 5. System redirects immediately to the Home page. 6. Postcondition: User is on the Home page. E3_Navigation Interrupted ● Trigger: Occurs at Normal Flow Step 5. ● Sequence: 1. System deletes the token. 2. System attempts to redirect to the Home page but fails (e.g., routing error or page not found). 3. System falls back to the Login page. 4. Postcondition: User is on the Login page.
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2.1.3 Register

Primary Actors	Guest	Secondary Actors	Email Service, Google OAuth Service
Description	Register allows new users to create an account in the system. After registration, users receive a verification email to activate their account.		
Preconditions	● The user does not have an account in the system ● The system is operating normally ● A stable internet connection is available ● The email address is not already registered in the system		
Postconditions	● New account is created ● Verification email is sent ● User can login after email verification ● For Lecturer: Account pending approval, activated after moderator acceptance		
Normal Sequence/Flow	Register: 1. User navigates to Register page 2. System shows the User Registration screen with tab selector (Học viên/Giảng viên) 3. User selects role type (Học viên/Giảng viên) 4. User types in registration details ("Email", "Mật khẩu", "Xác nhận mật khẩu", "Họ và tên") 5. User clicks the "Đăng ký ngay" button 6. System validates input data (email format, password match, password length >= 6 characters, all fields required) 7. System checks if email already exists 8. System creates a new account with "Inactive" status 9. System generates verification token and sends verification email		

	10. System displays success message: "Đăng ký thành công! Vui lòng kiểm tra email để xác thực tài khoản." (MSG05) 11. System redirects user to login page after 2 seconds
Alternative Sequences/Flows	Step 6.1. Invalid password format: 1. System displays "Mật khẩu phải có ít nhất 6 ký tự" 2. Return to step 4 6.2. Password confirmation doesn't match: 1. System displays "Mật khẩu không khớp" 2. Return to step 4 6.3. Required fields missing: 1. System highlights missing fields 2. Return to step 4
Exception	E1_Email Service Failure ● Trigger: Occurs at Normal Flow Step 9 (System attempts to send verification email). ● Sequence: 1. System creates the account in the database but fails to connect to the Email Service (SMTP timeout or service down). 2. System logs the error. 3. System displays a warning message: "Tài khoản đã được tạo nhưng không thể gửi email xác thực lúc này. Vui lòng liên hệ quản trị viên hoặc thử lại chức năng 'Gửi lại email' sau." (MSG-ERR-EMAIL). 4. Postcondition: Account is created with status "Inactive"; User is on the Login page or a specific notification page. E2_System/Database Error ● Trigger: Occurs at Normal Flow Step 7 or 8. ● Sequence: 1. System encounters a database write error or connection timeout. 2. System displays message: "Đã có lỗi xảy ra trong quá trình xử lý. Vui lòng thử lại sau." (MSG-SYS-01). 3. Postcondition: Account is not created; User remains on the Registration screen with their data preserved in the fields. E3_Google Registration Error ● Trigger: Occurs if the user uses the Google OAuth Service (implied by Secondary Actors list, though not detailed in Normal Flow). ● Sequence: 1. User attempts to register via Google.

	- Google returns an error or the email is already linked to a manual account without support for merging. - System displays message: "Không thể đăng ký bằng Google. Vui lòng sử dụng đăng ký thủ công." (MSG-SYS-02). - Postcondition: User remains on the Registration screen.
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2.1.4 Forgot password

Primary Actors	User	Secondary Actors	Email Service
Description	User requests password reset when forgotten		
Preconditions	- User has an account in the system - User's email is valid		
Postconditions	- Password reset email is sent - Reset token is created and stored - User can set a new password		
Normal Sequence/Flow	Register: - User navigates to Login page - User clicks "Quên mật khẩu" link - System shows Forgot Password screen - User types in their email address - User clicks "Gửi email khôi phục" button - System validates email format - System checks if email exists and is verified - System generates password reset token - System sends password reset email with token link - System displays success screen with message: "Email khôi phục mật khẩu đã được gửi! Vui lòng kiểm tra hộp thư." - System shows instructions: - Kiểm tra hộp thư đến của bạn - Mở email từ SkillUp - Click vào link đặt lại mật khẩu - Nhập mật khẩu mới của bạn - User can click "thử lại" to resend email or "Quay lại đăng nhập" to return to login		
Alternative Sequences/Flows	Step 7_Email does not exist: - System detects email does not exist or is not verified - System displays error message: "Email không tồn tại hoặc chưa được xác thực" (MSG16) - Return to step 4 of normal flow Step 6.1_Invalid Email Format: - System detects invalid email format - System displays error message: "Email không đúng định dạng!" - Return to step 4 of normal flow		

	Step 6.2_Empty Email: 1. System detects email field is empty 2. System displays error message: "Vui lòng nhập email!" 3. Return to step 4 of normal flow
Exception	E1_Email Service Failure ● Trigger: Occurs at Normal Flow Step 9 or Step 13 (System attempts to send/resend email). ● Sequence: 1. System generates the token successfully but fails to connect to the SMTP/Email provider (Timeout or Service Unavailable). 2. System logs the critical error. 3. System displays an error message: "Hệ thống không thể gửi email lúc này. Vui lòng thử lại sau ít phút." (MSG-ERR-EMAIL). 4. Postcondition: User remains on the Forgot Password screen; No email is sent. E2_System/Database Error ● Trigger: Occurs at Normal Flow Step 7 or 8. ● Sequence: 1. System encounters a database connection error while checking the email or saving the reset token. 2. System displays a generic error message: "Email không tồn tại hoặc chưa được xác thực." (MSG-SYS-01). 3. Postcondition: User remains on the Forgot Password screen.

2.1.5 Reset password

Primary Actors	User	Secondary Actors	Email Service
Description	Reset Password allows users to set a new password using the reset link sent to their email.		
Preconditions	● User has received password reset email ● Reset token is valid and not expired ● The system is operating normally		
Postconditions	● User's password is updated ● User can login with new password ● User is redirected to login page		
Normal Sequence/Flow	Register: 1. User clicks password reset link from email 2. System redirects user to Reset Password page. 3. If email or token is missing, System displays error: "Link không hợp lệ!" and redirects to login		

	4. System shows Reset Password screen with email field (read-only) 5. User types in new password 6. System validates password in real-time and displays requirements with checkmarks: - Minimum 8 characters - At least one uppercase letter (A-Z) - At least one lowercase letter (a-z) - At least one number (0-9) 7. User types in confirm password 8. User clicks "Reset Password" button 9. System validates password meets all requirement 10. System displays success screen with message: "Đặt lại mật khẩu thành công!" 11. System shows message: "Bạn có thể đăng nhập với mật khẩu mới ngay bây giờ" 12. System automatically redirects to login page after 3 seconds
Alternative Sequences/Flows	Step 2_Missing Email or Token: 1. System detects email or token is missing from URL 2. System displays error message: "Link không hợp lệ!" 3. System redirects to login page Step 9.1_Password Does Not Meet Requirements: 1. System detects password does not meet requirements (length, uppercase, lowercase, number) 2. System displays error message: "Mật khẩu không đáp ứng yêu cầu!" 3. System disables submit button 4. Return to step 5 of normal flow Step 9.2_Invalid Password: 1. System detects password does not meet requirements or passwords don't match 2. System displays error message: "Dữ liệu không hợp lệ" (MSG14) 3. Return to step 5 of normal flow
Exception	E1_Token Expired or Invalid (Database Validation) ● Trigger: Occurs at Normal Flow Step 2 (When the page loads) or Step 8 (When submitting). ● Sequence: 1. The system checks the token against the database. 2. The system detects that the token has expired (time limit exceeded) or has already been used. 3. System displays an error message: "Đường dẫn đặt lại mật khẩu đã hết hạn hoặc không còn hiệu lực. Vui lòng yêu cầu lại." (MSG-ERR-TOKEN).

	- System redirects the user back to the Forgot Password page (Request screen) instead of the Login page, so they can easily request a new link. - Postcondition: User is on the Forgot Password page. E2_System/Database Error (Update Failed) - Trigger: Occurs at Normal Flow Step 9/10 (After clicking "Reset Password"). - Sequence: - System validates the inputs successfully but fails to update the password record in the database due to a connection error or server fault. - System displays a generic error message: "Đã có lỗi xảy ra khi cập nhật mật khẩu. Vui lòng thử lại sau." (MSG-SYS-01). - Postcondition: User remains on the Reset Password screen; The password is not changed. E3_Network Connection Lost - Trigger: Occurs at Normal Flow Step 8. - Sequence: - User clicks "Reset Password". - Client detects no internet connection. - System displays a toast/alert: "Không có kết nối mạng. Vui lòng kiểm tra đường truyền." (MSG-NET-01). - Postcondition: User remains on the Reset Password screen.
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2.2 Course Feature

2.2.1 View Lecturer's Courses

Primary Actors	Lecturer	Secondary Actors	None
Description	The Lecturer views a comprehensive list of all courses they have created, along with their current status, recent activity, and quick actions.		
Preconditions	- The user must be logged in as a Lecturer. - The Lecturer account must be active and verified. - The Lecturer must have at least one active course and have created sections. - The lecturer has navigated to the "Khóa học" (Courses) menu item.		
Postconditions	The Lecturer's Course Management list is fully displayed		
Normal Sequence/Flow	- The lecturer clicks the "Khóa học" option in the side menu. - The system retrieves all courses linked to the Lecturer's ID. - The system displays the "Quản lý khóa học" screen.		

	- The list is initially filtered to "Tất cả" (All), showing a summary card for each course (Thumbnail, Title, Status, Dates). - System displays filter tabs for status: Nháp (Draft), Đã xuất bản (Published), Đã ẩn (Unpublished), Chờ duyệt (Pending). - For each course, action buttons (Edit, Delete/Hide) are visible.
Alternative Sequences/Flows	A1. Filter Courses: Lecturer clicks a status tab (e.g., "Nháp"). The System refines the displayed list to only include courses with that status. A2. Search/Sort: Lecturer uses the search bar or sort dropdowns (Sắp xếp: Mới nhất) to reorganize the course list based on input criteria.
Exception	E1_System/Database Retrieval Error - Trigger: Occurs at Normal Flow Step 2 (System retrieves courses) or Step A1/A2 (Filtering/Searching). - Sequence: - The system attempts to query the database for the Lecturer's courses but encounters a timeout or internal server error (500). - System logs the error. - System displays a generic error message in the main content area: "Không thể tải danh sách khóa học lúc này. Vui lòng làm mới trang." (MSG-ERR-LOAD). - System displays a "Reload/Retry" button. - Postcondition: User remains on the Course Management page; The list is empty or shows a skeleton loader state. E2_Network Connection Failure - Trigger: Occurs at Normal Flow Step 1 or A1/A2. - Sequence: - User clicks the "Khóa học" menu or attempts to filter the list. - The client detects a loss of internet connection. - System displays a toast/alert: "Kết nối mạng bị gián đoạn. Dữ liệu có thể không được cập nhật." (MSG-NET-01). - System may display cached data if available, otherwise shows an empty state with a "No Connection" icon. - Postcondition: User remains on the current screen. E3_Session Expired (Unauthorized Access) - Trigger: Occurs at Normal Flow Step 1 or Step 2. - Sequence: - User clicks "Khóa học". - System checks the user's JWT/Session token and finds it has expired. - System redirects the user to the Login page immediately. - System displays message: "Phiên đăng nhập đã hết hạn. Vui lòng đăng nhập lại." (MSG-AUTH-01). - Postcondition: User is on the Login page.

2.2.2 Create New Course

Primary Actors	Lecturer	Secondary Actors	None
Description	The Lecturer initializes a new course shell by providing basic metadata. This creates a "Draft" course in the system.		
Preconditions	• The user must be logged in as a Lecturer. • The Lecturer account must be active and verified. • The Lecturer must have at least one active course and have created sections. • The lecturer has navigated to the "Khóa học" (Courses) menu item.		
Postconditions	• A new course ID is generated. • The course status is set to "Nháp" (Draft). • The Lecturer is redirected to the Course Editor interface.		
Normal Sequence/Flow	1. Lecturer clicks the yellow "+ Tạo khóa học mới" (Create New Course) button on the dashboard. 2. The system displays the creation form. 3. Lecturer enters the Tên khóa học (Course Name) and Mô tả khóa học (Course Description). 4. Lecturer selects a Danh mục (Category) and Danh mục con (Sub-category). 5. Lecturer clicks the "Tạo khóa học" (Create) button. 6. System saves the data and redirects the Lecturer to the Course Editor's "Trang giới thiệu" tab.		
Alternative Sequences/Flows	3) A1. Missing Information: If the Lecturer attempts to proceed without filling mandatory fields, the System displays an error message next to the required fields. 4) A2. Cancel Creation: Lecturer clicks "X" on the modal. The process is aborted.		
Exception	E1_System/Database Save Error • Trigger: Occurs at Normal Flow Step 5 (System attempts to save the new course). • Sequence: 1. System validates data successfully but fails to insert the record into the database (e.g., deadlock, storage full, service timeout). 2. System logs the critical error. 3. System displays an error message on the modal: "Không thể tạo khóa học lúc này. Vui lòng thử lại sau." (MSG-SYS-01). 4. Postcondition: User remains on the creation modal; Data entered is preserved so the user doesn't have to re-type it. E2_Network Connection Lost • Trigger: Occurs at Normal Flow Step 5. • Sequence:		

	1. Lecturer clicks "Tạo khóa học". 2. Client detects no internet connection or the request times out. 3. System displays a toast/alert: "Kết nối mạng không ổn định. Vui lòng kiểm tra lại đường truyền." (MSG-NET-01). 4. Postcondition: User remains on the creation modal. E3_Session Expired (Time-out) ● Trigger: Occurs at Normal Flow Step 5. ● Sequence: 1. Lecturer spends a long time filling out the description. 2. The session/JWT token expires in the background. 3. Lecturer clicks "Tạo khóa học". 4. System rejects the request with a 401 Unauthorized error. 5. System displays a modal: "Phiên làm việc đã hết hạn. Vui lòng đăng nhập lại để tiếp tục." 6. Postcondition: User is redirected to the Login page. (Ideally, the system should autosave the draft to local storage to prevent data loss, but in a standard exception flow, the data is often lost).
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2.2.3 Update Course & Manage General Settings

Primary Actors	Lecturer	Secondary Actors	None
Description	The Lecturer updates the core metadata, image, price, and coupon information for an existing course. This Use Case covers the actions within the Course Editor sidebar tabs.		
Preconditions	● The course exists in the database. ● Lecturer is inside the Course Editor mode (e.g., by clicking the Edit icon)		
Postconditions	● Course details, price, and coupon lists are updated in the database.		
Normal Sequence/Flow	1. Lecturer views the Course Editor interface. 2. Lecturer navigates using the sidebar links: "Trang giới thiệu", "Giá khóa học", or "Phiếu giảm giá". 3. Lecturer modifies the fields on the active tab (e.g., updates the course description, sets a price value, or creates a new coupon). 4. The lecturer clicks the "Lưu" (Save) button (implied action for saving the form). 5. System confirms the update and ensures the completion status checkmarks are updated in the "Tổng quan tiến độ" box.		
Alternative Sequences/Flows	5) A1. Unpublish Course: The Lecturer returns to the Dashboard and chooses to "Ẩn khóa học". If the course was published, the System confirms the action and sets the status to Unpublished. Students who previously bought the course retain access.		

	6) A2. Permanent Delete (Draft only): If the course status is "Nháp" (Draft), the System allows permanent deletion upon confirmation.
Exception	E1_Invalid Input Data (Validation Error) ● Trigger: Occurs at Normal Flow Step 4 (When clicking "Save"). ● Sequence: 1. System validates the input data on the server side. 2. System detects a violation of business rules (e.g., "Price" is negative, "Coupon Expiry Date" is in the past, or "Start Date" is after "End Date"). 3. System halts the save process. 4. System highlights the specific field causing the error. 5. System displays an error message: "Dữ liệu không hợp lệ. Vui lòng kiểm tra lại các trường được đánh dấu đỏ." (MSG-VAL-01). 6. Postcondition: User remains on the current tab; Data is not saved. E2_Media Upload Failure ● Trigger: Occurs at Normal Flow Step 3 (When uploading a Course Image/Thumbnail). ● Sequence: 1. Lecturer selects an image and attempts to upload. 2. System/Storage Service rejects the file (File size too large, unsupported format, or storage server timeout). 3. System displays an error message: "Tải ảnh thất bại. Vui lòng chọn ảnh .JPG/.PNG dưới 5MB." (MSG-ERR-UPLOAD). 4. System reverts the image preview to the previous state. 5. Postcondition: User remains on the "Trang giới thiệu" tab.

2.2.4 Manage Sections

Primary Actors	Lecturer	Secondary Actors	None
Description	The Lecturer structures the course content by creating, modifying, and deleting sections (chapters) within the " Khung chương trình " tab.		
Preconditions	● Lecturer selects the "Khung chương trình" sidebar tab		
Postconditions	● The course structure is updated with new, modified, or deleted sections.		
Normal Sequence/Flow	1. Lecturer clicks the "+ Chương mới" (New Chapter) button. 2. The system displays the input form. 3. The lecturer inputs the mandatory "Tên chương" (Chapter Title). 4. (Optional) Lecturer inputs the "Mô tả chương". 5. The lecturer clicks "Lưu" (Save). 6. The new chapter is rendered as a collapsible item in the curriculum list.		

Alternative Sequences/Flows	7) A1. Edit Section: Lecturer clicks the Pencil icon on a section row, modifies the title/description, and saves. 8) A2. Delete Section: Lecturer clicks the Trash Bin icon. The system prompts a warning before deletion.
Exception	E1_ Validation Error (Missing/Invalid Title) ● Trigger: Occurs at Normal Flow Step 5 (Lecturer clicks "Save"). ● Sequence: 1. System validates the input. 2. System detects that "Tên chương" is empty or exceeds the character limit (e.g., > 255 chars). 3. System displays an inline error message: "Tên chương là bắt buộc và không quá 255 ký tự." (MSG-VAL-02). 4. Postcondition: User remains in the input form mode; Section is not created. E2: Section contains active content (Cannot Delete) ● Trigger: Occurs in Alternative Flow A2 when Lecturer attempts to delete a section ● Sequence: 1. Lecturer clicks the Trash Bin icon on a section row (step 11 in Delete diagram) 2. System checks if the section has active content (quizzes, lectures, assignments with isActive = true) 3. System detects that section contains active/published content (step 17 in Delete diagram) 4. System displays error message: "Không thể xóa chương này vì còn chứa nội dung đang hoạt động. Vui lòng xóa hoặc vô hiệu hóa các nội dung bên trong trước." (MSG-ERR-404) 5. Deletion is cancelled ● Postcondition: Section remains unchanged; User stays on the "Khung chương trình" page E3: Section not found (Edit/Delete) ● Trigger: Occurs in A1 or A2 when the section has been deleted by another user/session ● Sequence: 1. Lecturer clicks Edit (Pencil icon) or Delete (Trash Bin icon) on a section 2. System attempts to retrieve section data from repository 3. System returns null/not found (step 14 in Edit diagram shows "Section Not Found") 4. System displays error message: "Chương không tồn tại hoặc đã bị xóa." (MSG-ERR-404) 5. System refreshes the section list to show current state

	● Postcondition: User remains on "Khung chương trình" page with updated section list
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2.2.5 Manage Lessons

Primary Actors	Lecturer	Secondary Actors	None
Description	The Lecturer populates sections with learning units, choosing between a standard Bài học (Video/Text) or a Quiz format, and managing supplementary material.		
Preconditions	● At least one Section exists. ● The lecturer has clicked the option to add content inside a section.		
Postconditions	● A new Lesson or Quiz is added to the section. ● Uploaded media (video, PDF) is stored.		
Normal Sequence/Flow	1. Lecturer selects the "Bài học" card from the "Chọn loại nội dung" section. 2. Lecturer inputs "Tên bài học" and "Mô tả bài học". 3. The lecturer selects "Video" from the type dropdown. 4. The lecturer clicks "Chọn video" to upload the media source file. 5. (Optional) Lecturer checks "Miễn phí" for a free preview lesson. 6. (Optional) Lecturer uploads "Tài liệu PDF". 7. Lecturer clicks "Thêm bài học" (Add Lesson).		
Alternative Sequences/Flows	9) A1. Create Text Lesson: In Step 3, Lecturer selects "Văn bản" (Text). A Rich Text Editor is displayed for inputting documentation. Steps 5-7 are followed. 10) A2. Create Quiz (Use case 2.4.1): In Step 1, Lecturer selects the "Quiz" card. The system redirects the user to the quiz creation interface. 11) A3. Edit/Delete Lesson: Lecturer clicks the Pencil or Trash Bin icon on an existing lesson item.		
Exception	E1_Invalid File Format or Size (Upload Error) ● Trigger: Occurs at Normal Flow Step 4 or 6 (User selects a file). ● Sequence: 1. Lecturer selects a file that violates system rules (e.g., Video > 2GB, Format is .exe instead of .mp4, or PDF > 50MB). 2. System immediately validates the file metadata before uploading. 3. System rejects the file. 4. System displays an error message: "File không hợp lệ. Vui lòng chọn Video MP4 dưới 2GB hoặc PDF dưới 50MB." (MSG-ERR-FILE). 5. Postcondition: User remains on the form; No file is attached.		

	E2_Upload Interrupted (Network Failure) ● Trigger: Occurs during Normal Flow Step 4 (While the progress bar is filling). ● Sequence: 1. The upload starts, but the internet connection is lost or unstable during the transfer. 2. The upload process stalls or fails. 3. System displays an error: "Quá trình tải lên bị gián đoạn. Vui lòng kiểm tra mạng và thử lại." (MSG-NET-UPLOAD). 4. System provides a "Resume" (Thử lại) button if supported, or resets the upload field. 5. Postcondition: User remains on the form. E3_Missing Core Content (Validation) ● Trigger: Occurs at Normal Flow Step 7 (Clicking "Add Lesson"). ● Sequence: 1. Lecturer clicks "Thêm bài học". 2. System detects that the "Lesson Type" is Video but no video file has been uploaded, or Type is Text but the editor is empty. 3. System blocks the creation. 4. System highlights the missing area. 5. System displays message: "Nội dung bài học không được để trống." (MSG-VAL-CONTENT). 6. Postcondition: User remains on the form.
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2.2.6 Submit Course for Publish

Primary Actors	Lecturer	Secondary Actors	Content Moderator
Description	The Lecturer submits the finalized course for review by a Content Moderator once all core prerequisites are met.		
Preconditions	● The "Tổng quan tiến độ" (Progress Overview) must show 3/3 bước completed. ● All tabs must be marked green as completed.		
Postconditions	● Course Status changes from "Nháp" (Draft) to "Chờ duyệt" (Pending). ● The course is locked against major edits by the Lecturer.		
Normal Sequence/Flow	1. Lecturer verifies that the progress box confirms "Tất cả đã hoàn thành! Có thể đề xuất khóa học". 2. Lecturer clicks the active green button "Đề xuất khóa học" (Submit Course). 3. The system performs final validation checks on course integrity. 4. The system updates the course status and displays a confirmation message.		

Alternative Sequences/Flows	12) A1. Prerequisites Not Met: If any section of the course (Intro, Curriculum, or Price) is incomplete, the "Đề xuất khóa học" button remains disabled/grayed out, and the Lecturer cannot proceed.
Exception	E1_Backend Validation Failure (Data Sync Error) ● Trigger: Occurs at Normal Flow Step 3 (System performs final validation). ● Sequence: 1. The User Interface indicates "100% Complete," but the Backend detects missing critical data (e.g., a video file was deleted in another tab, or a text description is null). 2. System rejects the submission request. 3. System reloads the "Progress Overview" box to reflect the actual missing items (e.g., turning a green checkmark back to red). 4. System displays an error message: "Dữ liệu không đồng bộ. Vui lòng hoàn tất các mục còn thiếu được đánh dấu đỏ." (MSG-ERR-SYNC). 5. Postcondition: Course remains "Draft"; User stays on the page to fix errors. E2_Minimum Content Policy Violation ● Trigger: Occurs at Normal Flow Step 3. ● Sequence: 1. System checks the course against specific quality policies (e.g., Total video duration must be > 30 minutes, or Course must have at least 5 lessons). 2. System detects the course is too short or empty despite technically having sections. 3. System displays a policy warning: "Khóa học chưa đạt tiêu chuẩn tối thiểu (VD: Tổng thời lượng > 30 phút). Vui lòng bổ sung nội dung." (MSG-POL-01). 4. Postcondition: Course remains "Draft". E3_Account Restricted/Banned ● Trigger: Occurs at Normal Flow Step 2. ● Sequence: 1. Lecturer clicks "Submit". 2. System checks the Lecturer's current account status. 3. System detects the account has been flagged, banned, or suspended by an Admin. 4. System denies the action. 5. System displays a modal: "Tài khoản của bạn đang bị hạn chế quyền. Vui lòng liên hệ hỗ trợ." (MSG-AUTH-BAN). 6. Postcondition: Course remains "Draft"; User might be logged out depending on severity. E4_System/Network Error

	● Trigger: Occurs at Normal Flow Step 4. ● Sequence: 1. System attempts to update the status in the database but fails. 2. System displays message: "Không thể gửi yêu cầu lúc này. Vui lòng thử lại sau." (MSG-SYS-01). 3. Postcondition: Course remains "Draft" and unlocked.
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2.2.7 Ban/unban course as Moderator Content

Primary Actors	Content Moderator	Secondary Actors	Email Service
Description	Content Moderator manages courses in the system, including viewing course list, approved/rejected pending courses, and ban/unban published courses.		
Preconditions	● Content Moderator has logged into the system ● Content Moderator has "Content Moderator" permission (RoleId = 3) ● Email services are operational ● At least one course exists in the system (for viewing list)		
Postconditions	● Course status is updated (Pending → Public/Draft, or IsActive changed) ● Email notification is sent to lecturer (when approved/rejected) ● Course list is displayed with updated information		
Normal Sequence/Flow	Ban course: 1. Content Moderator navigates to "Quản lý Khóa học" page 2. System displays a list of courses filtered "Đã công khai" 3. Content Moderator views a course (status: Public) 4. Content Moderator clicks "Cấm" button 5. System shows ban confirmation dialog 6. Content Moderator confirms ban action 7. System validates: - Course exists - Moderator has permission (RoleId = 3) 8. System displays success message: "Cấm khoá học thành công" 9. System refreshes course list		
Alternative Sequences/Flows	Unban course: 1. Content Moderator navigates to "Quản lý Khóa học" page 2. System displays a list of courses filtered "Đã công khai" 3. Content Moderator views a course (status: Public) 4. Content Moderator clicks "Bỏ Cấm" button 5. System shows ban confirmation dialog 6. Content Moderator confirms ban action 7. System validates: - Course exists - Moderator has permission (RoleId = 3) 8. System displays success message: "Bỏ cấm khoá học thành công" 9. System refreshes course list		

Exception	E1_Backend Validation Failure (Data Sync Error) ● Trigger: Occurs at Normal Flow Step 3 (System performs final validation). ● Sequence: 1. The User Interface indicates "100% Complete," but the Backend detects missing critical data (e.g., a video file was deleted in another tab, or a text description is null). 2. System rejects the submission request. 3. System reloads the "Progress Overview" box to reflect the actual missing items (e.g., turning a green checkmark back to red). 4. System displays an error message: "Dữ liệu không đồng bộ. Vui lòng hoàn tất các mục còn thiếu được đánh dấu đỏ." (MSG-ERR-SYNC). 5. Postcondition: Course remains "Draft"; User stays on the page to fix errors. E2_Minimum Content Policy Violation ● Trigger: Occurs at Normal Flow Step 3. ● Sequence: 1. System checks the course against specific quality policies (e.g., Total video duration must be > 30 minutes, or Course must have at least 5 lessons). 2. System detects the course is too short or empty despite technically having sections. 3. System displays a policy warning: "Khóa học chưa đạt tiêu chuẩn tối thiểu (VD: Tổng thời lượng > 30 phút). Vui lòng bổ sung nội dung." (MSG-POL-01). 4. Postcondition: Course remains "Draft". E3_Account Restricted/Banned ● Trigger: Occurs at Normal Flow Step 2. ● Sequence: 1. Lecturer clicks "Submit". 2. System checks the Lecturer's current account status. 3. System detects the account has been flagged, banned, or suspended by an Admin. 4. System denies the action. 5. System displays a modal: "Tài khoản của bạn đang bị hạn chế quyền. Vui lòng liên hệ hỗ trợ." (MSG-AUTH-BAN). 6. Postcondition: Course remains "Draft"; User might be logged out depending on severity. E4_System/Network Error ● Trigger: Occurs at Normal Flow Step 4. ● Sequence:
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	1. System attempts to update the status in the database but fails. 2. System displays message: "Không thể gửi yêu cầu lúc này. Vui lòng thử lại sau." (MSG-SYS-01). 3. Postcondition: Course remains "Draft" and unlocked.
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2.2.8 Approve/Rejected Course as Moderator Content

Primary Actors	Content Moderator	Secondary Actors	Email Service
Description	Content Moderator manages courses in the system, including viewing course list, approved/rejected pending courses, and ban/unban published courses.		
Preconditions	• Content Moderator has logged into the system • Content Moderator has "Content Moderator" permission (RoleId = 3) • Email services are operational • At least one course exists in the system (for viewing list)		
Postconditions	• Course status is updated (Pending → Public/Draft, or IsActive changed) • Email notification is sent to lecturer (when approved/rejected) • Course list is displayed with updated information		
Normal Sequence/Flow	Approve course: 1. Content Moderator navigates to "Quản lý Khóa học" page 2. System displays a list of courses filtered "Đã công khai" 3. Content Moderator select "Đang chờ duyệt" 4. System shows only courses with status "Đang chờ duyệt" (status="pending") 5. Content Moderator clicks "Duyệt" button 6. System shows approval confirmation dialog 7. Content moderator fills in "Duyệt khoá học" confirmation reason and click "Xác nhận duyệt" button 8. System validates: - Course exists - Course status is "Pending" - Moderator has permission (RoleId = 3) 9. System sends approval email to lecturer 10. System displays success message: "Khóa học đã được duyệt" 11. System refreshes course list		
Alternative Sequences/Flows	Rejected course: 1. Content Moderator navigates to "Quản lý Khóa học" page 2. System displays a list of courses filtered "Đã công khai" 3. Content Moderator select "Đang chờ duyệt" 4. System shows only courses with status "Đang chờ duyệt" (status="pending") 5. Content Moderator clicks "Từ chối" button 6. System shows approval confirmation dialog 7. Content moderator fills in "Từ chối khoá học" confirmation reason and click "Xác nhận từ chối" button		

	8. System validates: - Course exists - Course status is "Pending" - Moderator has permission (RoleId = 3) 9. System sends rejected email to lecturer 10. System displays success message: "Từ chối khóa học thành công" 11. System refreshes course list
Exception	E1_Email Service Failure ● Trigger: Occurs at Normal Flow Step 9 or Alternative Flow "Rejected" Step 9. ● Sequence: 1. The System successfully updates the course status in the database (to Public or Draft). 2. The System attempts to send the notification email to the Lecturer but fails (SMTP timeout, service down, or invalid email address). 3. System logs the email error. 4. System displays a warning message: "Trạng thái khóa học đã được cập nhật, nhưng không thể gửi email thông báo đến giảng viên." (MSG-WARN-EMAIL). 5. Postcondition: Course status is updated (Success), but the Moderator remains on the list view with the warning. E2_Course Status Conflict (Concurrency Error) ● Trigger: Occurs at Normal Flow Step 7 or Alternative Flow Step 7 (Clicking "Xác nhận"). ● Sequence: 1. Moderator A attempts to process the course. 2. System checks the current status of the course in the database. 3. System detects that the status is no longer "Pending" (e.g., The Lecturer cancelled the submission, or Moderator B already approved/rejected it 5 seconds ago). 4. System prevents the update. 5. System displays an error message: "Trạng thái khóa học không hợp lệ hoặc đã được xử lý bởi người khác." (MSG-ERR-STATUS). 6. System auto-refreshes the course list. 7. Postcondition: User returns to the updated Course List. E3_Missing Rejection Reason (Validation) ● Trigger: Occurs at Alternative Flow "Rejected" Step 7. ● Sequence: 1. Moderator clicks "Xác nhận từ chối". 2. System validates that the "Lý do" (Reason) field is empty. (Rejection usually requires a reason, whereas Approval might not).

	3. System displays an inline error: "Vui lòng nhập lý do từ chối để gửi cho giảng viên." (MSG-VAL-REASON). 4. Postcondition: User remains in the Dialog box. E4_Database Transaction Failure ● Trigger: Occurs at Normal Flow Step 9. ● Sequence: 1. System attempts to write the status change to the database but fails due to a connection error or deadlock. 2. System rolls back the transaction (status remains "Pending"). 3. System displays an error message: "Đã có lỗi hệ thống. Vui lòng thử lại sau." (MSG-SYS-01). 4. Postcondition: User remains in the Dialog box; Course status is unchanged.
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2.2.9 View detail course

Primary Actors	Student	Secondary Actors	None
Description	The "View Detail Course" use case allows a logged-in Student to view the full information of a specific course, including the curriculum, instructor details, ratings, and pricing, to decide whether to enroll or start learning.		
Preconditions	● The course must exist in the system database. ● The course status must be "Published" (Active). ● The user has navigated to the SkillUp website.		
Postconditions	● The course details are displayed successfully. The system identifies if the user has already enrolled to show the appropriate action button (Buy vs. Go to Course).		
Normal Sequence/Flow	1. The Student clicks on a course thumbnail or title from the Homepage or Course List. 2. The system verifies the Student's login session. 3. The system retrieves the course data from the database. 4. The system checks the Student's enrollment status for this specific course. 5. The system displays the Course Detail Page with: ● Main Content: Course Title, Description, Curriculum, Instructor, Reviews. ● Sidebar Actions: ○ Case A (Not Enrolled): Displays "Mua ngay" (Buy Now) and "Thêm vào giỏ" (Add to Cart). ○ Case B (Enrolled): Displays "Vào học" (Go to Course). 6. The Student views the course information.		

Alternative Sequences/Flows	Unauthenticated Access (User Not Logged In): At step 1, if a user tries to access the Course Detail page without logging in: 1. The system detects no active session. 2. The system displays a message: "Vui lòng đăng nhập để xem chi tiết khóa học." 3. The system redirects the user to the Login Page. 4. After successful login, the system redirects the user back to the Course Detail page. Course Not Found: At step 3, if the course ID is invalid or the course is disabled: 1. The system displays an error message: "Khóa học không tồn tại." 2. The system redirects the Student to the Course List. Session Expired: During step 6, if the Student's session expires while viewing the page and they try to click "Buy Now" or "Go to Course": 1. The system displays a session timeout message. 2. The system redirects the Student to the Login page.
Exception	E1_System/Database Retrieval Error ● Trigger: Occurs at Normal Flow Step 3 (System retrieves course data). ● Sequence: 1. The system attempts to query the database for course details (Curriculum, Reviews, Instructor). 2. The request times out or encounters a server-side error (500 Internal Server Error). 3. System displays a generic error page: "Hệ thống đang bảo trì hoặc gặp sự cố. Vui lòng quay lại sau." (MSG-ERR-LOAD). 4. System provides a "Quay lại trang chủ" (Back to Home) button. 5. Postcondition: User is not on the Course Detail page; No data is shown. E2_Course Status Changed (Race Condition) ● Trigger: Occurs at Normal Flow Step 3. ● Sequence: 1. User clicks the course card from the list (where it looked "Active"). 2. In the milliseconds between the list load and the click, an Admin/Lecturer changed the status to "Draft", "Hidden", or "Banned". 3. System retrieves the course but sees Status != Published.

	4. System denies access. 5. System displays a modal/page: "Khóa học này hiện không khả dụng hoặc đã bị ẩn." (MSG-ERR-ACCESS). 6. System redirects to Course List. 7. Postcondition: User is returned to the Course List. E3_External Media Failure (Partial Load) ● Trigger: Occurs at Normal Flow Step 5 (Displaying content). ● Sequence: 1. The page text loads successfully, but the Promo Video (hosted on YouTube/Vimeo/Cloudinary) fails to load due to a third-party API error. 2. System detects the media failure. 3. System replaces the video player with a placeholder image or a "Video unavailable" icon. 4. System displays a toast notification: "Không thể tải video giới thiệu. Vui lòng kiểm tra kết nối mạng." (MSG-WARN-MEDIA). 5. Postcondition: The user can still view the text details and buy the course, but the rich media is missing (Graceful Degradation). E4_Enrollment Check Failure ● Trigger: Occurs at Normal Flow Step 4. ● Sequence: 1. Course data loads, but the service checking "Did this user buy this?" fails. 2. System defaults to the "Safe" state (Assuming Not Enrolled) to prevent unauthorized access. 3. System displays the "Mua ngay" button. 4. If the user (who actually owns the course) clicks "Mua ngay", the system catches the duplicate order at the Checkout stage (Next Use Case). 5. Postcondition: Page loads with "Buy" buttons.
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2.2.10 view course

Primary Actors	Student, Guest	Secondary Actors	None
Description	The "View Course List by Category" use case allows a Student or Guest to browse a specific category of courses (e.g., "Học IT"). Users can view sub-categories, filter courses by rating, sort results, and select a specific course to view its details.		

Preconditions	• The system database must contain active courses and categories. • The "Học IT" category (and others) must be defined in the system. • The user has accessed the SkillUp website.
Postconditions	• A list of courses matching the selected category and filters is displayed. • The user can proceed to the "View Detail Course" use case by clicking a course card.
Normal Sequence/Flow	1. The Student clicks on a Category tab (e.g., "Học IT") from the main navigation or Homepage. 2. The system retrieves the category information, available sub-categories (e.g., C#, C++), and the list of associated courses. 3. The system displays the Course List Page with: • Header: Category Title ("Học IT") and Description. • Sub-navigation: List of sub-categories (Lập trình game, Kinh tế, C#, C++). • Sidebar Filter: "Đánh giá" (Rating) with checkboxes for star ratings (3 to 5 stars). • Sorting Tool: A dropdown to sort courses (e.g., "Nổi bật nhất"). • Course Grid: A list of course cards showing the Thumbnail, Title, Instructor Name, Rating, and Price (or "Miễn phí"). 4. The Student views the list of courses. 5. <i>Optional:</i> The Student clicks on a specific sub-category (e.g., "C#"). • The system refreshes the grid to show only courses with the "C#" tag. 6. <i>Optional:</i> The Student selects a rating filter (e.g., "4 sao trở lên"). • The system filters the list to show only courses with an average rating of 4.0+. 7. The Student clicks on a specific course card. 8. The system redirects the Student to the Course Detail Page.
Alternative Sequences/Flows	No Courses Found: • If the selected category or applied filter yields no results (e.g., selecting "C++" combined with "5 stars" where none exist): 1. The system displays an empty state message: "Không tìm thấy khóa học nào phù hợp." (No matching courses found). 2. The system allows the user to clear filters or select a different category. Pagination (Many Courses): • If the number of courses exceeds the page limit (e.g., more than 12 courses): 1. The system displays pagination controls at the bottom of the grid.

	- The Student clicks "Next" or a page number. - The system loads the next set of courses without reloading the entire page. Database Connection Error: - At step 2, if the system fails to retrieve course data: - The system displays an error message: "Đã xảy ra lỗi khi tải danh sách khóa học. Vui lòng thử lại sau." - The system provides a "Reload" button.
Exception	E1_Invalid Input Data (Validation Error) - Trigger: Occurs at Normal Flow Step 4 (When clicking "Save"). - Sequence: - System validates the input data on the server side. - System detects a violation of business rules (e.g., "Price" is negative, "Coupon Expiry Date" is in the past, or "Start Date" is after "End Date"). - System halts the save process. - System highlights the specific field causing the error. - System displays an error message: "Dữ liệu không hợp lệ. Vui lòng kiểm tra lại các trường được đánh dấu đỏ." (MSG-VAL-01). - Postcondition: User remains on the current tab; Data is not saved. E2_Media Upload Failure - Trigger: Occurs at Normal Flow Step 3 (When uploading a Course Image/Thumbnail). - Sequence: - Lecturer selects an image and attempts to upload. - System/Storage Service rejects the file (File size too large, unsupported format, or storage server timeout). - System displays an error message: "Tải ảnh thất bại. Vui lòng chọn ảnh .JPG/.PNG dưới 5MB." (MSG-ERR-UPLOAD). - System reverts the image preview to the previous state. - Postcondition: User remains on the "Trang giới thiệu" tab.

2.2.11 View my course

Primary Actors	Student	Secondary Actors	None
Description	The "View My Courses" use case allows a logged-in Student to access their personal learning dashboard. Here, they can track their learning statistics (total		

	courses, hours spent), search through their enrolled courses, resume learning, and view recent activities or learning tips.
Preconditions	- The User must be logged in as a Student. - The User navigates to the "Khóa học của tôi".
Postconditions	The dashboard is displayed with up-to-date statistics and course list. The student can navigate to a specific course or return to the main control panel.
Normal Sequence/Flow	- The Student accesses the My Courses page. - The system retrieves the Student's enrollment data and learning history from the database. - The system calculates and displays Learning Statistics: - Total enrolled courses (Khóa học đã đăng ký). - Active courses (Đang học). - Completed courses (Đã hoàn thành). - Total learning hours (Tổng giờ học). - The system displays the Course List section with a search bar. - The system displays the Sidebar Widgets: "Hoạt động gần đây" (Recent Activity) and "Mẹo học tập" (Learning Tips). - The Student views their progress or searches for a specific course to continue learning.
Alternative Sequences/Flows	No Courses Found: At step 3 or after filtering, if no courses match the criteria: - The system displays an empty state message: "Không tìm thấy khóa học nào phù hợp." - The system keeps the structure visible so the user can clear filters or choose another category. Database Connection Error: At step 2, if the system cannot retrieve course data: - The system displays an error message: "Đã xảy ra lỗi khi tải danh sách. Vui lòng thử lại sau."
Exception	E1_System/Database Retrieval Error (Global Failure) - Trigger: Occurs at Normal Flow Step 2 (System attempts to retrieve enrollment and history data). - Sequence: - The system attempts to query the database but fails due to a connection timeout or server error (500). - System logs the error.

	3. System displays a "Global Error" state: "Đã xảy ra lỗi khi tải dữ liệu. Vui lòng tải lại trang." (MSG-ERR-LOAD). 4. System provides a "Tải lại" (Reload) button. 5. Postcondition: User remains on the My Courses page (empty state); No data is displayed. E2_Session Expired (Unauthorized Access) • Trigger: Occurs at Normal Flow Step 2. • Sequence: 1. System validates the user's session token while fetching data. 2. System detects the token is expired or invalid. 3. System interrupts the data loading. 4. System displays a modal: "Phiên đăng nhập đã hết hạn. Vui lòng đăng nhập lại." (MSG-AUTH-01). 5. Postcondition: System redirects the user to the Login page.
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2.2.12 View Enrolled Students

Primary Actors	Lecturer	Secondary Actors	None
Description	Allows Lecturers to view a list of all students enrolled in their courses. Lecturers can search and filter the list by specific course to track learning progress.		
Preconditions	• The lecturer has logged into the system. • The course has been published and has at least one registered student.		
Postconditions	• The student list is displayed fully and accurately.		
Normal Sequence/Flow	1. The instructor clicks on the "Students" menu on the main navigation bar (Sidebar). 2. The system displays the "Student Management" page. 3. The system lists all students from all courses. 4. The instructor views the overview list.		
Alternative Sequences/Flows	A1: Filter by Course - The lecturer clicks on the "Dropdown list" filter. - The system displays a list of the lecturer's courses. - The lecturer selects a specific course. - The system filters and displays only students in that course. A2: Search Student: - The instructor enters the name in the search box.		

	- The system filters and displays the appropriate results immediately (or after pressing Enter). A3: View Student Progress Detail: - The instructor clicks on the student's name or the "Xem chi tiết" button. - The system displays a popup/subpage detailing: - Completed lessons. - Quiz scores.
Exception	E1: No Data If the lecturer does not have any students (or the filter/search results are not found).The system displays the message: "Không tìm thấy học viên."

2.3 Section Feature

2.3.1 View section

Primary Actors	Student	Secondary Actors	None
Description	The "View Section Detail" use case allows an enrolled Student to view the contents of a specific chapter (section) inside the learning interface. It displays the list of lessons, current progress within that chapter, and allows the student to navigate to specific lessons.		
Preconditions	• The Student is logged in. • The Student is enrolled in the course. • The Student is inside the Learning Interface of the course.		
Postconditions	• The details of the selected section are displayed, including lesson list and progress status.		
Normal Sequence/Flow	1. The Student navigates to the course learning interface. 2. The Student clicks on a specific Section/Chapter from the course outline. 3. The system displays the Section Detail Screen containing: • Header: Section Title (e.g., "Chương 1 - Chương 1") and description. • Progress Indicator: "Tiến độ chương x/y" (Chapter progress). • Lesson List: A numbered list of lessons within this section with their types (e.g., Bài đọc/Reading). • Progress Card: A summary card showing "% Hoàn thành" and remaining lessons. 4. The Student reviews the list and selects a lesson to start learning.		

Alternative Sequences/Flows	Back to Overview: At step 3, the Student clicks "Quay lại tổng quan" (< Back to overview): 1. The system redirects the Student back to the Course General Information or Syllabus page.
Exception	E1_Section Locked (Prerequisite Not Met) ● Trigger: Occurs at Normal Flow Step 2 (Student clicks on a Section). ● Sequence: 1. System checks the settings for the selected section. 2. System detects that the section is "Locked" because the previous section is not 100% complete (Drip Content logic). 3. System prevents opening the detail view. 4. System displays a toast/tooltip: "Bạn cần hoàn thành chương trước để mở khóa nội dung này." (MSG-LOCK-01). 5. Postcondition: User remains on the current screen/Course Outline. E2_Section Empty or Deleted (Data Integrity) ● Trigger: Occurs at Normal Flow Step 2 or 3. ● Sequence: 1. Student clicks a section. 2. System attempts to load the content but finds the Section ID is invalid (deleted by Lecturer) or the Section contains 0 lessons. 3. System displays an empty state message: "Chương này chưa có bài học nào." (MSG-INFO-EMPTY) or "Nội dung không tồn tại." 4. Postcondition: Section opens but shows an empty list, or redirects back to the outline.

2.4 Lesson Feature

2.4.1 View lesson

Primary Actors	Student	Secondary Actors	None
Description	The "View Lesson Content" use case allows an enrolled Student to view the specific content of a lesson (Video, Article, or Quiz). The student can read or watch the material, mark the lesson as completed, and navigate to the next or previous lesson.		
Preconditions	● The Student is logged in and enrolled in the course.		

	The Student has navigated to the Learning Interface.
Postconditions	The lesson content is displayed. If the student clicks "Mark as completed", their progress is updated in the database.
Normal Sequence/Flow	- The Student clicks on a specific lesson from the Sidebar Curriculum List or clicks "Start Learning" from the dashboard. - The system retrieves the lesson details (Title, Type, Content URL/Body). - The system displays the Learning Screen with: Sidebar: List of lessons in the current chapter, highlighting the active lesson. Main Content: The video player (if video) or text content (if article). Action Area: "Mark as completed" button and Navigation buttons (Previous/Next). - The Student consumes the content (reads text or watches video). - The Student clicks "Đánh dấu hoàn thành" (Mark as completed). - The system updates the progress status to 100% for this lesson. - The Student clicks "Tiếp theo" (Next) to move to the next lesson.
Alternative Sequences/Flows	Navigate via Sidebar: At step 4, the Student clicks a different lesson in the left sidebar: - The system immediately loads the selected lesson content. Lesson Locked (Sequential Learning): If the course requires sequential learning and the previous lesson is not done: - The system keeps the next lesson locked (grayed out). - The Student cannot access it until the current lesson is marked complete.
Exception	E1_Media Resource Unavailable (404/Broken Link) - Trigger: Occurs at Normal Flow Step 2 or 3 (System attempts to load the video URL). - Sequence: - System retrieves the lesson metadata, but the video hosting service (e.g., YouTube/Vimeo) returns an error (Video deleted, Private video, or Service Down). - System loads the page structure but replaces the video player with an error placeholder. - System displays a message: "Video bài học không khả dụng. Vui lòng báo cáo với hỗ trợ viên." (MSG-ERR-MEDIA). - Postcondition: User can read the description/text but cannot watch the video.

	E2_Progress Save Failure (Network/Sync Error) ● Trigger: Occurs at Normal Flow Step 5 (Student clicks "Mark as completed"). ● Sequence: 1. Student clicks the button. 2. Client attempts to send the API request to update progress. 3. Request fails due to internet disconnection or server timeout. 4. System displays a toast notification: "Không thể lưu tiến độ. Vui lòng kiểm tra kết nối mạng và thử lại." (MSG-ERR-SAVE). 5. The button state reverts to "Mark as completed" (not checked). 6. Postcondition: Lesson remains "Incomplete" in the database. E3_Access Denied (Direct URL Access) ● Trigger: Occurs at Normal Flow Step 2. ● Sequence: 1. Student attempts to access a specific lesson ID directly via the browser URL (bypassing the UI). 2. System checks permissions. 3. System detects that the user is Enrolled but has not completed the prerequisite previous lesson (in a Sequential Course). 4. System redirects the user to the latest unlocked lesson. 5. System displays a warning: "Bạn cần hoàn thành bài học trước để truy cập nội dung này." (MSG-LOCK-01). 6. Postcondition: User is redirected to the correct safe lesson. E4_Concurrent Deletion (Lesson Not Found) ● Trigger: Occurs at Normal Flow Step 1 or 7. ● Sequence: 1. Student clicks "Next Lesson". 2. In the time between the student loading the course and clicking next, the Lecturer deleted that specific lesson. 3. System returns a "404 Not Found" error. 4. System displays a modal: "Bài học này không còn tồn tại." 5. System refreshes the Curriculum Sidebar to show the updated structure. 6. Postcondition: User remains on the current lesson or is moved to the Section view.
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2.5 Quiz Feature

2.5.1 Take quiz

Primary Actors	Student	Secondary Actors	None
Description	The "Take Quiz" use case allows an enrolled Student to attempt a quiz assigned in a course. The student can view questions, select answers, navigate between questions, save progress, and submit the quiz for grading.		
Preconditions	- The Student is logged in and enrolled in the course. - The quiz is available (within the allowed time frame or attempts). - The Student has clicked "Start Quiz" (Làm bài).		
Postconditions	User answers are saved to the database. The system calculates the score (for auto-graded quizzes) and updates the student's progress.		
Normal Sequence/Flow	- The Student initiates the quiz. - The system retrieves the quiz questions and configuration (time limit, passing grade). - The system displays the Quiz Interface containing: Left Sidebar: Question navigation grid (Question Map), Timer (if applicable), and Action buttons (Submit, Save). Main Content: The current question text and answer options. - The Student reads the question (e.g., "API là viết tắt của gì?"). - The Student selects one or more options (e.g., "Application Program Interface"). - The Student clicks "Câu sau" (Next) to move to the next question. - The Student repeats steps 4-6 for all questions. - The Student clicks "Nộp bài" (Submit). - The system asks for confirmation. - The Student confirms. - The system grades the quiz and displays the Result Screen.		
Alternative Sequences/Flows	Jump to Question: At step 6, instead of clicking "Next", the Student clicks a specific number (e.g., "5") in the Sidebar Question Grid. - The system immediately displays Question 5. Save Draft (Lưu bài làm): During the attempt, the Student clicks "Lưu bài làm". - The system saves the current selected answers without submitting/grading. - The Student can resume the quiz later (if the time limit allows). Time Limit Exceeded: If the quiz has a timer and it reaches 00:00: - The system automatically submits the current answers.		

	2. The system redirects the Student to the Result Screen.
Exception	E1_Submission Network Failure (Data Rescue) ● Trigger: Occurs at Normal Flow Step 9 (Student confirms submission). ● Sequence: 1. Student clicks "Submit". 2. Client attempts to send the answer payload to the server. 3. Connection fails (Internet lost or Server timeout). 4. System displays a blocking alert: "Mất kết nối mạng. Không thể nộp bài!" (MSG-NET-SUBMIT). 5. System caches the final answers locally (in LocalStorage/SessionStorage) to prevent data loss. 6. System changes the "Submit" button to "Retry" (Thử lại). 7. Postcondition: User remains on the Quiz interface with all answers preserved; Quiz is not submitted yet.

2.5.2 View detail result

Primary Actors	Student	Secondary Actors	None
Description	The "View Quiz Result Detail" use case allows a Student to review their performance after submitting a quiz. The system displays an overall summary (score, pass/fail status) and a detailed breakdown of each question, showing the user's selected answers versus the correct answers with specific feedback.		
Preconditions	● The Student has submitted a quiz. ● The quiz grading process is complete. ● The Student is on the Quiz Result page.		
Postconditions	● The student understands their score and which questions they answered correctly or incorrectly.		
Normal Sequence/Flow	1. The system calculates the score immediately after submission. 2. The system displays the Quiz Result Screen containing two main areas: ● Left Sidebar (Summary): ○ Overall Score Percentage (e.g., 22%). ○ Pass/Fail Status (e.g., "Chưa vượt qua" - Not Passed). ○ Statistics: Correct answers, Wrong answers, Total questions. ○ Question Map: A grid of numbers colored by result (Green = Correct, Red = Wrong). ● Right Panel (Detail View): ○ Displays the first question (or selected question) details.		

	○ Shows the Question Text. ○ Shows Answer Options with visual feedback (Correct, Wrong, Missed). 3. The Student reviews the summary statistics. 4. The Student clicks on a specific question number in the Question Map or uses "Câu sau" (Next) buttons. 5. The system updates the Right Panel to show the selected question. 6. The Student reviews the explanation for the answer. 7. The Student clicks "Đóng" (Close) or "Làm lại" (Retry) (if attempts allowed).
Alternative Sequences/Flows	Retry Quiz: If the quiz allows multiple attempts: 1. The Student clicks the "Làm lại" (Retry) button (if available). 2. The system resets the attempt and redirects to the Take Quiz screen.
Exception	E1_Session Expired ● Trigger: Occurs at Precondition validation (Student's session expires). ● Sequence: 1. Student's session expires (JWT token expired). 2. Student attempts to access Quiz Result page. 3. System validates session token. 4. System detects expired authentication token. 5. System prevents page access. 6. System displays toast: "Phiên đăng nhập đã hết hạn. Vui lòng đăng nhập lại." (MSG-SESSION-EXPIRED). 7. System redirects to login page. 8. After re-authentication, the system redirects back to Quiz Result page (if quiz ID preserved). Postcondition: Result NOT displayed; Student must re-authenticate; Redirected to login; Returns to results after login.

2.5.3 Manage Quiz

Primary Actors	Lecturer	Secondary Actors	None
Description	Allows Lecturers to create new quizzes, update existing test information (title , quiz time , passing score) , or delete quizzes in a specific section to assess student knowledge.		

Preconditions	- The user must be logged in as a Lecturer. - The Lecturer account must be active and verified. - The Lecturer must have at least one active course and have created sections. - The Lecture is on the “Quản lý nội dung khóa học” page.
Postconditions	- Create a new quiz: A new quiz is added to the selected section and saved to the database, displayed inside the section. - Edit quiz: Quiz details are updated and saved. - Delete quiz: The quiz changes the status and is no longer visible to students.
Normal Sequence/Flow	- The lecturer logs into the system. - The system authenticates the Lecturer and displays the dashboard. - Lecturer navigates to the “Quản lý khóa học” page. - The Lecturer selects a specific course and navigates to the “Khung chương trình” tab. - The system displays a list of Sections and their current content. - The lecturer clicks the “Thêm nội dung” button in a desired Section and selects the content type as “Quiz”. - The system displays the form to create a quiz (Tên , Mô tả , Điểm đạt , Thời gian). - The lecturer enters information to complete the quiz. - The lecturer clicks the “Tạo quiz” button. - The System validates the input data. - The system saves the new quiz to the data, notifies that the quiz creation was successful, and displays the new quiz.
Alternative Sequences/Flows	13) A1: Update quiz: - At step 6 , instead of adding new , the Lecturer clicks the “Edit” icon on an existing Quiz item. - The system displays the “Chỉnh sửa Quiz” form pre-filled with current data. - The lecturer modifies the information and clicks “Tạo quiz”. - The system validates and saves changes. - The system saves the new information of the quiz to the data, notifies that the quiz update was successful, and displays the quiz. 14) A2 : Delete quiz: - At step 6, the Lecturer clicks the “Delete” icon on an existing Quiz item. - The system displays a confirmation popup: “Bạn có chắc chắn muốn xóa quiz này? Tất cả câu hỏi trong quiz sẽ bị xóa.” - The Lecturer clicks the “Xóa” button to confirm action. - The system removes the quiz from the Section list view.
Exception	E1_ Validation Error (Missing Required Fields) - Trigger: Occurs at Normal Flow Step 10 (System validates input data). - Sequence: - Lecturer enters quiz information and clicks "Tạo quiz" button. - System validates the input data. - System detects validation violations: "Tên Quiz" field is empty

	5. "Điểm đạt" is invalid or out of range 6. "Thời gian" is invalid 7. System displays inline error message: (MSG55) 8. System highlights invalid fields. The "Tạo quiz" button remains disabled until all fields are valid. E2_Lecturer Not Found • Trigger: Occurs at Normal Flow Step 11 during quiz creation/update/delete. • Sequence: 1. System checks if Lecturer record exists for the logged-in account. 2. System detects no Lecturer profile found. 3. System displays error message
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2.5.4 Add question to quiz

Primary Actors	Lecturer	Secondary Actors	None
Description	Allows Lecturers to add questions to a specific quiz to build the content. Lecturers can either create new questions manually or select existing questions from the Question Bank.		
Preconditions	• The user must be logged in as a Lecturer. • A Quiz must already exist within a Section. • The Lecturer is viewing the details/content of that specific Quiz.		
Postconditions	• New questions are successfully linked to the Quiz. • The total score/number of questions of the Quiz is updated.		
Normal Sequence/Flow	1. The Lecturer navigates to the “Khung chương trình” page of a specific course. 2. The Lecturer clicks on an existing Quiz item to view details. 3. The System displays the Quiz interface with a list of current questions (if any). 4. The Lecturer clicks the “Thêm câu hỏi” button. 5. The system transfers to the Question Bank interface. 6. The lecturer searches and selects questions to add. 7. The lecturer clicks the “Thêm câu hỏi” button to confirm. 8. The system links the selected questions to the Quiz, returns to the "Khung chương trình" page and notifies that the question has been successfully added.		
Alternative Sequences/Flows	A1: Create new question : ○ At step 5 (Question Bank interface), the Lecturer clicks the "Tạo mới" button. ○ The System displays the "Tạo câu hỏi" form including: ■ Question Type (Single/Multiple Choice). ■ Question Text/Content.		

	■ Answers/Options (and mark the correct one). ○ The Lecturer fills in the required information and clicks "Tạo câu hỏi". ○ The System validates the data, creates the new question, adds it to the Question Bank, and links it to the current Quiz. ○ The flow merges back to Step 8.
Exception	E1: Validation error ● In A1, if the Lecturer leaves mandatory fields blank (e.g., no question text, no correct answer selected) ● The System displays the error and prevents saving. E2: Quiz not found or deleted ● At step 2, if the Quiz has been deleted by another user or no longer exists in the system ● The System displays an error message "Quiz không tồn tại hoặc đã bị xóa" and redirects the Lecturer back to the "Khung chương trình" page E3: No questions selected ● At step 7, if the Lecturer clicks "Thêm câu hỏi" without selecting any questions from the Question Bank ● The System displays an error message "Vui lòng chọn ít nhất một câu hỏi" and remains on the Question Bank interface

2.6 Cart Feature

2.6.1 Add cart item

Primary Actors	Student	Secondary Actors	None
Description	The "Add Course to Cart" use case allows a logged-in Student to select a specific course and add it to their shopping cart database. This allows the student to collect multiple courses before proceeding to a single checkout process.		
Preconditions	● The User must be logged in as a Student. ● The User is viewing a Course Detail Page. ● The User has not enrolled in the course yet. ● The course is not already in the User's cart.		
Postconditions	● The course is successfully added to the Cart table in the database. The Cart Icon counter in the navigation header increments by 1. A success notification (Toast) is displayed.		

Normal Sequence/Flow	1. The Student navigates to a Course Detail page. 2. The Student clicks the "Thêm vào giỏ hàng" (Add to Cart) button in the sidebar. 3. The system verifies the Student's session and checks if the course is already in their cart or enrolled. 4. The system saves the Course ID to the Student's Cart record in the database. 5. The system updates the Cart Item Count in the Header. 6. The system displays a toast message: "Thêm vào giỏ hàng thành công!". 7. The button text may update to "Đã thêm" or remain available but disabled/changed.
Alternative Sequences/Flows	● Item Already in Cart: At step 3, if the course is already in the cart: ○ The system displays a warning message: "Khóa học này đã có trong giỏ hàng." ○ The system prevents duplicate addition. ● User Not Logged In: At step 2, if the session is invalid or expired: ○ The system redirects the user to the Login Page. ○ After login, the system redirects back to the Course Detail page (User must click Add again, or system auto-adds depending on implementation).
Exception	E1_User Not Authenticated / Session Invalid ● Trigger: Occurs at Normal Flow Step 2 (Student clicks "Thêm vào giỏ hàng" button). ● Sequence: 1. The user (not logged in or session expired) is viewing the Course Detail page. 2. User clicks "Thêm vào giỏ hàng" (Add to Cart) button. 3. The system checks user authentication status. 4. System detects no valid session/authentication (JWT token missing or expired). 5. The system prevents add-to-cart operation. 6. System displays toast message: "Vui lòng đăng nhập để thêm khóa học vào giỏ hàng." (MSG-AUTH-REQUIRED). 7. The system redirects the user to the Login Page after 2-3 seconds. 8. System stores intended action (add-to-cart) and course ID in session/localStorage. 9. After successful login, the system redirects back to the Course Detail page.

	Option A: The user must click "Add to Cart" again manually. Option B: System automatically adds course to cart after login (better UX). Postcondition: Course NOT added to cart; User redirected to login; After authentication, returns to course page; Can complete add-to-cart after login. E2_Course Already in Cart (Duplicate Prevention) ● Trigger: Occurs at Normal Flow Step 3 (System checks if course already in cart). ● Sequence: 1. The student is logged in and viewing the Course Detail page. 2. The course is already present in the student's cart from previous action. 3. Student clicks "Thêm vào giỏ hàng" button again. 4. The system validates cart contents. 5. The system detects that the course ID already exists in the student's cart. 6. The system prevents duplicate addition. 7. System displays warning toast: "Khóa học này đã có trong giỏ hàng." (MSG-ALREADY-IN-CART). 8. Add to Cart button may show "Đã thêm" (Already Added) or "Xem giỏ hàng" (View Cart). 9. Cart count does NOT increment. 10. Student can click "Xem giỏ hàng" to navigate to cart page. Postcondition: Duplicate course NOT added; Student informed; Cart unchanged; Button state reflects already-added status; Can view cart. E3_Student Already Enrolled in Course ● Trigger: Occurs at Normal Flow Step 3 (System checks enrollment status). ● Sequence: 1. Students are logged in and viewing the Course Detail page. 2. Students have already purchased and enrolled in this course. 3. Student attempts to click "Thêm vào giỏ hàng" button. 4. The system validates enrollment status. 5. The system detects the existing enrollment record for this student + course. 6. The system prevents add-to-cart operation. 7. Add to Cart button replaced with "Học ngay" (Start Learning) button.
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	8. System may display message: "Bạn đã đăng ký khóa học này." (MSG-ALREADY-ENROLLED). 9. Clicking "Học ngay" redirects to the learning page. Postcondition: Already-enrolled course NOT added to cart; Button shows "Start Learning"; Student redirected to learning page if clicked; Good UX prevents re-purchase.
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2.6.2 Remove cart item

Primary Actors	Student	Secondary Actors	None
Description	The "Remove Item from Cart" use case allows a logged-in Student to remove a specific course from their shopping cart if they decide not to purchase it. The system updates the cart total immediately after removal.		
Preconditions	• The Student is logged in. • The Student is viewing the Shopping Cart page. • The cart contains at least one course.		
Postconditions	• The selected course is removed from the Student's cart database. The cart's Total Price is recalculated. The Cart Item Count in the header is updated.		
Normal Sequence/Flow	1. The Student views the list of courses in the Cart. 2. The Student identifies the course they want to remove. 3. The Student clicks the "Xóa" (Remove) button/link associated with that course. 4. The system sends a request to remove the item from the database. 5. The system recalculates the Subtotal and Total amounts. 6. The system removes the item row from the interface. 7. The system updates the displayed Total Price and the Cart Counter in the navigation bar. 8. The system displays a toast message: "Đã xóa khóa học khỏi giỏ hàng."		

Alternative Sequences/Flows	Remove Last Item (Cart Empty): At step 6, if the removed item was the last one in the cart: 1. The system clears the cart list. 2. The system displays the "Empty Cart" state (Image/Icon + Message "Giỏ hàng trống"). 3. The "Thanh toán" (Checkout) button becomes disabled. Confirmation (Optional - UI dependent): At step 3, if the system requires confirmation: 1. System shows a popup: "Bạn có chắc muốn xóa khóa học này?". 2. Student clicks "Đồng ý" -> Proceed to step 4. 3. Student clicks "Hủy" -> Cancel action.
Exception	E1_Item Already Removed (Concurrent Deletion) ● Trigger: Occurs at Normal Flow Step 4 (System attempts to remove item from database). ● Sequence: 1. Student views cart with Course A. 2. Student (or another session) removes Course A from cart in different tab/device. 3. Student clicks "Xóa" (Remove) button on original cart view. 4. System sends removal request to backend. 5. System queries database to find cart item. 6. System detects that item no longer exists in cart (already deleted). 7. System prevents error and refreshes cart state. 8. System displays message: "Khóa học đã được xóa trước đó. Giỏ hàng đã được làm mới." (MSG-ITEM-ALREADY-REMOVED). 9. Cart list refreshes to show current state. 10. Total price recalculated based on remaining items. Postcondition: No error thrown; Cart refreshed with current state; Student sees updated cart; Graceful handling of concurrent deletion.

2.6.3 Apply voucher

Primary Actors	Student	Secondary Actors	None
Description	The "Apply Voucher" use case allows a Student to enter a promotional code during the shopping cart review process. If the code is valid, the system calculates the discount amount and deducts it from the total price.		
Preconditions	- The Student is logged in and viewing the Shopping Cart page. - The cart contains at least one course. - The Student possesses a valid voucher code.		
Postconditions	The voucher is applied to the cart. The discount amount is displayed. The Total Price is updated.		
Normal Sequence/Flow	- The Student locates the Voucher section in the Shopping Cart screen. - The Student enters a code (e.g., "SKILLUP20") into the input field. - The Student clicks the "Áp dụng" (Apply) button. - The system sends the code to the server for validation. - The system verifies validity (Exists? Not expired? Min spend met? Course eligible?). - The system calculates the discount amount (Percentage or Fixed amount). - The system updates the Total Price and displays the discount row. - The system displays a success message: "Áp dụng mã giảm giá thành công." - The UI updates to show the applied voucher tag with a "Remove" (X) option.		
Alternative Sequences/Flows	Invalid/Expired Code: At step 5, if the code is invalid: - The system displays an error message: "Mã giảm giá không hợp lệ hoặc đã hết hạn." - The Total Price remains unchanged. Remove Voucher: After applying, if the Student wants to change the code: - The Student clicks the "X" (Remove) icon on the applied voucher tag. - The system removes the discount. - The system recalculates the Total Price back to the original subtotal.		
Exception	E1_Validation Error (Empty Voucher Code) - Trigger: Occurs at Normal Flow Step 3 (Student clicks "Áp dụng" button). - Sequence:		

	1. Student locates voucher section in shopping cart. 2. Student leaves voucher code input field empty. 3. Student clicks "Áp dụng" (Apply) button. 4. System validates input. 5. System detects that voucher code field is empty. 6. System prevents API request. 7. System displays inline error message below input field: "Vui lòng nhập mã giảm giá." (MSG-VAL-VOUCHER-EMPTY). 8. Input field highlighted in red. 9. Cart total remains unchanged. Postcondition: Voucher NOT applied; Student must enter code; Cart unchanged; No API call made. E2_Invalid Voucher Code (Does Not Exist) ● Trigger: Occurs at Normal Flow Step 5 (System verifies voucher validity). ● Sequence: 1. Student enters voucher code "INVALID123". 2. Student clicks "Áp dụng" button. 3. System sends code to server for validation. 4. System queries database for voucher code. 5. System detects that voucher code does not exist in database. 6. System returns validation failure. 7. System displays error message: "Mã giảm giá không hợp lệ hoặc không tồn tại." (MSG-VOUCHER-NOT-FOUND). 8. Input field may show error styling. 9. Cart total remains unchanged. Postcondition: Voucher NOT applied; Student informed code invalid; Cart unchanged; Can retry with different code. E3_Expired Voucher Code ● Trigger: Occurs at Normal Flow Step 5 (System verifies expiration date). ● Sequence: 1. Student enters valid voucher code format. 2. Student clicks "Áp dụng" button. 3. System sends code to server. 4. System retrieves voucher record from database. 5. System checks validity period (start date and end date). 6. System detects that current date/time is after voucher end date. 7. System marks voucher as expired. 8. System displays error message: "Mã giảm giá đã hết hạn. Vui lòng sử dụng mã khác." (MSG-VOUCHER-EXPIRED). 9. May show expiration date: "Mã này hết hạn vào: 10/12/2025".
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	10. Cart total remains unchanged. Postcondition: Expired voucher NOT applied; Student informed of expiration; Cart unchanged; Can try different code.
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2.7 Lecturer Application Feature

2.7.1 Create Lecturer Application

Primary Actors	Lecturer (New/Pending Approval)	Secondary Actors	moderator system
Description	The "Submit Lecturer Application" use case allows a newly registered Lecturer to complete their profile by submitting professional details (experience, CV, degrees). This step is mandatory to activate the teaching privileges (create courses).		
Preconditions	• The User has registered a Lecturer account successfully. • The User is logged into the Lecturer Dashboard. • The User's account status is "Pending" or "Not Verified".		
Postconditions	• The application data (CV, Description) is saved. The application status updates to "Đang chờ duyệt" (Pending Approval). The User sees the submission confirmation screen.		
Normal Sequence/Flow	1. Registration Phase: • Guest selects "Lecturer" tab on Register screen . • Guest fills basic info (Email, Name, Password) and clicks Register. • System creates account with Role = Lecturer. 2. Application Phase: • User logs in and is directed to the Lecturer Dashboard. • User navigates to the "Giảng dạy" (Teaching) or "Đơn ứng tuyển" (Applications) section. • System displays the Application Form. • User fills in "Mô tả / Kinh nghiệm" (Description/Experience). • User uploads "Ảnh bằng cấp / CV" (Credentials). • User clicks "Nộp đơn" (Submit). 3. Completion Phase: • System validates files and text content. • System saves the application record. • System displays the Success Screen With message "Đơn ứng tuyển đã được gửi!".		

Alternative Sequences/Flows	● Re-submission (If Rejected): If a previous application was rejected: ● User views the rejection reason on the dashboard. ● User clicks "Nộp đơn mới" (Submit New) or "Chỉnh sửa" (Edit). ● User updates the info and resubmits.
Exception	E1_Unauthorized Access (Non-Lecturer Role) ● Trigger: Occurs at Precondition validation (User attempts to access application form). ● Sequence: 1. User (Student, Moderator, or Admin) attempts to navigate to lecturer application submission page. 2. System validates user role. 3. System detects that logged-in user does not have 'Lecturer' role. 4. System prevents access to application form. 5. System displays error message: "Bạn không có quyền truy cập trang này. Chỉ tài khoản giảng viên mới có thể nộp đơn ứng tuyển." (MSG-UNAUTHORIZED-ACCESS). 6. System redirects user to appropriate dashboard. 7. System may log unauthorized access attempt. Postcondition: Unauthorized user cannot access application form; User redirected; Access attempt logged; Security maintained. E2_Missing_Required_Files ● Trigger: User does not upload CV or degree files ● System: Display error "Vui lòng tải Cv lên" or "Vui lòng tải bằng cấp lên" ● Result: Application submission blocked E3_Invalid_File_Format ● Trigger: CV is not PDF or degree is not image (JPG/PNG) ● System: Display "CV chỉ chấp nhận dạng PDF" or "Chỉ ảnh JPG / PNG được chấp nhận" ● Result: File rejected, must reselect E4_File_Too_Large ● Trigger: File size > 5MB ● System: Display "File không vượt quá 5 MB" ● Result: File rejected E5_Already_Has_Pending_Application ● Trigger: User already has application with Status = "Pending" ● System: Block new submission, redirect to view existing application ● Result: Must wait for current application to be approved/rejected E6_Unauthenticated ● Trigger: Token expired or not logged in

	• System: Return 401 "Unauthorized" • Result: Must login again E7_Upload_Service_Failed • Trigger: Cloudinary error (network issue, quota exceeded, server down) • System: Return 500 Internal Server Error • Result: Application not saved, retry required E8_Database_Error • Trigger: Error during SaveChanges() to database • System: Return "Không thể nộp CV , làm ơn thử lại" • Result: Application not saved, must resubmit
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2.7.2 View detail Lecturer Application

Primary Actors	Lecturer	Secondary Actors	none
Description	The "View My Application Detail" use case allows a Lecturer (who has submitted a registration form) to review the details of their application. They can check the current approval status (Pending, Approved, Rejected), review the submitted experience text and credentials, and access the "Edit" function if the application is still pending or was rejected.		
Preconditions	• The User is logged in as a Lecturer. • The User has already submitted an application. • The User navigates to the "Teaching" or "My Application" section in the Dashboard.		
Postconditions	• The application details and current status are displayed. The User knows whether they need to update information or wait for approval.		
Normal Sequence/Flow	1. The User accesses the Lecturer Dashboard. 2. The system retrieves the latest application record for this user. 3. The system displays the Application Detail Screen containing: • Status Badge: e.g., "Đang chờ duyệt" (Pending) or "Từ chối" (Rejected). • Description: The text previously entered by the user. • Attachments: List of uploaded images/files (CV, Degrees). • Actions: "Chỉnh sửa" (Edit) button (if applicable). 4. The User reviews the information to ensure accuracy. 5. <i>Optional</i>: The User clicks on an attachment to preview it. 6. <i>Optional</i>: If the status allows, the User clicks "Chỉnh sửa" to update details.		
Alternative Sequences/Flows	Application Rejected: At step 3, if the status is "Rejected": 1. The system displays the status in Red.		

	- The system displays the Rejection Reason provided by the Admin (e.g., "Ảnh bằng cấp bị mờ"). - The system enables the "Chỉnh sửa" or "Nộp đơn mới" button. Application Approved: At step 3, if the status is "Approved": - The system displays the status in Green. - The fields become Read-only (cannot be edited). - The system displays a message: "Bạn đã trở thành giảng viên chính thức."
Exception	E1_No Application Found - Trigger: Occurs at Normal Flow Step 2 (System retrieves latest application record). - Sequence: - Lecturer navigates to "My Application" section in Dashboard. - System attempts to retrieve application record for logged-in user. - System detects that no application exists (user never submitted application). - System displays empty state screen: - Empty state illustration (e.g., document icon) - Message: "Bạn chưa nộp đơn ứng tuyển nào." (MSG-NO-APPLICATION) - Subtitle: "Hãy nộp đơn ứng tuyển để trở thành giảng viên chính thức." - Call-to-action button: "Nộp đơn ứng tuyển" (Submit Application) - Application detail view not displayed. Postcondition: Empty state displayed; User informed no application exists; Clear guidance to submit new application; Good onboarding experience. E2_Unauthorized_Access - Trigger: User not logged in or token expired - Sequence: - User attempts to access application details - System checks userId from token - System detects userId == null - System returns Code 401 - "Unauthorized" - Postcondition: Access denied, redirect to login required

	E3_Non_Lecturer_Role ● Trigger: Logged-in user does not have Lecturer role (RoleId ≠ 4) ● Sequence: 1. User navigates to application details page 2. System validates account.RoleId 3. System detects RoleId ≠ 4 4. System throws UnauthorizedAccessException: "Bạn không có quyền truy cập." Postcondition: Access denied, error displayed E4_Application_Not_Found_By_ID ● Trigger: Specific application ID does not exist in database ● Sequence: 1. User clicks on specific application link 2. System queries GetApplicationByIdAsync(applicationId) 3. System returns null (application not found) 4. System returns Code 404 - "Không tìm thấy CV" Postcondition: Detail view not displayed, "Not found" message shown E5_Server_Error Trigger: Database error, network failure, or unexpected exception Sequence: 1. User requests application details 2. System encounters exception during retrieval 3. System catches Exception 4. System returns Code 500 + exception message Postcondition: Error screen displayed, user prompted to retry
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2.7.3 Manage Lecturer Application

Primary Actors	System Moderator	Secondary Actors	none
Description	The "Manage Lecturer Application" use case allows a System Moderator to review the details of lecturer applications. They can check the current approval status (Pending, Approved, Rejected), review the submitted experience text and credentials, and choose to reject or accept the application.		
Preconditions	• The moderator logged into the system with an account. • The account Role must be "System Moderator". • The account status must be "Active". • The Moderator clicked on the "Quản lý ứng tuyển" tab on the side menu and navigate to the "Quản lý đơn ứng tuyển của giảng viên" page.		
Postconditions	• View Application List: The system successfully displays the list of all Lecturer's applications. • Approve Application: The application status is updated successfully to "Đã chấp nhận", and an email is sent afterward. • Reject Application: The application status is updated successfully to "Đã từ chối", and an email is sent afterward.		
Normal Sequence/Flow	View Application List: 1. The moderator accesses the "Quản lý đơn ứng tuyển của giảng viên" page. 2. The system successfully display the list of all Lecturer's applications, consist of following informations: • Họ tên (Full name) • Email • Tiêu đề (Title) • Chức danh (Profession) • Mô tả (Short Description) • Trạng thái (Status) • Ngày tạo (Date of Create) • CVs and Degrees buttons. • Thao tác (Action: Accept or Reject)		

Alternative Sequences/Flows	A1: Approve Application: ○ The moderator accesses the “Quản lý đơn ứng tuyển của giảng viên” page. ○ The system successfully displays the list of all Lecturer’s applications, consist of following informations: ■ Họ tên (Full name) ■ Email ■ Tiêu đề (Title) ■ Chức danh (Profession) ■ Mô tả (Short Description) ■ Trạng thái (Status) ■ Ngày tạo (Date of Create) ■ CVs and Degrees buttons. ■ Thao tác (Action: Accept or Reject) ○ The moderator clicks the “Duyệt” button on the Action column to accept the application. ○ The system updates the application status to “Đã chấp nhận”, and sends an email to notify the lecturer of approval. A2: Reject Application: ○ The moderator accesses the “Quản lý đơn ứng tuyển của giảng viên” page. ○ The system successfully displays the list of all Lecturer’s applications, consist of following informations: ■ Họ tên (Full name) ■ Email ■ Tiêu đề (Title) ■ Chức danh (Profession) ■ Mô tả (Short Description) ■ Trạng thái (Status) ■ Ngày tạo (Date of Create) ■ CVs and Degrees buttons. ■ Thao tác (Action: Accept or Reject) ○ The moderator clicks the “Từ chối” button on the Action column to reject the application. ○ The system updates the application status to “Đã từ chối” and sends an email to notify the lecturer of disapproval.
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Exception	E1_No Application Found ● Trigger: Occurs at Normal Flow Step 2 (System retrieves latest application record). ● Sequence: 1. Lecturer navigates to "My Application" section in Dashboard. 2. System attempts to retrieve application record for logged-in user. 3. System detects that no application exists (user never submitted application). 4. System displays empty state screen: a. Empty state illustration (e.g., document icon) b. Message: "Bạn chưa nộp đơn ứng tuyển nào." (MSG-NO-APPLICATION) c. Subtitle: "Hãy nộp đơn ứng tuyển để trở thành giảng viên chính thức." d. Call-to-action button: "Nộp đơn ứng tuyển" (Submit Application) 5. Application detail view not displayed. Postcondition: Empty state displayed; User informed no application exists; Clear guidance to submit new application; Good onboarding experience. E2_Unauthorized_Access ● Trigger: Moderator not logged in or token expired ● Sequence: 1. User attempts to access application management page 2. System validates token/userId 3. System detects userId == null 4. System returns Code 401 - "Unauthorized" Postcondition: Access denied, redirect to login E3_Non_Moderator_Role ● Trigger: Logged-in user is not System Moderator (RoleId ≠ 2) ● Sequence: 1. User attempts to view or update applications 2. System validates _currentUserService.RoleId 3. System detects RoleId ≠ 2 4. System returns Code 400 - "Không thể cập nhật trạng thái ứng tuyển! Kiểm tra quyền truy cập."
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	Postcondition: Access denied, moderator privileges required E4_Application_Not_Found_For_Update ● Trigger: Moderator tries to approve/reject non-existent application ● Sequence: 1. Moderator clicks "Duyệt" or "Từ chối" button 2. System queries GetByIdAsync(applicationId) 3. System returns null 4. System returns Code 400 - "Không thể cập nhật trạng thái ứng tuyển! Kiểm tra quyền truy cập." Postcondition: Status not updated, application not found E5_Status_Update_Failed ● Trigger: Database fails to update application status ● Sequence: 1. Moderator submits approve/reject action 2. System calls UpdateStatusAsync() 3. System encounters database error or SaveChanges() returns false 4. System returns Code 400 - "Không thể cập nhật trạng thái ứng tuyển! Kiểm tra quyền truy cập." Postcondition: Status unchanged, error displayed E6_Email_Notification_Failed ● Trigger: Email service fails after successful status update ● Sequence: 1. Application status updated successfully 2. System attempts SendLecturerApplicationStatusEmailAsync() 3. Email service throws exception 4. System throws Exception: "Lỗi cập nhật trạng thái đơn ứng tuyển giảng viên !" 5. System returns Code 500 Postcondition: Status updated in DB but email not sent, error logged E7_Server_Error ● Trigger: Unexpected system error during retrieval or update ● Sequence: 1. Moderator performs any action 2. System encounters unexpected exception
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	3. System catches Exception 4. System returns Code 500 + exception message Postcondition: Operation failed, user prompted to retry
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2.8 Shopping cart & payment feature

2.8.1 Buy course

Primary Actors	Student	Secondary Actors	none
Description	The "Buy Course" use case allows a logged-in Student to purchase courses. There are two entry points: Direct Purchase (Mua ngay): Buying a single course directly from the detail page. Cart Checkout (Thanh toán giỏ hàng): Buying multiple courses saved in the cart. Upon success, the system enrolls the student in the course(s) and records the transaction.		
Preconditions	- The Student is logged in. Case A (Buy Now): Student is on Course Detail page, course is not enrolled.		

	● Case B (Cart): Student is on Cart page, cart is not empty.
Postconditions	● Transaction status is "Success". Student is enrolled in the purchased course(s). If coming from Cart, the purchased items are removed from the Cart. System displays the Payment Success screen.
Normal Sequence/Flow	1. Entry Points: ● Case A: Student clicks "Mua ngay" on Course Detail page. ● Case B: Student clicks "Tiến hành thanh toán" on Cart page. 2. The system displays the Checkout Screen (image_c361d7.jpg) showing: ● List of course(s) to be purchased. ● Total amount (Tổng thanh toán). ● Payment method selection (PayOS/QR). 3. The Student selects the Payment Method (e.g., PayOS). 4. The Student clicks the "Thanh toán" (Pay) button. 5. The system redirects User to the Payment Gateway (or displays QR Code). 6. The Student completes the payment via Banking App. 7. The Payment Gateway returns the transaction status to the system. 8. The system validates the payment and updates the database (Enroll user, create invoice). 9. The system displays the Payment Success Screen. 10. The Student clicks "Vào học ngay" or "Về trang chủ".
Alternative Sequences/Flows	Payment Failed / Cancelled: At step 7, if the payment fails or user cancels: 1. The system displays a Payment Failed Screen. 2. Message: "Thanh toán thất bại hoặc đã bị hủy." 3. Action: "Thử lại" (Retry) or "Quay lại giỏ hàng". Voucher Application (Optional): At step 2, the User may enter a Voucher code (if not applied in Cart). 1. System validates and updates the "Total Amount".
Exception	E1: Authentication Error ● Trigger: In Normal flow, if the user is not logged into the system with an account. ● Sequence: 1. The user is not logged into the system 2. The user adds items to the cart. 3. The user proceeds to checkout. ● System: Displays a toast message informing the user to login. ● Result: The user is redirected to the login page. E2: Validation Error

	● Trigger: In Normal flow, if the user already owns one or more courses in the cart. ● Sequence: 1. The user is logged into the system 2. The user adds items including owned courses to the cart. 3. The user proceeds to checkout. ● System: Displays a toast message “Bạn đã sở hữu khoá học này rồi”. ● Result: The user is notified of an owned course in the cart. E3: Payment Gateway Error ● Trigger: In Normal flow, if the payment gateway encounters an unexpected error. ● Sequence: 1. The user is logged into the system. 2. The user adds items to the cart. 3. The user proceeds to checkout. 4. The 3rd party payment gateway fails unexpectedly. ● System: Displays a toast message “Có lỗi xảy ra. Vui lòng thử lại sau”. ● Result: The user is notified of a payment gateway error.
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2.8.2 View history transaction

Primary Actors	Student	Secondary Actors	none
Description	The "View Transaction History" use case allows a logged-in Student to view a chronological list of their financial activities within the platform. This includes wallet top-ups, course purchases, and refunds (if any). It helps users track their spending and verify payment statuses.		
Preconditions	● The User is logged in as a Student. ● The User has navigated to the "Wallet" or "Profile" section.		
Postconditions	● A tabular list of transactions is displayed. The user can see the status (Success/Failed) and amount of each transaction.		

Normal Sequence/Flow	1. The Student clicks on "Lịch sử giao dịch" (Transaction History) in the sidebar menu or Wallet page. 2. The system retrieves the transaction records associated with the Student's User ID from the database. 3. The system sorts the records by "Created Date" (Newest first). 4. The system displays the Transaction History Screen containing a table with: ● Transaction Code: Unique identifier (e.g., DH64169). ● Date: Timestamp of creation. ● Amount: Money value (Red for payment, Green for top-up). ● Type: Description (e.g., Course Payment, Wallet Top-up). ● Status: Current state (e.g., Success). 5. The Student views the list. 6. Optional: If the list is long, the Student clicks the pagination arrows ("<" or ">") to view older transactions.
Alternative Sequences/Flows	No Transactions Found: At step 2, if the user has never made a transaction: 1. The system displays an empty state message: "Bạn chưa có giao dịch nào." (No transactions found). 2. The table is hidden or shows an empty row.
Exception	E1_No Transactions Found (Empty State) ● Trigger: Occurs at Normal Flow Step 2 (System retrieves transaction records). ● Sequence: 1. Student navigates to "Lịch sử giao dịch" (Transaction History) page. 2. System queries database for transactions associated with Student's User ID. 3. System detects that no transaction records exist (count = 0). 4. System displays empty state on Transaction History page: a. Empty state illustration/icon (e.g., empty wallet icon) b. Message: "Bạn chưa có giao dịch nào." (MSG-EMPTY-TRANSACTION-HISTORY) c. Subtitle: "Các giao dịch của bạn sẽ xuất hiện ở đây sau khi bạn nạp tiền hoặc mua khóa học." 5. Transaction table is hidden or shows empty placeholder.

	6. System may show call-to-action button: "Khám phá khóa học" (Explore courses). Postcondition: Empty state displayed; Student informed no transactions exist; Can navigate to browse courses; No errors; Good user experience for new users.
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2.8.3 Feedback course

Primary Actors	Student	Secondary Actors	none
Description	The "Submit Course Feedback" use case allows an enrolled Student to rate a course (using a star system) and write a textual review regarding their learning experience. This feedback helps prospective students make informed decisions and provides valuable insights to the lecturer.		
Preconditions	• The Student is logged in. • The Student is enrolled in the course. • The Student has not submitted a review for this course previously (or is updating an existing one).		
Postconditions	• The review data (stars + comment) is saved to the database. The course's Average Rating and Total Reviews count are recalculated. The review becomes visible on the Course Detail page.		
Normal Sequence/Flow	1. The Student navigates to the Learning Screen or Course Detail Page (Enrolled view). 2. The Student clicks the "Đánh giá" (Rate Course) button/link. 3. The system displays the Feedback Modal containing: • Star rating input (1 to 5 stars). • Text area for comments. 4. The Student selects a Star Rating (e.g., 5 stars). 5. The Student enters a Review Comment (e.g., "Khóa học rất hay, dễ hiểu"). 6. The Student clicks the "Gửi đánh giá" (Submit) button. 7. The system validates the input (Stars are mandatory). 8. The system saves the review and updates the course statistics. 9. The system displays a success notification: "Cảm ơn bạn đã đánh giá!". 10. The modal closes.		
Alternative Sequences/Flows	Missing Rating: At step 6, if the Student clicks Submit without selecting stars: 1. The system displays an error message: "Vui lòng chọn số sao đánh giá."		

	2. The form remains open. Edit Review: If the user already reviewed the course: 1. The system pre-fills the modal with the previous rating and comment. 2. The Student modifies the content and clicks "Cập nhật" (Update).
Exception	E1_Session Expired During Submission ● Trigger: Occurs at Normal Flow Step 7 (Student clicks "Gửi đánh giá" after session expires). ● Sequence: 1. Student opens feedback modal and begins writing review. 2. Student's session expires while modal is open (e.g., idle for 30+ minutes). 3. Student completes form and clicks "Gửi đánh giá" button. 4. System validates session token. 5. System detects expired JWT authentication token. 6. System prevents review submission. 7. System displays error toast: "Phiên đăng nhập đã hết hạn. Vui lòng đăng nhập lại." (MSG-SESSION-EXPIRED). 8. System may save form data to localStorage for recovery. 9. System redirects Student to login page. 10. After re-authentication, system may restore form data. Postcondition: Review not submitted; Student must re-authenticate; Form data may be preserved for recovery; System prompts login.

2.9 Profile Management Feature

2.9.1 View Profile

Primary Actors	Lecturer	Secondary Actors	none
Description	The "View Lecturer Profile" use case allows a logged-in Lecturer to view their account details. This includes Public Information (visible to students like Title, Profession, Bio) and Private Information (Financial details for payouts like Bank Account).		
Preconditions	● The User is logged in as a Lecturer. ● The User has navigated to the "Profile" or "Account Settings" page		

Postconditions	The Lecturer's full profile information is displayed. Sensitive financial data is displayed (possibly masked for security).
Normal Sequence/Flow	- The Lecturer clicks on their Avatar or "Tài khoản" in the navigation menu. - The system retrieves the user's basic info (from User table) and extended info (from Lecturer table). - The system displays the Profile Screen divided into sections: - Basic Info: Avatar, Name, Email. - Professional Info: Title (Học hàm/học vị), Profession (Nghề nghiệp), Bio. - Financial Info: Bank Name, Bank Number, Receiver Name. - The Lecturer reviews the information. - Optional: The Lecturer clicks "Chỉnh sửa" (Edit) to update information.
Alternative Sequences/Flows	Missing Financial Info: At step 3, if the Lecturer hasn't set up payment details yet: The system displays a warning or prompt in the Financial section: "Vui lòng cập nhật thông tin ngân hàng để nhận thanh toán."
Exception	E1_Session Expired - Trigger: Occurs at Normal Flow Step 1 (User navigates to Profile page). - Sequence: - Lecturer's session expires (e.g., JWT token expired). - The lecturer attempts to access the Profile page. - System validates session token. - System detects expired authentication tokens. - System prevents profile page access. - System displays toast message: "Phiên đăng nhập đã hết hạn. Vui lòng đăng nhập lại." (MSG-SESSION-EXPIRED). - System redirects Lecturer to login page. - After re-authentication, system redirects back to Profile page. Postcondition: Profile not displayed; Lecturer must re-authenticate; Redirected to login; Returns to profile after login.

2.9.2 Update Profile

Primary Actors	Lecturer	Secondary Actors	none
Description	The "Update Lecturer Profile" use case allows a logged-in Lecturer to modify their personal, professional, and financial details. Note that profile picture updates are not supported in this function; users can only update text-based information.		
Preconditions	• The User is logged in as a Lecturer. • The User is viewing the Profile screen. • The User clicks the "Chỉnh sửa" (Edit) button.		
Postconditions	• The Lecturer's text information is updated in the database. The User interface reflects the new changes immediately.		
Normal Sequence/Flow	1. The Lecturer clicks "Chỉnh sửa hồ sơ" on the Profile page. 2. The system displays the Edit Profile Form with current data pre-filled (Avatar is read-only or hidden). 3. The Lecturer modifies the allowed fields: • Updates Contact Info (Full Name, Phone). • Updates Professional Info (Title, Profession, Bio). • Updates Financial Info (Bank Name, Number, Receiver). 4. The Lecturer clicks the "Lưu thay đổi" (Save Changes) button. 5. The system validates the input (e.g., checks required fields, validates phone/bank number format). 6. The system updates the record in the database. 7. The system displays a success notification: "Cập nhật hồ sơ thành công." 8. The system redirects back to the View Profile mode.		
Alternative Sequences/Flows	• Validation Error: At step 5, if the Lecturer enters an invalid phone number (e.g., contains letters): ○ The system highlights the "Phone" field. ○ The system displays error: "Số điện thoại không hợp lệ." ○ The update is blocked until corrected.		
Exception	E1_Validation Error (Invalid Phone Number Format) • Trigger: Occurs at Normal Flow Step 5 (System validates input after Lecturer clicks "Lưu thay đổi"). • Sequence: 1. System validates phone number field. 2. System detects that phone number contains: 3. Non-numeric characters (letters, special characters)		

	4. Invalid format (not matching Vietnamese phone pattern) 5. Incorrect length (not 10 digits) 6. System prevents profile update. 7. System highlights "Phone" field in red. 8. System displays inline error message: "Số điện thoại không hợp lệ. Vui lòng nhập 10 chữ số." (MSG-VAL-INVALID-PHONE). Postcondition: Profile not updated; Lecturer remains on edit form to correct phone number; No database changes made. E2_Validation Error (Missing Required Field - Full Name) • Trigger: Occurs at Normal Flow Step 5 (System validates input after Lecturer clicks "Lưu thay đổi"). • Sequence: 1. System validates required fields. 2. System detects that "Full Name" field is empty or contains only whitespace. 3. System prevents profile update. 4. System highlights "Full Name" field in red. 5. System displays inline error message: "Họ và tên là bắt buộc." (MSG-VAL-NAME-REQUIRED). Postcondition: Profile not updated; Lecturer remains on edit form to enter full name; No database changes made.
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2.10 Question Feature

2.10.1 Manage Question

Primary Actors	Lecturer	Secondary Actors	None
Description	Allows Lecturer to view a list of existing questions in a Quiz, make edits to question content, or delete questions from the Quiz.		
Preconditions	• The Lecturer must be logged in to the system. • Quiz has been created. • The instructor is on the Quiz detail.		
Postconditions	• List of questions updated (edited or deleted).		
Normal Sequence/Flow	1. The Lecturer clicks on a Quiz in the section list. 2. The Lecturer clicks on a specific Quiz item.. 3. The System displays the Quiz interface, including the list of questions (Question Text, Type). 4. The Lecturer reviews the list of questions.		
Alternative Sequences/Flows	A1. Edit Question :		

	● At Step 4, the Lecturer clicks the "Edit" icon on a specific question. ● The System displays the "Chỉnh sửa câu hỏi" form with current data pre-filled. ● The Lecturer modifies the details (e.g., Text, Answers, Correct Option) and clicks "Cập nhật". ● The System validates the new data. ● The System updates the question and refreshes the list. A2. Delete Question from Quiz: ● At Step 4, the Lecturer clicks the "Delete" icon on a specific question. ● The System prompts for confirmation: "<i>Bạn có muốn xóa câu hỏi này không?</i>" ● The Lecturer confirms. ● The System removes question from quiz. ● The System displays a success message.
Exception	E1. Validation Error (Edit): ● In A1, if the Lecturer clears the "Câu hỏi".The system display "Vui lòng nhập câu hỏi". ● If the Lecturer does not select any correct answer. The system will report an error "Vui lòng ít nhất có 1 đáp án đúng". E2_Unauthorized_Access Trigger: Lecturer not logged in or token expired Sequence: User attempts to access Quiz question management System validates authentication token System detects userId == null System returns Code 401 - "Unauthorized" Postcondition: Access denied, redirect to login required E3_Non_Lecturer_Role Trigger: Logged-in user does not have Lecturer role Sequence: User attempts to edit/delete questions System validates user role System detects role is not Lecturer

	System denies access Postcondition: Action blocked, unauthorized access prevented E4_Quiz_Not_Found Trigger: Quiz ID does not exist in database Sequence: Lecturer navigates to Quiz detail page System queries quiz by ID System returns null (quiz not found) System displays error message Postcondition: Quiz detail not displayed, error shown E5_Question_Not_Found Trigger: Question ID does not exist when attempting edit/delete Sequence: Lecturer clicks Edit/Delete on a question System queries question by ID System returns null System displays error message Postcondition: Operation cancelled, question not found E6_Permission_Denied Trigger: Lecturer tries to manage questions in Quiz they don't own Sequence: Lecturer attempts to edit/delete question System validates Quiz ownership (Quiz.CourseId → Course.LecturerId ≠ current userId) System detects permission mismatch System denies access Postcondition: Operation blocked, ownership protected
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	E7_Database_Update_Failed Trigger: Database fails to save changes during edit/delete Sequence: Lecturer submits edit or confirms delete System attempts SaveChanges() Database operation fails System displays error message Postcondition: Changes not saved, user prompted to retry E8_Cannot_Delete_Question_Constraint Trigger: Question cannot be deleted due to dependencies (e.g., student submissions exist) Sequence: Lecturer confirms delete question System attempts to delete Database constraint violation occurs System displays error message Postcondition: Question not deleted, constraint preserved E9_Invalid_Answer_Configuration Trigger: Edit validation fails for answer options Sequence: Lecturer modifies question answers System validates answer configuration System detects issues (e.g., duplicate answers, empty options) System displays "Vui lòng kiểm tra các đáp án" Postcondition: Update blocked, validation errors shown
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2.10.2 Manage Question Bank

Primary Actors	Lecturer	Secondary Actors	None
Description	Allows Lecturers to manage a shared question bank for each Section of the course. Functions include: viewing the list, creating new questions , editing, deleting, and importing in bulk from Excel files.		
Preconditions	- The Lecturer must be logged in to the system. - The Lecturer has at least one Course and Section created.		
Postconditions	Question bank data is updated.		
Normal Sequence/Flow	- Lecturers navigate the "Ngân hàng đề" page. - The system displays a list of the lecturer's courses in the left column. - Lecturer selects a Course. - The system loads and displays a list of Sections belonging to that course. - The lecturer selects a specific Chapter. - The system displays a list of questions in that chapter (including columns:Câu hỏi, Ngày tạo, Ngày cập nhật, Hành động).		
Alternative Sequences/Flows	A1: Create New Question : - At the question list interface (Step 6), the Lecturer clicks the "Add New" button. - The system displays the Modal "Create New Question". - The Lecturer enters the Question Content (using Rich Text Editor). - The Lecturer selects the Question Type (Single Choice / Multiple Choice). - The Lecturer enters the Answers and selects the correct answer.(Optional) The Lecturer can upload a photo for the question or each answer. - The Lecturer clicks the "Create Question" button. - The system checks the data (Validation). - The system saves the new question to the bank and updates the displayed list. A2: Edit Question : - At the question list, the Lecturer clicks the "Edit" button (pencil icon) on the line of the question he wants to edit. - The system displays the Modal "Chỉnh sửa câu hỏi" with the old data already filled in. - The lecturer changes the information. - The lecturer clicks the "Lưu" button. - The system updates the question and notifies success. A3: Delete Question - In the question list, the lecturer clicks the "Delete" button (trash can icon). - The system displays a confirmation dialog box (Popconfirm). - The lecturer confirms the deletion.		

	- The system deletes the question and removes it from the display list. A4: Import from Excel - In the list interface, the lecturer clicks the "Nhập từ Excel" button. - The system displays the Modal "Nhập câu hỏi từ excel". - The lecturer clicks the "Tải file mẫu" button to download the standard template file (quiz_questions_template.xlsx) provided by the system. - The lecturer fills in the question information into the Excel file according to the correct template format. - The lecturer returns to the modal and uploads the Excel file with the data filled in. - The system reads the file, checks the format and adds valid questions to the bank. - The system notifies the input result and refreshes the question list.
Exception	E1: Validation Error - Trigger: In A1, A2, A4 flow, if the Lecturer submitted invalid Question Bank field values(too many characters, blank fields, invalid number of correct questions). - Sequence: - The Lecturer opens the Create/Update form or Import window. - The Lecturer enters/imports invalid field values(too many characters, blank fields, invalid number of correct questions). - The Lecturer clicks “Lưu”. - System: Displays a toast message informing the Lecturer of the error. - Result: The Question Bank managing operations is blocked. E2: Authentication Error - Trigger: In Normal, A1, A2, A3, A4 flow, if the user is not logged into the system with an account. - Sequence: - An user accesses the system without logging in. - The user tries to access the managing page of a course with links. - System: Redirect the user to the login page. - Result: The Question Bank view page is blocked, the user is redirected to the login page. E3: Authorization Error - Trigger: In Normal, A1, A2, A3, A4 flow, if the logged in user role is not Lecturer. - Sequence: - An user logged into the system - The user tries to access the managing page of an unowned course with links. - System: Display the notice “Truy cập bị từ chối”.

	- Result: The Question Bank view page is blocked. E4: Question Not Found - Trigger: In A3 flow, if the deleting question is not found. - Sequence: - The Lecturer logged into the system. - The Lecturer accesses the Question Bank manage page. - The Lecturer clicks Delete on an already deleted question. - System: Display a toast message “Không tìm thấy câu hỏi”. - Result: The Question Bank view page is refreshed, with the deleted Question Banks disappeared from the list. E5: Invalid File Format Type - Trigger: In A1, A2, A4 flow, if the uploaded file format is not valid (A1, A2: JPEG, PNG; A4: Excel). - Sequence: - The Lecturer logged into the system. - The Lecturer accesses the Question Bank manage page. - The Lecturer opens the create/update or import Question Bank window. - The Lecturer uploads invalid files (A1, A2: JPEG, PNG; A4: Excel). - System: Display a toast message “Chỉ được phép nhập file hình ảnh” or “Chỉ được phép nhập file Excel”. - Result: The Question Bank creation is blocked.
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2.11 Answer Feature

2.11.1 Manage Answer

Primary Actors	Lecturer	Secondary Actors	None
Description	Allows Lecturers to manage answer options for a specific question. Includes adding answers, editing content, deleting answers, uploading answer illustrations, and configuring correct answers.		
Preconditions	- The Lecturer must be logged in to the system. - The Lecturer is in the "Tạo câu hỏi mới" or "Chỉnh sửa câu hỏi" interface .		
Postconditions	The list of answers to the question is updated validly.		
Normal Sequence/Flow	- In the question form, the system displays a list of input boxes for the answers (by default, there are usually 2 blank boxes available). - The lecturer enters text content for each answer. - The lecturer determines the correct answer: - If it is a Single Choice question: Check the radio button next to the only correct answer.		

	+ If it is a Multiple Choice question: Check the checkbox next to the correct answers. - The lecturer clicks the "Add answer" button (plus sign +) if they want more options. - The system adds a new input line to the list.
Alternative Sequences/Flows	A2: Delete Answer - The lecturer clicks the "Delete" icon (trash can or minus sign -) next to the answer he wants to delete. - The system removes that answer line from the list.
Exception	E1: Validation Error - Trigger: In Normal, A4 flow, if the Lecturer submitted invalid Answer field values (too many characters, blank fields, invalid number of correct answers). - Sequence: - The Lecturer is creating/updating a question. - The Lecturer submitted invalid Answer field values (too many characters, blank fields, invalid number of correct answers). - System: Displays a toast message informing the Lecturer of the error. - Result: The Question creation is blocked.

2.12 User Feature

2.12.1 Manage Profile

Primary Actors	Student , Lecturer	Secondary Actors	None
Description	The "View Profile" use case allows a student & lecturer to view their personal information and account details on the SkillUp platform		
Preconditions	- Students or lecturers logged into the system with an account. - The account Role must be "Student" or "Lecturer". - The account status must be "Active".		
Postconditions	The user's profile information is successfully displayed.		
Normal Sequence/Flow	View Profile: - The user navigates to their profile section by clicked on the "Hồ sơ của tôi" tab while hovering on their Profile Picture on the Header menu. - The system displays the profile information, including: Profile Picture, Full name, Description, Date of Birth, Join date, Email, Phone number, Gender.		

	3. The user can review their details on the profile page.
Alternative Sequences/Flows	A1: Update Profile: • The user navigates to their profile section by clicked on the “Hồ sơ của tôi” tab while hovering on their Profile Picture on the Header menu. • The system displays the profile information, including: Profile Picture, Full name, Description, Date of Birth, Join date, Email, Phone number, Gender. All along with the “Đổi mật khẩu” section. • The student modifies the editable fields in the “Cài đặt hồ sơ” section: Họ và tên (Full name), Tiểu sử (Description), Ngày sinh (Date of Birth), Số điện thoại (Phone number), Giới tính (Gender). • The student clicks “Lưu thay đổi” button. • The system saves the changes to the database. • The system displays a success message: “Cập nhật hồ sơ thành công”. A2: Change Password: • The user navigates to their profile section by clicked on the “Hồ sơ của tôi” tab while hovering on their Profile Picture on the Header menu. • The system displays the profile information, including: Profile Picture, Full name, Description, Date of Birth, Join date, Email, Phone number, Gender. All along with the “Đổi mật khẩu” section. • The student input fields in the “Đổi mật khẩu” section: Mật khẩu cũ (Current Password), Mật khẩu mới (New Password), Xác nhận mật khẩu mới (Confirm new password). • The student clicks “Cập nhật mật khẩu” button. • The system saves the changes to the database. • The system displays a success message: “Đổi mật khẩu thành công”.
Exception	E1: Validation Errors: • In Alternative Sequence A1(Step 3), A2(Step 3): ○ If the user leaves any of the input fields as blank, the system will give out error messages under the blank fields.

2.13 Ticket Feature

2.13.1 Manage Ticket for Student and Lecturer

Primary Actors	Student, Lecturer	Secondary Actors	None
Description	The “Manage Ticket” feature allows a student or lecturer to create, view and update their support ticket, which will be sent to a Moderator for consideration.		
Preconditions	- Students or lecturers logged into the system with an account. - The account Role must be “Student” or “Lecturer”. - The account status must be “Active”. - The Student or Lecturer clicked on the “Phiếu hỗ trợ” tab on the Header menu and navigate to the “Phiếu hỗ trợ” page.		
Postconditions	Create Ticket: A Ticket is created in the ticket list of both Student/Lecture and System Moderator. View Ticket List: The list of Tickets is displayed successfully. View Ticket Details: The details of a Ticket is displayed successfully. Update Ticket: The Ticket details are updated successfully corresponding to the user inputs.		
Normal Sequence/Flow	Create Ticket: - The user clicks on the “Tạo phiếu mới” button. - The user inputs the “Tiêu đề” for Title and describes their problem in the “Nội dung” input box for Content. - The user then clicks “Tạo phiếu”. - The Ticket will now appear in the user’s and Moderator’s Ticket List.		
Alternative Sequences/Flows	A1: View Ticket List: - The user redirected to the “Phiếu hỗ trợ” page. - The List of Student’s/Lecturer’s Tickets is displayed successfully. A2: View Ticket Details: - The user redirected to the “Phiếu hỗ trợ” page. - The user finds the needed Ticket in the Ticket List. - The user clicks on the “Xem” button. - The Ticket’s information is displayed successfully for review. A3: Update Ticket: - The user then redirected to the “Phiếu hỗ trợ” page. - The user finds the needed Ticket in the Ticket List. - The user clicks on the “Xem” button. - The Ticket’s information is displayed successfully for review. - The user clicks on the “Cập nhật phiếu” button.		

	○ The user edits the “Tiêu đề” for Title and “Nội dung” field for Content in the popup window. ○ The user clicks on the “Lưu thay đổi” button. ○ The system displays the “Cập nhật phiếu thành công” message and the Ticket information is updated.
Exception	E1: Validation Errors: ● In Normal Sequence (Step 2), Alternative Sequence A2,(Step 5); A3,(Step 6): ○ If the user leaves any of the “Tiêu đề” or “Nội dung” field as blank, the system will give out error messages under the blank fields.

2.13.2 Manage Ticket for System Moderator

Primary Actors	System Moderator	Secondary Actors	None
Description	The "Manage Ticket" feature allows a System Moderator to view and handle all Tickets submitted by the Students and Lecturers.		
Preconditions	● The System Moderator logged into the system with an account. ● The account role must be “System Moderator” ● The account status must be “Active”. ● The System Moderator clicked on the “Quản lý phiếu hỗ trợ” tab on the side menu.		
Postconditions	● View All Ticket List: The list of all Student’s/Lecturer’s Tickets is displayed successfully. ● View Ticket Details: The details of a Ticket is displayed successfully. ● Resolve Ticket: The Ticket Status is updated corresponding to the System Moderator choice.		
Normal Sequence/Flow	View All Ticket List: 1. The moderator is redirected to the “Quản lý phiếu hỗ trợ” page. 2. The List of all Student’s/Lecturer’s Tickets is displayed successfully.		
Alternative Sequences/Flows	A1: View Ticket Details: ○ The moderator is redirected to the “Quản lý phiếu hỗ trợ” page. ○ The moderator finds the needed Ticket in the Ticket List. ○ The user clicks on the Eye Icon in the Ticket’s row, on the “Hành động” column.		

	○ The Ticket's information is displayed successfully for View. A2: Resolve Ticket: ○ The moderator is redirected to the “Quản lý phiếu hỗ trợ” page. ○ The moderator finds the needed Ticket in the Ticket List. ○ The user clicks on the Eye Icon in the Ticket's row, on the “Hành động” column. ○ The Ticket's information is displayed successfully for View. ○ The moderator clicks on the “Chấp nhận” button to accept the Ticket's proposal, or “Từ chối” button and enters the “Lý do từ chối” field for Rejecting Reason to decline the Ticket under the “Quyết định” option. ○ The moderator clicks on the “Lưu” button to save the response. ○ The Ticket status is updated correspondingly to the System Moderator decision.
Exception	E1: Validation Errors: ● In Alternative Sequence A2, Step 5: ○ If the Moderator leaves the “Lý do từ chối” input blank, the system will give out an error message telling the user to input the field .

2.14 News Feature

2.14.1 Manage News for Content Moderator

Primary Actors	Content Moderator	Secondary Actors	None
Description	Allows a moderator to create, edit, publish, and delete news articles that will be displayed to students and other users on the platform.		
Preconditions	● The user must be authenticated with the system. ● The user must have the role of 'Content Moderator'.		
Postconditions	● A new article is successfully created, updated, or deleted from the system's database. ● The list of news articles reflects the changes made.		
Normal Sequence/Flow	1. The Moderator logs into the platform and navigates to the administrative dashboard. 2. The Moderator selects the "Quản lý tin tức" option from the menu. 3. The system displays a list of all existing news articles, showing their Title, Status (e.g., Published, Draft), Creation Date, and action buttons (Add New, Edit, Delete).		

Alternative Sequences/Flows	Sub-flow 1: Create a New Article 1. The Moderator clicks the "Add New" button. 2. The system presents a form with fields for Title, Body/Content (using a rich text editor), Feature Image Upload, and a Status dropdown (Published/Draft). 3. The Moderator fills in all required fields and clicks "Save changes" or "Publish". 4. The system validates the input data. 5. The system saves the new article to the database. 6. The system displays a success message (e.g., "Tạo tin tức thành công!") and redirects back to the news management list. Sub-flow 2: Update an Existing Article 1. From the news list, the Moderator clicks the "Edit" button next to the desired article. 2. The system presents the same form as the creation flow, pre-populated with the selected article's data. 3. The Moderator modifies the Title, Body, Image, or Status. 4. The Moderator clicks the "Save changes" button. 5. The system validates and saves the changes to the database. 6. The system displays a success message (e.g., "Cập nhật tin tức thành công!") and redirects back to the list. Sub-flow 3: Delete an Existing Article 1. From the news list, the Moderator clicks the "Delete" button next to the desired article. 2. The system displays a confirmation pop-up (e.g., "Bạn có chắc muốn xóa tin tức này ?"). 3. The Moderator confirms the deletion. 4. The system permanently removes the article from the database. 5. The system displays a success message and refreshes the news list.
Alternative Sequences/Flows	● Invalid Data Entry: If the Moderator submits the create/update form with missing or invalid data (e.g., no title), the system displays an error message next to the corresponding field(s) and does not save the article. The user remains on the form to correct the errors. ● Cancel Action: The user can click a "Cancel" button on the create/edit form or the delete confirmation prompt. The system will discard any changes and return the user to the news management list. ● Unauthorized Access: If a user without Moderator or Admin privileges attempts to access the News Management page, the system will display a "403 Access Denied" error.

Exception	E1_Validation Error (Missing Required Title) ● Trigger: Occurs at Sub-flow 1 Step 4 (Create Article) or Sub-flow 2 Step 5 (Update Article) - Moderator clicks "Save changes". ● Sequence: 1. System validates form input data. 2. System detects that "Title" field is empty or contains only whitespace. 3. System prevents article creation/update. 4. System displays inline error message next to Title field: "Tiêu đề là bắt buộc." (MSG-VAL-TITLE-REQUIRED). 5. System highlights Title field in red. 6. Form remains open with all entered data preserved. Postcondition: Article is not saved to database; Moderator remains on form to correct error; No data changes made.
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2.14.2 View News

Primary Actors	Guest, Student	Secondary Actors	None
Description	Allows students and guests to view a list of published news articles and click on one to read its full content.		
Preconditions	● The user has navigated to SkillUp. ● The user navigates to the "Tin tức" section. ● There is at least one news article with the status "Published" in the system.		
Postconditions	● The student has successfully viewed the news content. No system data is changed.		
Normal Sequence/Flow	1. The Student logs into their account. 2. The Student navigates to the "News" or "Announcements" section, possibly from their dashboard or a main navigation menu. 3. The system retrieves and displays a list of all published news articles, typically showing the article's Image, Title, and a short summary or snippet. Articles are usually sorted by the most recent first. 4. The Student finds an article of interest and clicks on its title. 5. The system navigates to the News Detail page. 6. The system displays the full content of the selected article, including its title, feature image, publication date, and full body text.		

Alternative Sequences/Flows	● No News Articles: If there are no published news articles in the system, the news list page will display a message like, "No news to display at the moment." ● Invalid Article Link: If a student tries to access a news detail page with an invalid or non-existent article ID (e.g., through a bad link), the system will display a "404 Page Not Found" error.
Exception	E1_No News Articles Available (Empty State) ● Trigger: Occurs at Normal Flow Step 3 (System retrieves and displays list of news articles). ● Sequence: 1. User navigates to "Tin tức" (News) section. 2. System queries database for news articles with status = "Published". 3. System detects that no published news articles exist in database (count = 0). 4. System displays empty state message on news list page: "Hiện tại chưa có tin tức nào." or "No news to display at the moment." (MSG-EMPTY-NEWS). 5. System may show: a. Empty state illustration/icon b. Message: "Vui lòng quay lại sau để xem tin tức mới nhất." c. Button to return to homepage or refresh page Postcondition: News list page displays empty state; No articles shown; User can navigate to other sections; No system data changed.

2.15 Post Feature

2.15.1 View List Post

Primary Actors	Content Moderator	Secondary Actors	None
Description	Content Moderator is authenticated and inside "Quản lý bài đăng".		
Preconditions	● Forum posts are available in the system ● Content Moderator has a valid session.		
Postconditions	Content Moderator has viewed the list of posts; no data is modified.		

Normal Sequence/Flow	7. The Student logs into their account. 8. The Student navigates to the "News" or "Announcements" section, possibly from their dashboard or a main navigation menu. 9. The system retrieves and displays a list of all published news articles, typically showing the article's Image, Title, and a short summary or snippet. Articles are usually sorted by the most recent first. 10. The Student finds an article of interest and clicks on its title. 11. The system navigates to the News Detail page. 12. The system displays the full content of the selected article, including its title, feature image, publication date, and full body text.
Alternative Sequences/Flows	● No News Articles: If there are no published news articles in the system, the news list page will display a message like, "No news to display at the moment." ● Invalid Article Link: If a student tries to access a news detail page with an invalid or non-existent article ID (e.g., through a bad link), the system will display a "404 Page Not Found" error.
Exception	E1_No News Articles Available ● Trigger: Occurs at Normal Flow Step 3 (System retrieves and displays list of news articles). ● Sequence: 1. System queries database for published news articles. 2. System detects that no published news articles exist (count = 0). 3. System displays empty state message on news list page: "Hiện tại chưa có tin tức nào." or "No news to display at the moment." (MSG-EMPTY-NEWS). 4. System may show placeholder image or icon indicating empty state. 5. System provides option to refresh or navigate back to dashboard. Postcondition: News list page displays empty state; No articles shown; Student can navigate to other sections.

2.16 Voucher Feature

2.16.1 Manage voucher

Primary Actors	Lecturer	Secondary Actors	None
Description	The Lecturer is authenticated and is on the course update screen, at the "Phiếu giảm giá" step. The Lecturer can view the list of vouchers for the course, create new vouchers, edit existing ones, or delete them.		
Preconditions	• The Lecturer has successfully logged into the system. • The Lecturer has selected a specific course.		
Postconditions	All voucher changes are stored in the system		
Normal Sequence/Flow	Create voucher 1. The Lecturer logs into the system. 2. The Lecturer selects an existing course and click update. 3. The system displays the course configuration steps, including "Trang giới thiệu", "Khung chương trình", "Giá khóa học" and "Phiếu giảm giá". 4. The Lecturer selects the "Phiếu giảm giá" step. 5. The system displays the voucher list. 6. The Lecturer clicks the "Tạo phiếu giảm giá" button. 7. The system shows a voucher creation form, allowing the Lecturer to enter: • Voucher code. • Discount type • Validity period: start date/time and end date/time. 8. The Lecturer fills in all required fields with valid data and clicks the "Tạo phiếu giảm giá" button. 9. The system redisplays the voucher list		
Alternative Sequences/Flows	A1: Update voucher 1. At step 5, the Lecturer clicks the edit icon on a voucher. 2. The system opens the voucher form pre-filled with the current data. 3. The Lecturer updates one or more fields and clicks "Cập nhật". 4. The system updates the voucher and redisplays the list with the new information. A2: Delete voucher 1. At step 5, the Lecturer clicks the delete icon on a voucher. 2. The system shows a confirmation dialog such as "Bạn có chắc chắn muốn xóa voucher này?".		

	3. If the Lecturer click "xóa" button, the system deletes the voucher and refreshes the list. If the Lecturer click "hủy" button, no changes are made A3: Voucher expired 1. The system automatically calculates voucher status based on the current date and time. 2. If the voucher's end time has passed, the system marks it as Expired and does not allow students to apply for it at checkout, though it remains visible to the Lecturer for history or possible deletion.
Exception	E1: Authorization Error ● Trigger: In every flow, if the Lecturer does not own the course. ● Sequence: 1. The Lecturer is logged into the system. 2. The Lecturer tries to navigate to the course managing page by links. ● System: Displays a toast message “Bạn không có quyền chỉnh sửa khóa học này”. ● Result: The Lecturer is notified of the authorization error. E2: Validation Error ● Trigger: In every flow, if the Lecturer inputs invalid field information (exceeding character length, blank input, impossible dates, invalid type). ● Sequence: 1. The Lecturer is logged into the system. 2. The Lecturer navigates to the course update page. 3. The Lecturer navigates to the Voucher tab. 4. The Lecturer tries to input invalid fields information (exceeding character length, blank input, impossible dates, invalid type). ● System: Displays a toast message informing the Lecturer to check the inputs again. ● Result: The Voucher managing operations are blocked.

2.17 AI Feature

2.17.1 AI Lesson Q&A

Primary Actors	Student	Secondary Actors	AI(Gemini), Qdrant
Description	Student is logged in and viewing a lesson. Student uses the AI Q&A feature to ask questions about the lesson content and receives answers based on the video subtitles of that lesson.		
Preconditions	• Student has logged into the system • Student has enrolled in the course containing this lesson • Course has AI support enabled • Lesson exists and is active • AI(Gemini) and Qdrant Service are operational		
Postconditions	• Student receives an AI answer based on the lesson content		
Normal Sequence/Flow	AI Lesson Q&A 1. Student is viewing the lesson (Lesson detail page) 2. Student enters a question in the input field "Trợ lý AI cho bài học này" 3. Student clicks the "Gửi câu hỏi" button 4. System validates request: checks that the question is not empty 5. System checks authentication: verifies Student is logged in 6. System checks authorization: verifies user is a Student 7. System checks that lesson exists and is active 8. System checks that course has AI support enabled 9. System checks that Student has enrolled in this course 10. System sends question to AI (gemini) to create embedding vector 11. System searches in Qdrant Vector Database with lessonId filter and score threshold 12. System retrieves top-k results (default topK = 5) with score >= threshold 13. System sorts chunks by chunkIndex to maintain order 14. System combines context from chunks into a text string 15. System sends question and context to AI (gemini) to generate answer 16. System receives answer from AI (gemini) 17. System formats response including answer 18. System displays answer to Student		
Alternative Sequences/Flows	Question empty 1. System detects request. Question is null or empty 2. System returns error 400: "Please enter a question." 3. System displays error message to Student 4. Student must re-enter the question Course AI Support Not Enabled		

	1. System detects course.IsAiSupport != true 2. System returns error 400: "Course has not enabled AI support." 3. System displays error message to Student 4. Student cannot use this feature for this course
Exception	E1_AI Service Error • Trigger: Normal Flow Step 10 (Create embedding vector) or Step 15 (Generate answer) • Sequence: 1. System sends request to AI Service (Gemini) for embedding or chat completion 2. System encounters an exception: a. Network error (connection timeout, no response) b. API error (invalid API key, quota exceeded, rate limit) c. Service unavailable (Gemini API down) d. Timeout (request exceeds timeout threshold) 3. System catches the exception and logs error details (error message, timestamp, lessonId, userId) 4. System returns error 400: "An error occurred: [error message]" 5. System displays error message to Student: "Sorry, the AI service is temporarily unavailable. Please try again later." 6. System does not create embedding or generate answer Postcondition: Student remains on the lesson detail page, question is not processed, no answer is displayed, student can retry with the same or a different question after the service recovers

3. Functional Requirements

3.1 Common

3.1.1 User Login

SkillUp

Đăng nhập

Email

Mật khẩu

☐ Ghi nhớ đăng nhập [Quên mật khẩu?](#)

hoặc tiếp tục với

[Chưa có tài khoản? Đăng ký](#)

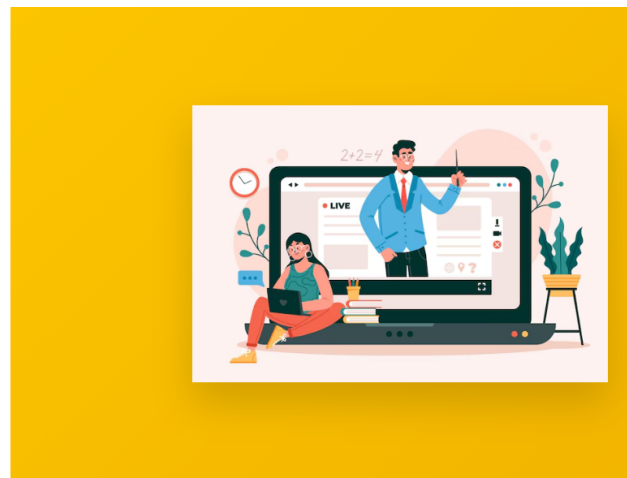


Figure 3.1.1 - login screen interface

Field Description

Field Name	Description
Email	Data type: email format, required, placeholder “Nhập địa chỉ email”.
Mật khẩu	Data type: password string, required (≥ 6 chars), includes show/hide icon, placeholder “Nhập mật khẩu”.
Ghi nhớ đăng nhập	Checkbox: save login session.
Quên mật khẩu?	Red hyperlink to the password recovery page
Đăng nhập	Primary yellow button; disabled while form is invalid or submitting.
Tiếp tục sử dụng dịch vụ bằng Google	Google button triggers OAuth, receives Google token, and signs the user in.
Đăng ký	linking to the register page .

3.1.2 User Register

SkillUp

Đăng ký tài khoản

Học viên

Giảng viên

Email

Nhập địa chỉ email

Mật khẩu

Nhập mật khẩu

Xác nhận mật khẩu

Nhập lại mật khẩu

Họ và tên

Nhập họ và tên

Đăng ký ngay

Đã có tài khoản? [Đăng nhập](#)

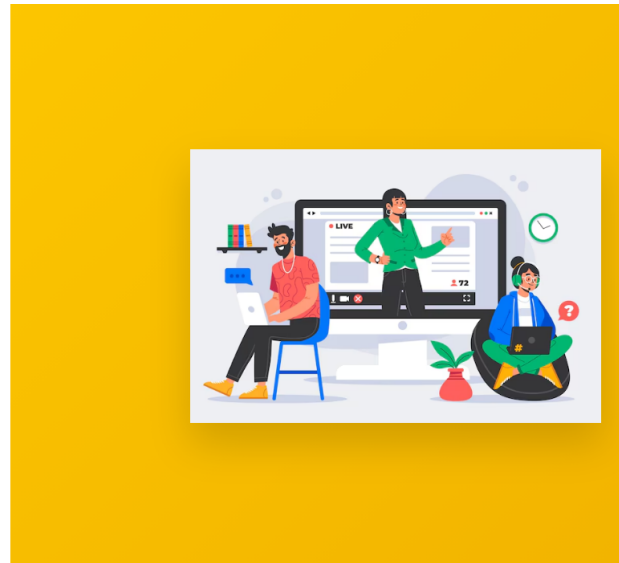


Figure 3.1.2 - register screen interface

Field Name	Description
Toggle	default Student. Determines roleId (5 for “Học viên”, 4 for “Giảng viên”).
Email	Data type: email string, required, placeholder “Nhập địa chỉ email”.
Mật khẩu	Data type:Password string, required, min 6 characters, placeholder “Nhập mật khẩu”.
Xác nhận mật khẩu	Data type:Password string, required, must match Password, placeholder “Nhập lại mật khẩu”.
Họ và tên	Data type: Text string, required, placeholder “Nhập họ và tên”.
Đăng ký ngay	Primary yellow button labeled “Đăng ký ngay” disabled during submission or when validations fail
Đăng nhập	Link pointing to `/login`

3.1.3 Forgot password

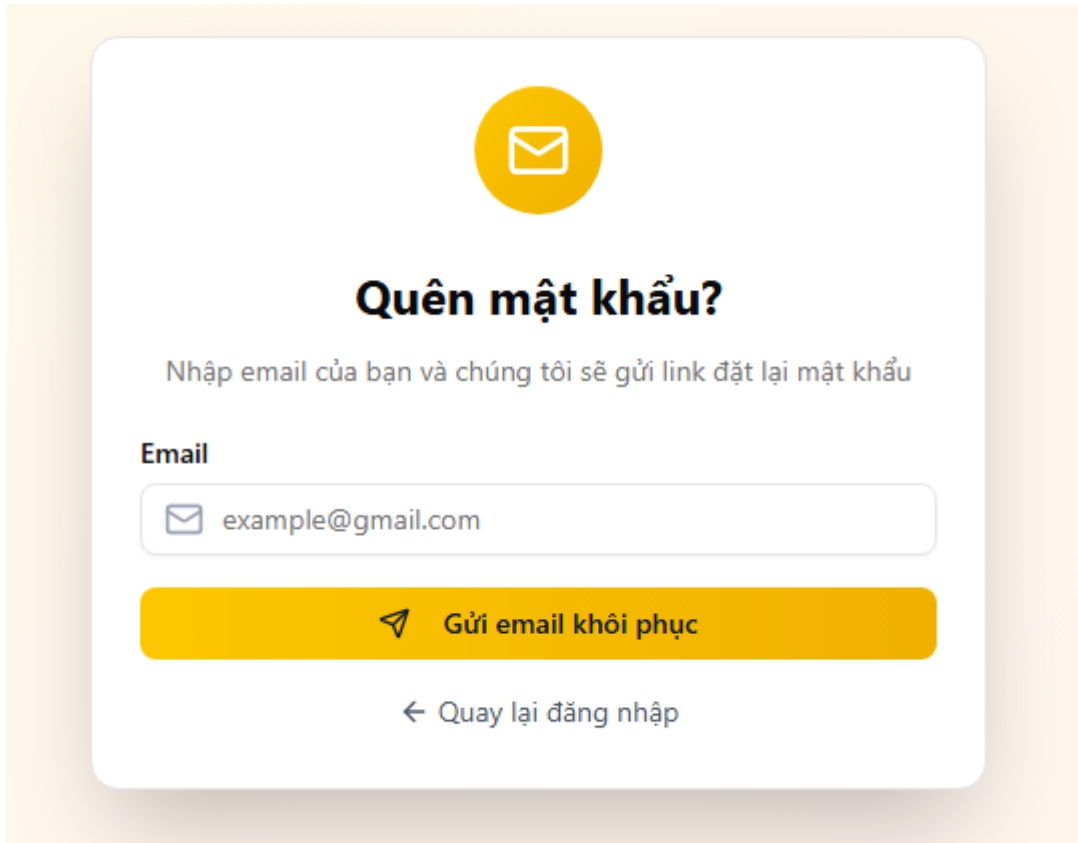
The image shows a 'Forgot password' screen. At the top, there is a yellow circular icon with a white envelope symbol. Below this, the title 'Quên mật khẩu?' is displayed in bold black text. Underneath the title, a subtitle reads 'Nhập email của bạn và chúng tôi sẽ gửi link đặt lại mật khẩu'. There is a text input field labeled 'Email' containing the text 'example@gmail.com'. Below the input field is a large yellow button with a white paper plane icon and the text 'Gửi email khôi phục'. At the bottom, there is a link that says '← Quay lại đăng nhập'.

Figure 3.1.3.1 - forgot password screen interface

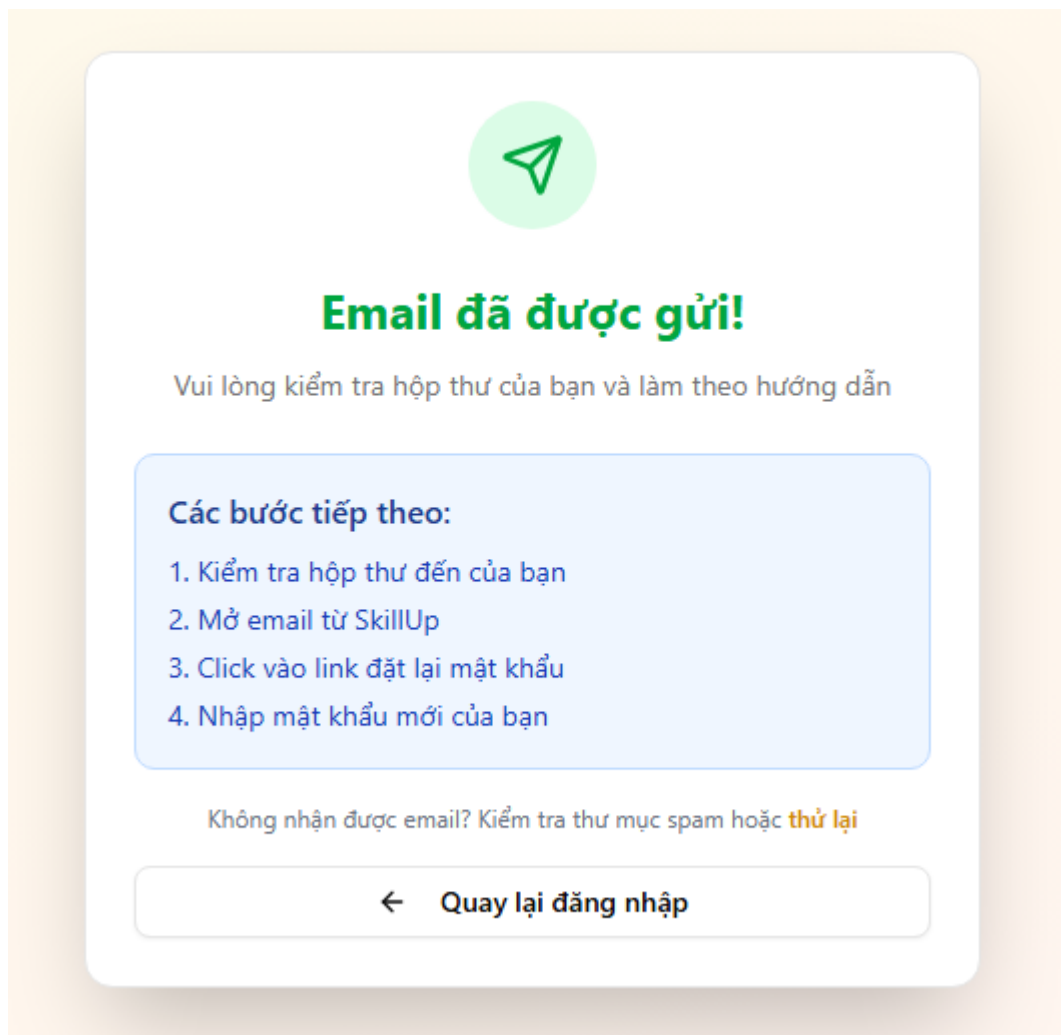


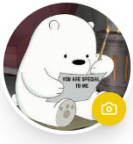
Figure 3.1.3.2 - forgot password screen interface

Field Description

Field Name	Description
Email	Data type: text field for user email; required, must pass email-format validation (email regex).
Gửi email khôi phục	Data type: action button; submits the recovery request, disabled while sending; no user-entered data stored.
Quay lại đăng nhập	Data type: navigation link; routes the user back to the login screen, no data entry.
thử lại	Data type: text hyperlink; “thử lại” triggers resend action but collects no user data.

3.1.4 Profile Manage

3.1.4.1 View and Update Profile Details



quatbt
hocfptkk@gmail.com

"top 1 account"

Ngày sinh
19/11/2025

Ngày tham gia
02/11/2025

Cài đặt hồ sơ

 Thông tin liên hệ

 Email

 Số điện thoại

 Họ và tên *

 Giới tính

Chọn giới tính

 Ngày sinh

 Tiểu sử

Lưu thay đổi

Figure 3.1.4.1.1 - profile screen interface

Đổi mật khẩu

 Mật khẩu cũ

 Mật khẩu mới

 Xác nhận mật khẩu mới

Cập nhật mật khẩu

Figure 3.1.4.2.2 - change password screen interface

This screen allows the User to:

- View and Update Profile: View their current account information and update it as needed.
- Change password: Input the old password and new password to change it.

Field Name	Description
Số điện thoại	Data type: string (of number only), max length of 10 characters.
Họ và tên	Data type: string, max length of 30 characters.
Giới tính	Initial values: Male, Female, Other.
Ngày sinh	Data type: date(string).
Tiểu sử	Data type: string, max length of 255 characters.
Lưu thay đổi	Type: Button, save the changes to the database.
Camera icon	Type: Button, change the profile picture with an uploaded image.
Mật khẩu cũ	Data type: string, max length of 255 characters.
Mật khẩu mới	Data type: string, max length of 255 characters.
Xác nhận mật khẩu mới	Data type: string, max length of 255 characters.
Cập nhật mật khẩu	Type: Button, save the password change to the database.

3.2 Forum

3.2.1 Post feature

3.2.1.1 View Post List

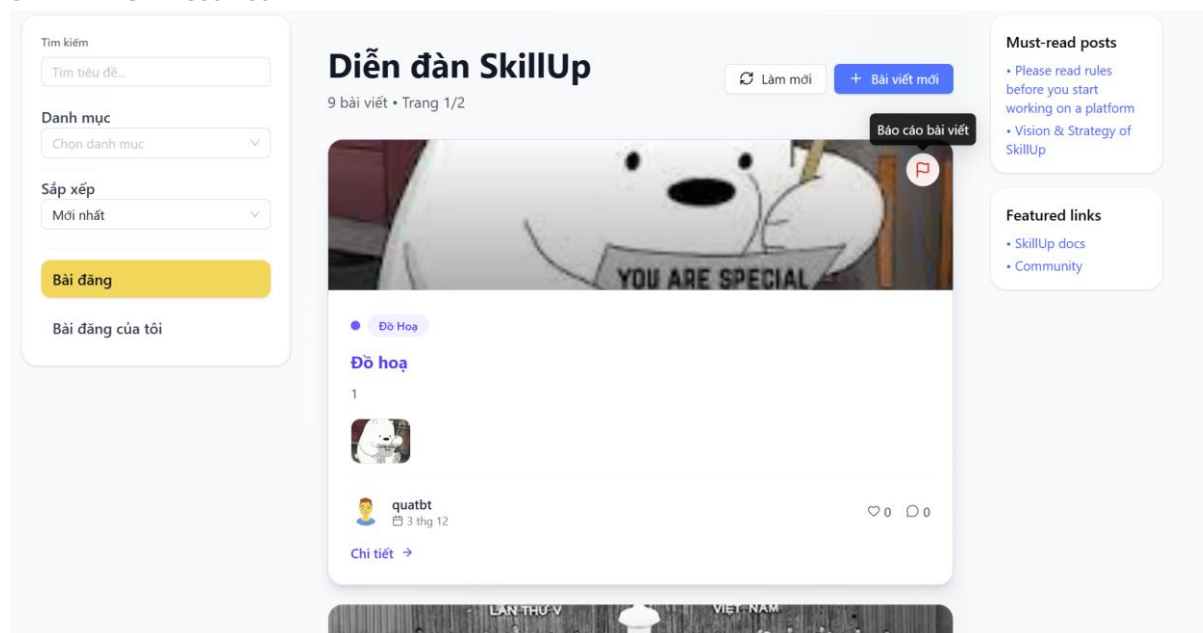


Figure 3.2.1.1 - post list screen interface

This screen allows the Student to:

- View Post List: view list of all active Posts or the User posts.
- Sort by category: select a category to display relevant posts.
- Search and filter content: search for relevant posts and filter by Newest or Oldest.

Field Description

Field Name	Description
Tìm kiếm	Data type: String, max length of 30 characters.
Danh Mục	Values: all active post categories.
Sắp xếp	Values: Newest and Oldest
Bài đăng	Type: Button, display all active posts.
Bài đăng của tôi	Type: Button, display all current student's active posts.
Làm mới	Type: Button, refresh the post list.
Bài viết mới	Type: Button, navigate to the post creating page.
Báo cáo bài viết	Type: Button, open the report post popup window.
Chi tiết	Type: Button, navigate to the post details page.

3.2.1.2 View Post Details

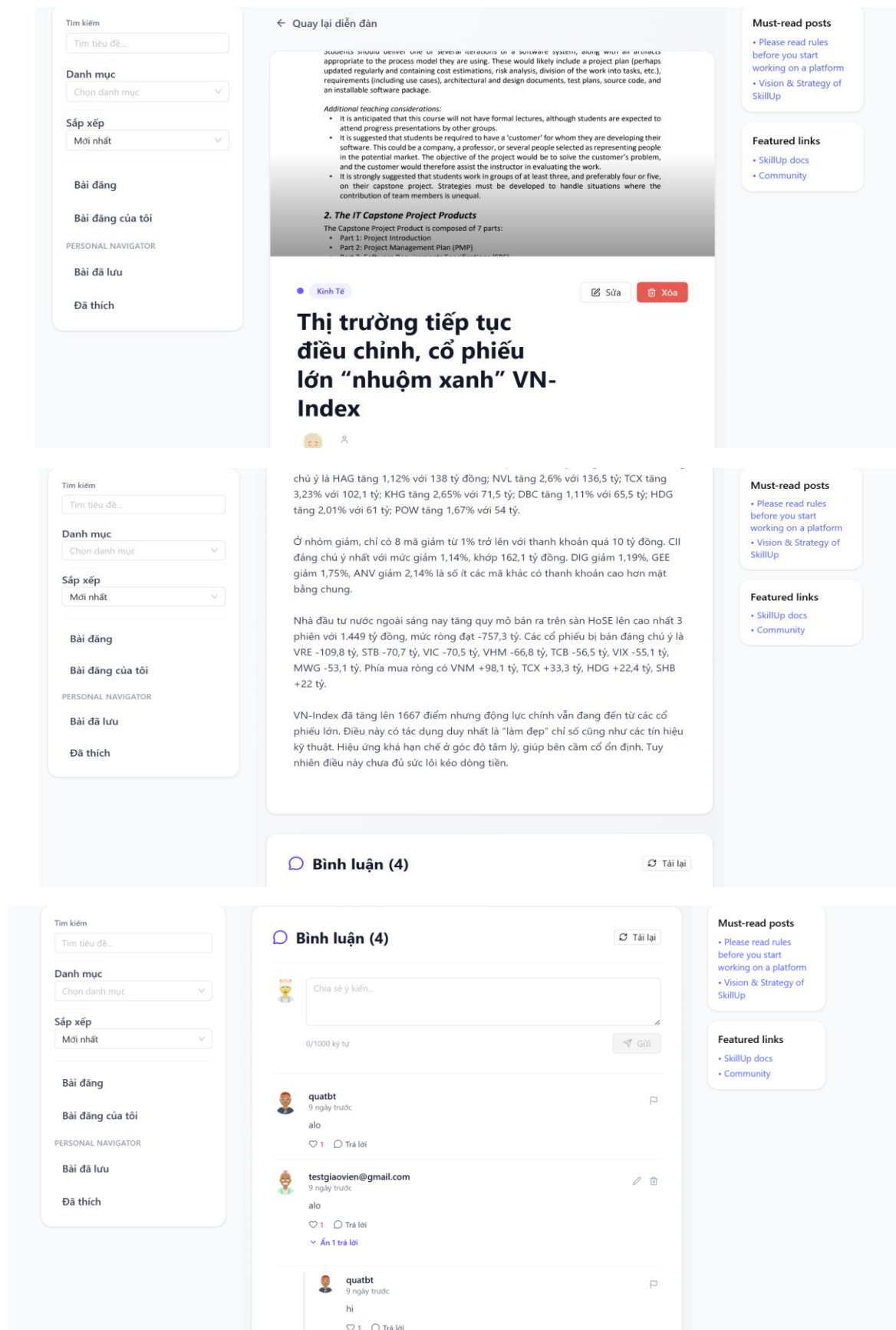


Figure 3.2.1.2 - post details screen interface

This screen allows the Student to:

- View Post Details: view details of the post and the number of reactions.
- Edit or Delete the post: edit the post details or delete it.
- View Author Details: view details of the post's author.
- View Post Replies: view the replies of other students who have commented on the post.
- React and Reply: give likes and replies on the post or other comments.

Field Name	Description
Sửa	Type: Button, navigate to the post update page.
Xoá	Type: Button, open the confirmation popup window, delete the post on user confirm.
Tải lại	Type: Button, reload the comment section.
Chia sẻ ý kiến	Data type: string, max 1000 characters.
Gửi	Type: Button, send the inputted comment.
Flag icon	Type: Button, open the report comment popup window.
Pencil icon	Type: Button, edit the user comment.
Bin icon	Type: Button, open the confirmation popup window, delete the post comment on user confirm.
Heart icon	Type: Button, increase the like number by 1 (decrease by 1 if already liked)
Trả lời	Type: Button, open the reply comment input box.

3.2.1.3 Update Post Details

Cập nhật bài viết

Chỉnh sửa nội dung bài viết của bạn

Danh mục *

Kinh Tế

Tiêu đề *

Thị trường tiếp tục điều chỉnh, cổ phiếu lớn "nhuộm xanh" VN-Index

66/200 ký tự

Nội dung *

Trạng thái "xanh vỏ đỏ lòng" diễn ra gần như trọn phiên giao dịch sáng nay. VN-Index được VIC và VHM kéo lên mạnh mẽ trong khi độ rộng lại thể hiện sắc đỏ áp đảo. Thanh khoản khớp lệnh sàn HoSE tiếp tục rơi xuống mức thấp kỷ lục mới như một tín hiệu "phản ứng ngược" của nhà đầu tư.

3458 ký tự

Hình ảnh (Tối đa 8 ảnh, mỗi ảnh < 10MB)

Thêm ảnh

1/8

Hủy

Cập nhật bài viết

Figure 3.2.1.3 - update post details screen interface

This screen allows the Student to:

- Update Post Details: update the title, content and images of the post.
- Navigate Back: navigate back to the post list.

Field Description

Field Name	Description
------------	-------------

Tiêu đề	Data type: String, max length of 200 characters.
Nội dung	Data type: String, max length of 5000 characters.
Hình ảnh	Data type: Image, max number of 8 images, with 10MB or less each.
Hủy	Type: Button, cancel all modifications and return to the post.
Cập nhật bài viết	Type: Button, save the updated post to the database.

3.2.1.4 Delete Post

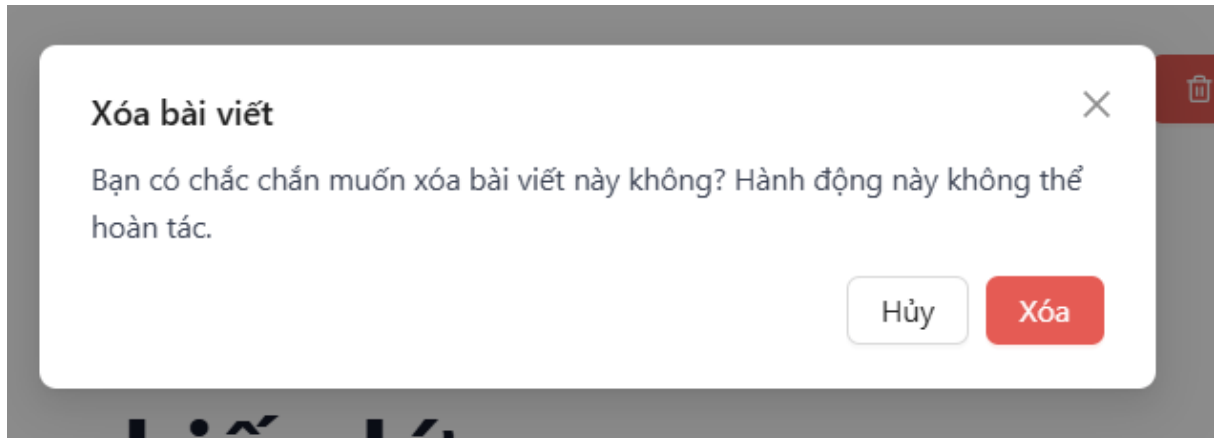


Figure 3.2.1.4 - delete post screen interface

This screen allows the Student to:

- Delete Post: delete the post from the post list.
- Navigate Back: navigate back to the post list.

Field Description

Field Name	Description
Xoá	Type: Button, delete the post from the post list.
Hủy	Type: Button, cancel post deletion and close the popup window.

3.2.1.5 Create Post

Tạo bài viết mới

Chia sẻ điều gì đó có giá trị với cộng đồng

Danh mục *

Chọn danh mục cho bài viết của bạn

Tiêu đề *

Nhập tiêu đề hấp dẫn cho bài viết...

0/200 ký tự

Nội dung *

Viết nội dung chi tiết của bài viết... Bạn có thể sử dụng Markdown để định dạng.

0 ký tự

Hình ảnh (Tối đa 8 ảnh, mỗi ảnh < 10MB)

Thêm ảnh

0/8

Hủy

Đăng bài viết

Figure 3.2.1.5 - create post screen interface

This screen allows the Student to:

- Create Post: create a new post in the database and add it to the post list.
- Navigate Back: navigate back to the post list.

Field Description

Field Name	Description
Danh mục	Values: the post category list.
Tiêu đề	Data type: string, max length of 200 characters
Nội dung	Data type: string, max length of 5000 characters
Hình ảnh	Data type: image, max number of 8 images, with each being 10MB or lower.
Hủy	Type: Button, cancel post creation and return to the post list.
Đăng bài viết	Type: Button, save post data to the database and add it to the post list.

3.2.2 Forum Report Feature

3.2.2.1 Report Post

Figure 3.2.2.1 - report post screen interface

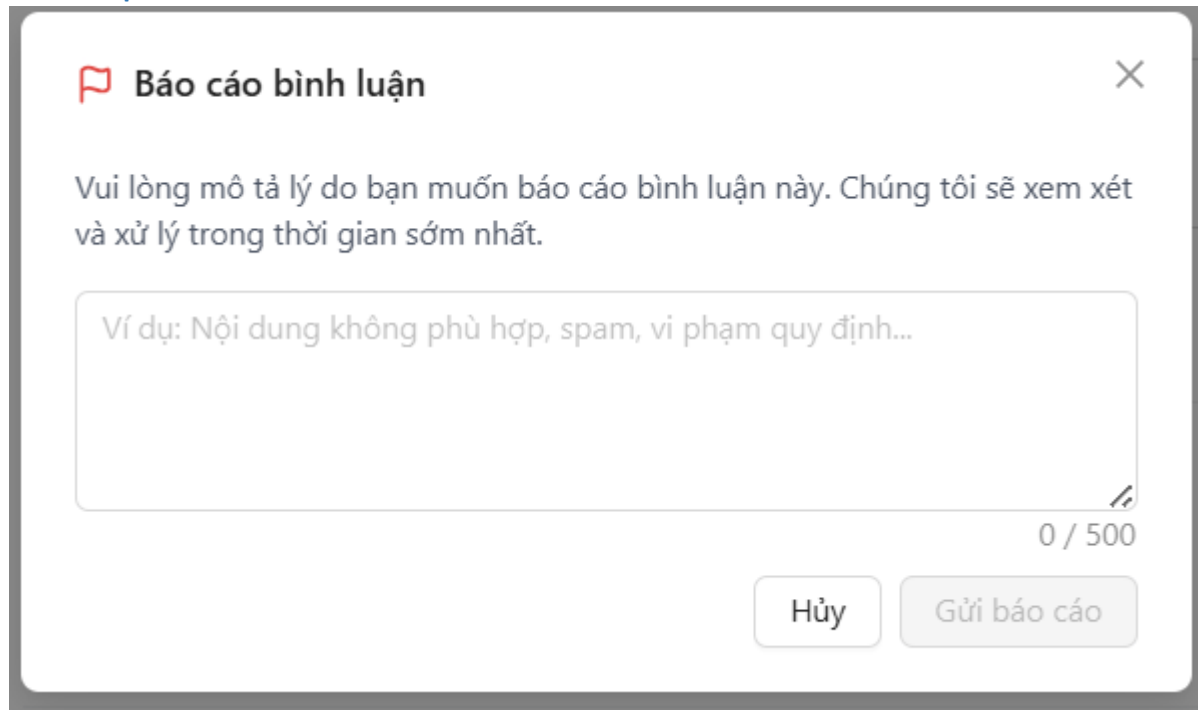
This screen allows the Student to:

- Report the Post: create a post report and send it for the Moderator to review.
- Navigate Back: navigate back to the post list.

Field Description

Field Name	Description
Lý do	Data type: string, max length of 500 characters
Hủy	Type: Button, cancel post report and close the popup window.
Gửi báo cáo	Type: Button, send post report to the Moderator for review.

3.2.2.2 Report Post Comment



Báo cáo bình luận

Vui lòng mô tả lý do bạn muốn báo cáo bình luận này. Chúng tôi sẽ xem xét và xử lý trong thời gian sớm nhất.

Ví dụ: Nội dung không phù hợp, spam, vi phạm quy định...

0 / 500

Hủy Gửi báo cáo

Figure 3.2.2.2 - report post comment screen interface

This screen allows the Student to:

- Report the Post Comment: create a post comment report and send it for the Moderator to review.
- Navigate Back: navigate back to the post.

Field Description

Field Name	Description
Lý do	Data type: string, max length of 500 characters
Hủy	Type: Button, cancel post comment report and close the popup window.
Gửi báo cáo	Type: Button, send post comment report to the Moderator for review.

3.2.2.3 View Post Comment Report List

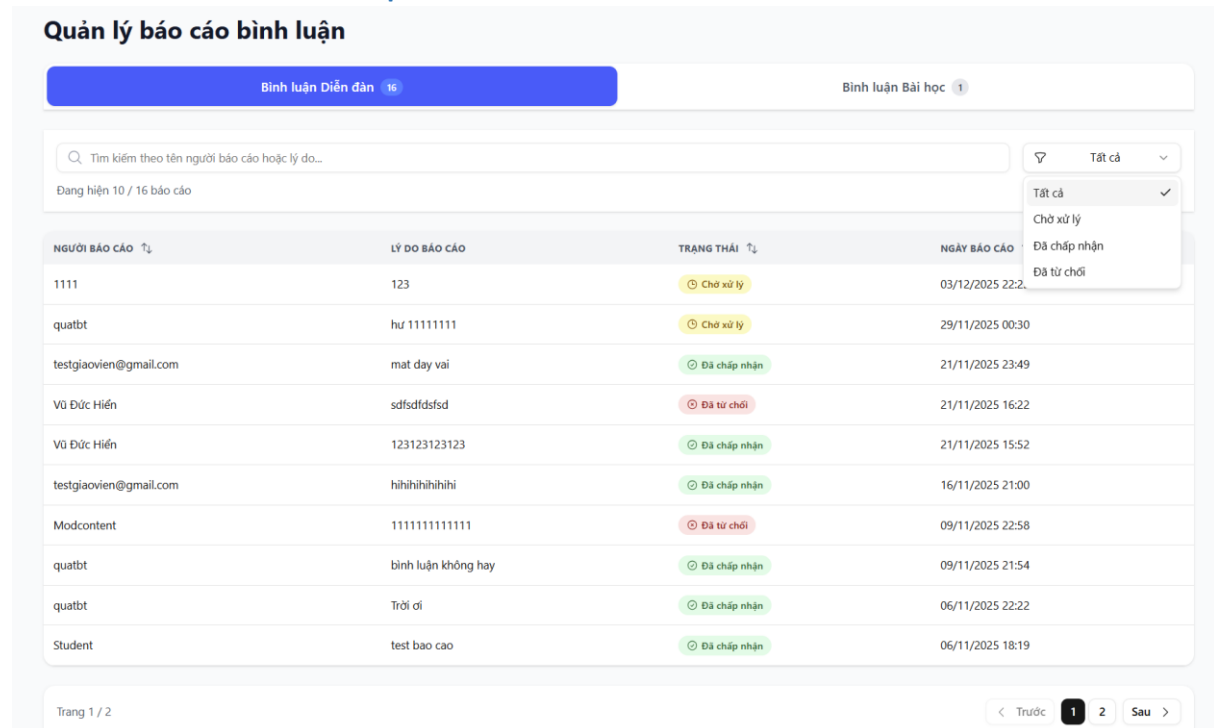


Figure 3.2.2.3 - post comment report list screen interface

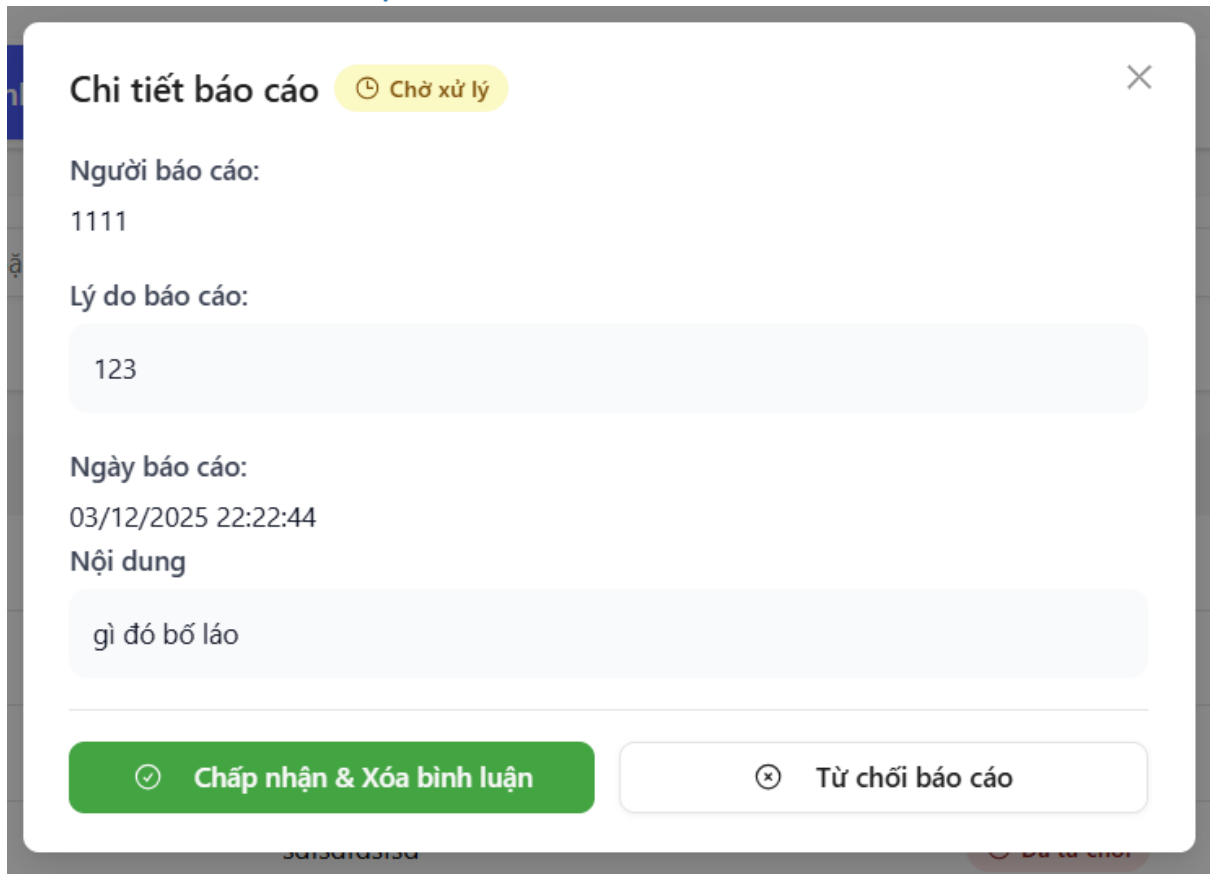
This screen allows the Content Moderator to:

- View all post comment report: View all comment reports made in the forum, submitted by the students.
- Filter and Search: Filter reports by its status and search for matching report submitter or reason.

Field Description

Field Name	Description
Bình luận diễn đàn	Type: Button, display all of the comment reports in the Forum.
Bình luận bài học	Type: Button, display all of the comment reports in the Course's Lessons.
Tìm kiếm	Data type: string, max length of 20 characters.
Filter Select Box	Initial values: Pending, Accepted, Rejected.

3.2.2.4 View Post Comment Report Details



Chi tiết báo cáo ⌚ Chờ xử lý ✕

Người báo cáo:
1111

Lý do báo cáo:
123

Ngày báo cáo:
03/12/2025 22:22:44

Nội dung
gì đó bố lão

⏹ Chấp nhận & Xóa bình luận ⌛ Từ chối báo cáo

Figure 3.2.2.4 - post comment report details screen interface

This screen allows the Content Moderator to:

- View the report details: View the report details and choose to accept and delete the comment or delete the report.

Field Description

Field Name	Description
Chấp nhận & Xóa bình luận	Type: Button, mark the report as Accepted and delete the violating comment.
Từ chối báo cáo	Type: Button, mark the report as Rejected.

3.3 Course Feature

3.3.1 View Detail Course

3.3.1.1 Screen: Course Detail Page

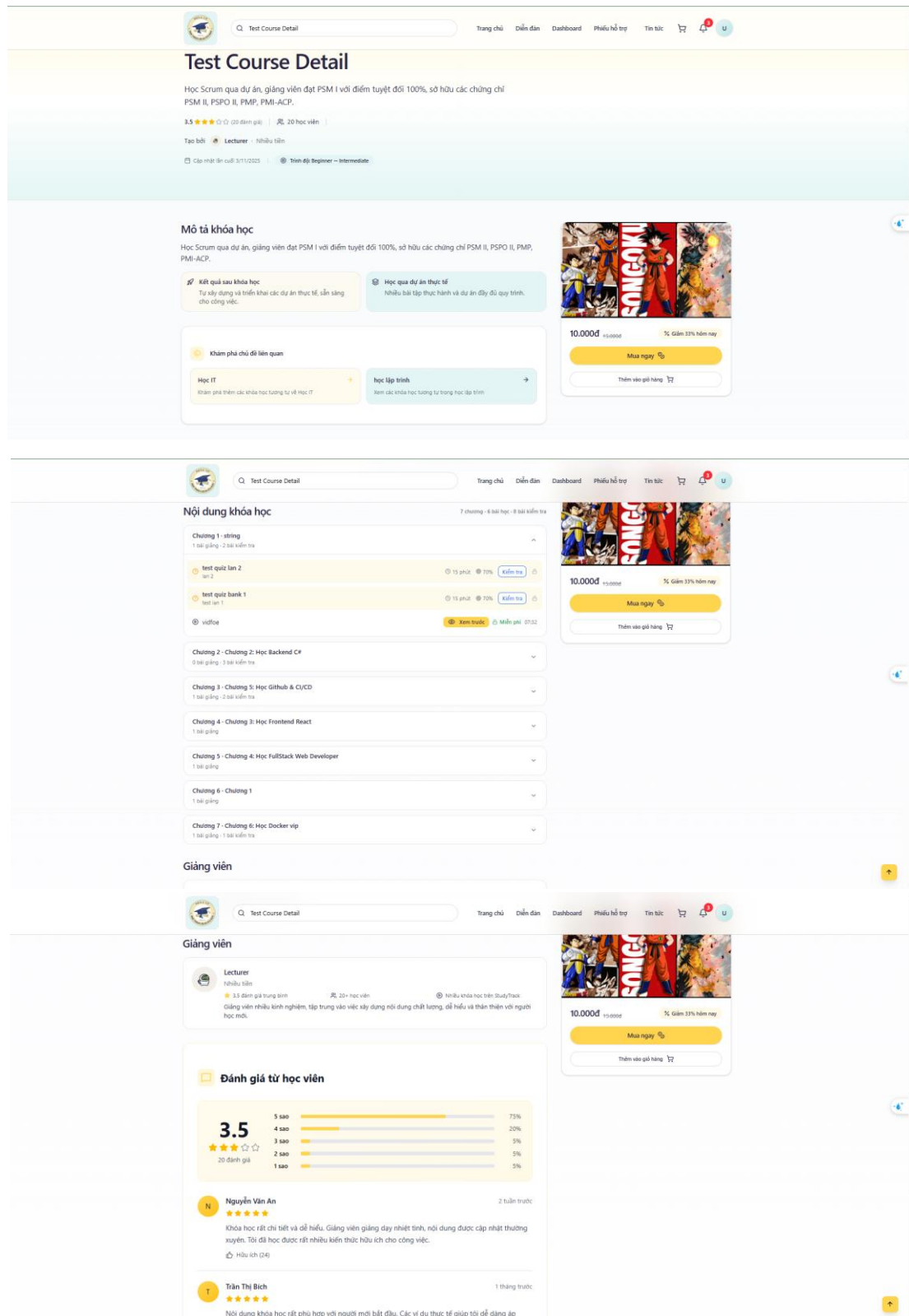


Figure 3.3.1.1 - course details page screen interface

This screen allows the Student to:

- **View Course Overview:** view comprehensive details of the course including title, description, learning outcomes, and average ratings.
- **Explore Syllabus:** browse the curriculum structure (chapters and lessons), check durations, and watch preview videos (if available).
- **View Instructor Details:** view the instructor's profile information, bio, number of students, and their reputation.
- **View Reviews:** view ratings and feedback from other students who have taken the course.
- **Enroll or Purchase:** add the course to the shopping cart or purchase it immediately to gain access to the full content.

Mapped Use Cases: UC-ViewCourseDetail, UC-AddToCart, UC-EnrollCourse (Buy Now), UC-ViewReviews.

Field Name	Description
"Created By" Link	Link. Action: Clicking the instructor's name redirects the user to the Instructor Profile page.
"Mua ngay" Button	Button. Action: Adds the course to the cart and immediately redirects the user to the Checkout/Payment page.
"Thêm vào giỏ hàng" Button	Button. Action: Adds the course to the cart, updates the cart badge, and displays a success toast notification. User stays on the current page.
Related Tags	Link. Action: Clicking a tag redirects the user to a search result page filtered by that specific keyword/topic.
Chapter Accordion	Button/Control. Action: Toggles the chapter section to expand or collapse the list of lessons inside.
"Xem trước" Button	Button. Action: Opens a video player popup to watch the trial lesson content (only visible for "Trial" lessons).
"Hữu ích" Button	Button. Action: Allows the user to "Like" or mark a review as helpful.

3.3.2 View Course List

3.3.2.1 Screen: Course Catalog Page

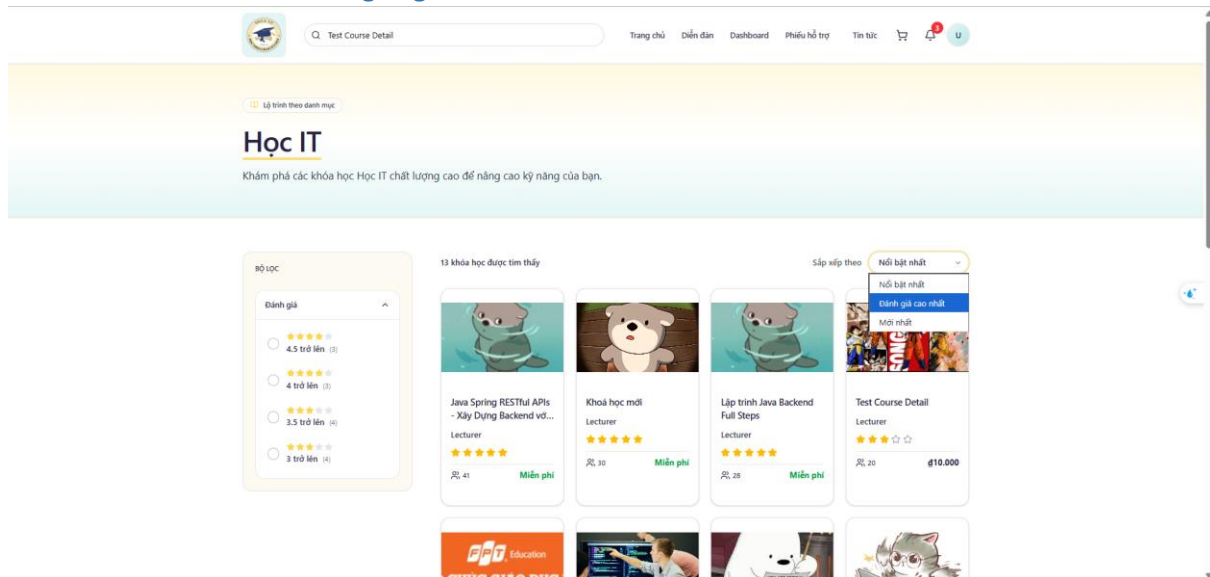


Figure 3.3.2.1 - course catalog page screen interface

This screen allows the Student to:

- **View Category Information:** see the title and description of the selected category (e.g., IT, Business) to understand the learning path.
- **Filter Courses:** narrow down the course list using sidebar filters (e.g., by Rating) to find relevant content.
- **Sort Courses:** organize the displayed courses by criteria such as "Most Featured", "Highest Rated", or "Newest".
- **Browse Course Cards:** view a grid of course summaries including thumbnail, title, instructor, rating, student count, and price.
- **Navigate to Detail:** click on a specific course card to access the Course Detail Page.

Field Name	Description
Sidebar Filter	Left panel for filtering results
Tiêu đề Bộ lọc	Label. Section header "FILTER" (BỘ LỌC).
Bộ lọc Xếp hạng	Radio Button Group. Allows filtering by star rating (e.g., "4.5 stars & up"). Displays the count of courses next to each option.
Phần Thanh công cụ	Tools above the grid
Số lượng Kết quả	Label. Shows total number of courses found (e.g., "13 courses found").
Menu thả xuống Sắp xếp	Dropdown List. Options include: 1. Featured (Nổi bật nhất) 2. Highest Rated (Đánh giá cao nhất) 3. Newest (Mới nhất).
Lưới Khóa học	List of course items
Thẻ Khóa học	Container. Clickable area representing a single course.
Hình thu nhỏ	Image. Cover image of the course.
Tiêu đề Khóa học	Link. Title of the course. Max 2 lines.
Tên Giảng viên	Label. Name of the lecturer.
Số sao xếp hạng	Icon. Visual representation of the average rating (Yellow stars).
Số lượng Học viên	Icon & Label. Icon representing users and the number of enrolled students (e.g., 41).

3.3.3 View My Course

3.3.3.1 Screen: My Learning Dashboard

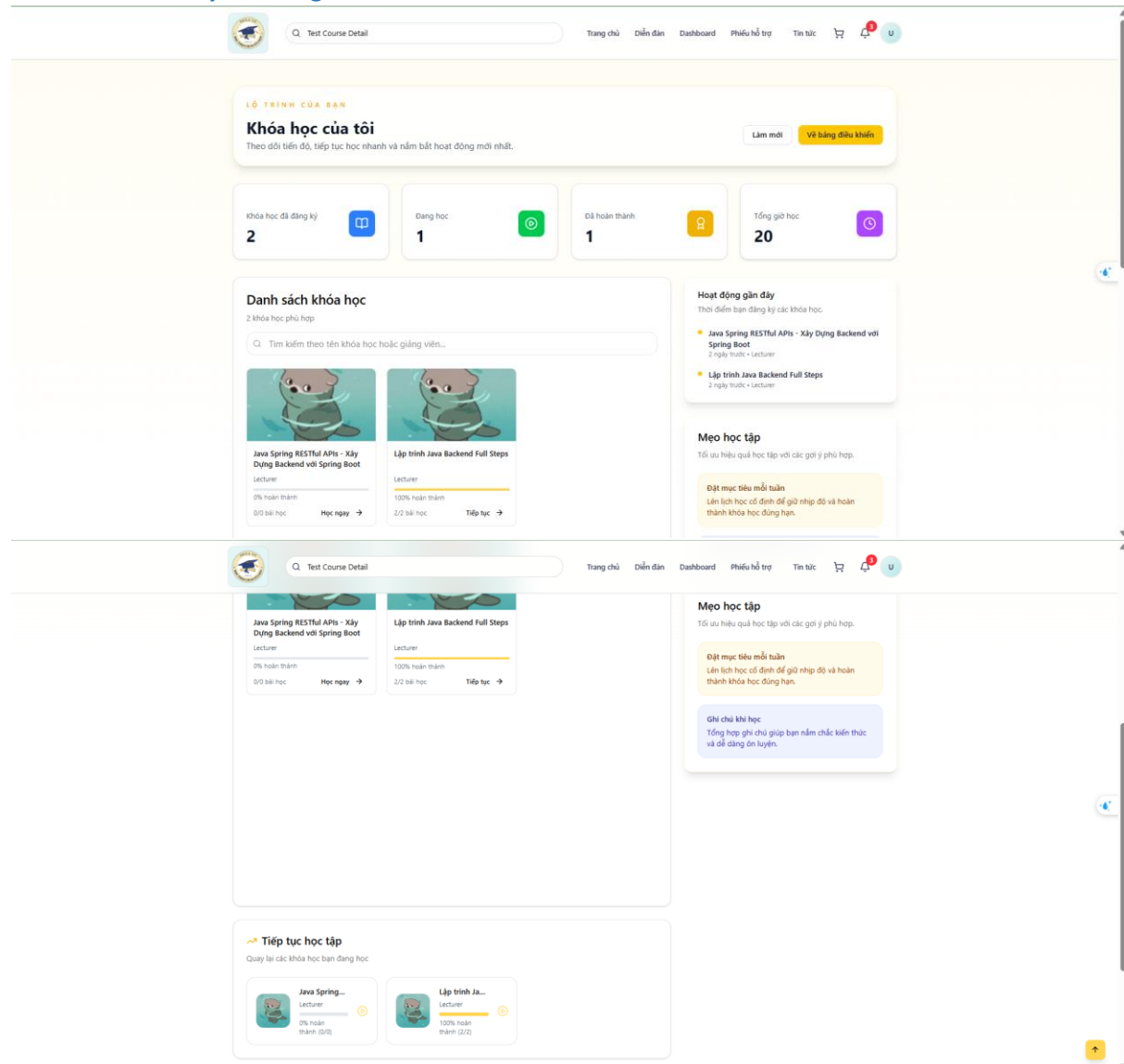


Figure 3.3.3.1 - my course dashboard screen interface

This screen allows the Student to:

- **Track Learning Overview:** view statistics on enrolled courses, courses in progress, completed courses, and total learning hours.
- **Search Enrolled Courses:** find specific courses within their personal library using keywords (course name or instructor).
- **Resume Learning:** view the progress of current courses and click "Continue" or "Start Now" to enter the learning player immediately.
- **View Recent Activities:** monitor recent interactions or updates related to their courses.
- **Get Study Tips:** view helpful advice (e.g., setting goals, taking notes) displayed on the sidebar to improve learning efficiency.

Field Name	Description
"Làm mới" Button	Button. Action: Reloads the latest learning progress and data from the server.
"Về bảng điều khiển" Button	Button. Action: Navigates back to the main user dashboard.
Ô tìm kiếm	Textbox. Action: Allows the user to type keywords to filter the course list by name or instructor.
Tiêu đề/Hình thu nhỏ khóa học	Link/Image. Action: Clicking either the title or the image redirects the user to the Course Player to view details or study.
Học ngay	Button. Label: "Start Now" or "Continue". Action: Immediately launches the Lesson Player. - "Start Now": For courses with 0% progress.- "Continue": For courses with >0% progress.
Trình chiếu khóa học	List/Swipe. Action: Allows horizontal scrolling/swiping to view more courses in the "Continue Learning" section.

3.4 Section Feature

3.4.1 View Section List

3.4.1.1 Screen: Course Section Component

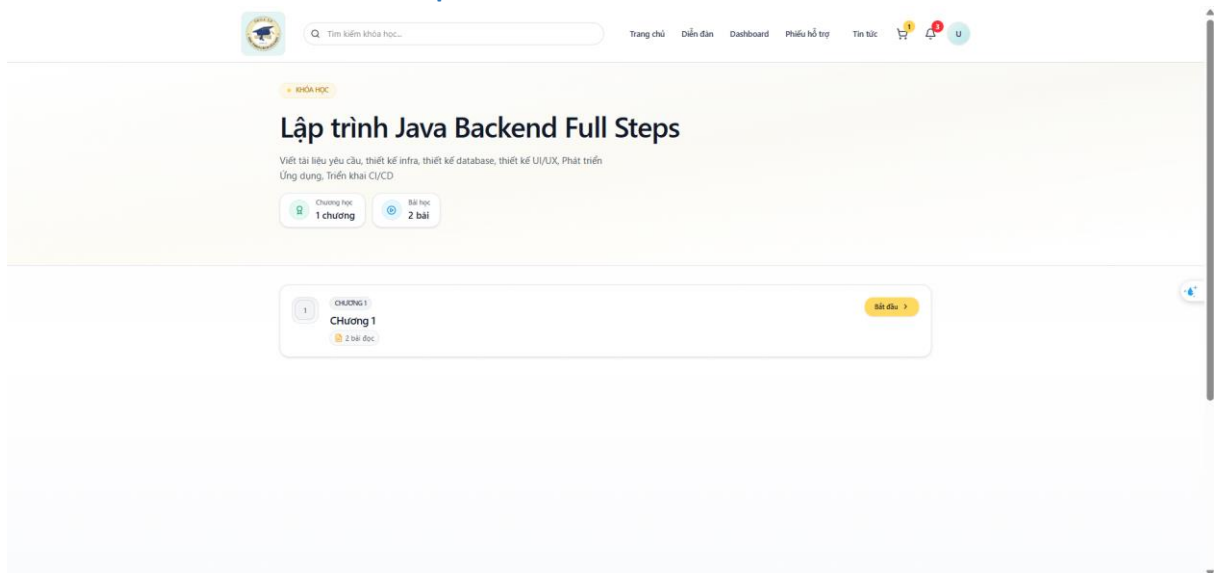


Figure 3.4.1.1 - section list screen interface

This function allows the Student (or Guest) to:

- View Course Structure: browse the organized list of sections (chapters) to understand the course outline.
- Expand/Collapse Details: click on section headers to toggle the visibility of detailed lesson items inside (Accordion interaction).
- Check Content Duration: view the total estimated time for each section and individual lessons.
- Identify Lesson Types: distinguish between video lectures (Play icon) and quizzes (Question icon).
- Access Preview: identify and access "Trial/Preview" lessons if available before purchasing.

Field Name	Description
1. Section Header	Interactive bar representing a chapter
Section Title	Label. Name of the section (e.g., "Chương 1 - String").
Content Summary	Label. Shows quantity (e.g., "1 lecture • 2 quizzes").
Toggle Icon	Icon. Chevron Up/Down. Indicates expanded/collapsed state.
Section Duration	Label. Total duration of content (e.g., "45 mins").
2. Lesson List Item	Items visible when section is expanded
Lesson Name	Label. Title of the lesson or quiz.
Type Icon	Icon. Play icon for videos, Question mark/Page icon for quizzes.
Lesson Duration	Label. Specific duration (e.g., "15:00") or status.
"Preview" Button	Button/Link. Visible on trial lessons. Opens preview popup.
Lock Status	Icon. Padlock icon. Shown if the user hasn't enrolled yet.

3.5 Lesson Feature

3.5.1 View Lesson Content

3.5.1.1 Screen: Lesson Player Interface

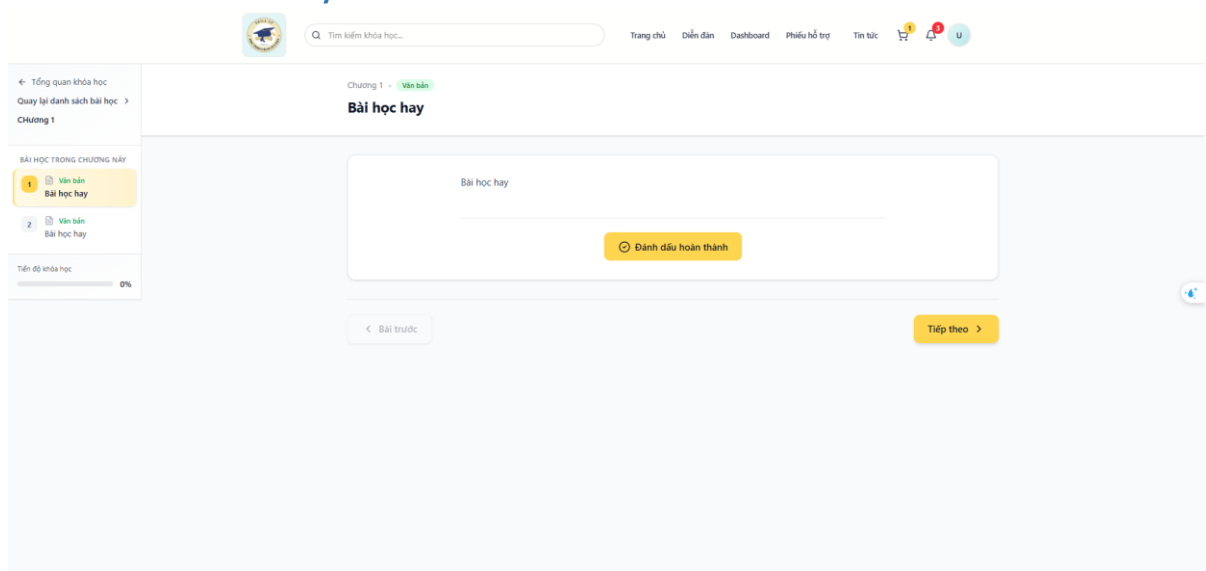


Figure 3.5.1.1 - lesson screen interface

This screen is the main learning interface. It allows the Student to:

- **Consume Content:** read text materials, watch videos, or view documents associated with the current lesson.
- **Navigate Syllabus:** use the sidebar to switch between different lessons within the current chapter or course.
- **Track Progress:** manually mark the lesson as completed to update their learning progress.
- **Navigate Linearly:** use "Previous" and "Next" buttons to move sequentially through the course.
- **Return to Overview:** easily navigate back to the course overview or home page.

Field Name	Description
1. Sidebar Navigation (Left)	Panel for navigating through course content
"Back to Overview" Link	Link. Action: Returns to the Course Detail or Dashboard.
Chapter Title	Label. Displays the current chapter name (e.g., "Chương 1").
Lesson List	List. Shows all lessons in the current chapter.
Active Lesson Highlight	Visual State. Highlights the lesson currently being viewed (Yellow background).
Lesson Icon	Icon. Indicates content type (Document icon for text, Play icon for video).
Course Progress Bar	Visual Bar. Located at the bottom left. Shows overall course completion percentage (e.g., 0%).
2. Main Content Area	Central area displaying the lesson material
Lesson Breadcrumb	Label. Shows context: "Chapter 1 • Text".
Lesson Title	Label. The main heading of the lesson (e.g., "Bài học hay").
Content Body	Container/Rich Text. Displays the actual content (Text, Images, Code blocks, or Video Player).
"Mark as Completed" Button	Button. Action: Updates status to "Completed". Visual change: Button turns green or label changes.
3. Footer Navigation	Bottom controls for sequential navigation
"Previous" Button	Button. Action: Navigates to the previous lesson. Disabled if this is the first lesson.
"Next" Button	Button. Action: Navigates to the next lesson.

3.5.2 View Lesson Comment Report List

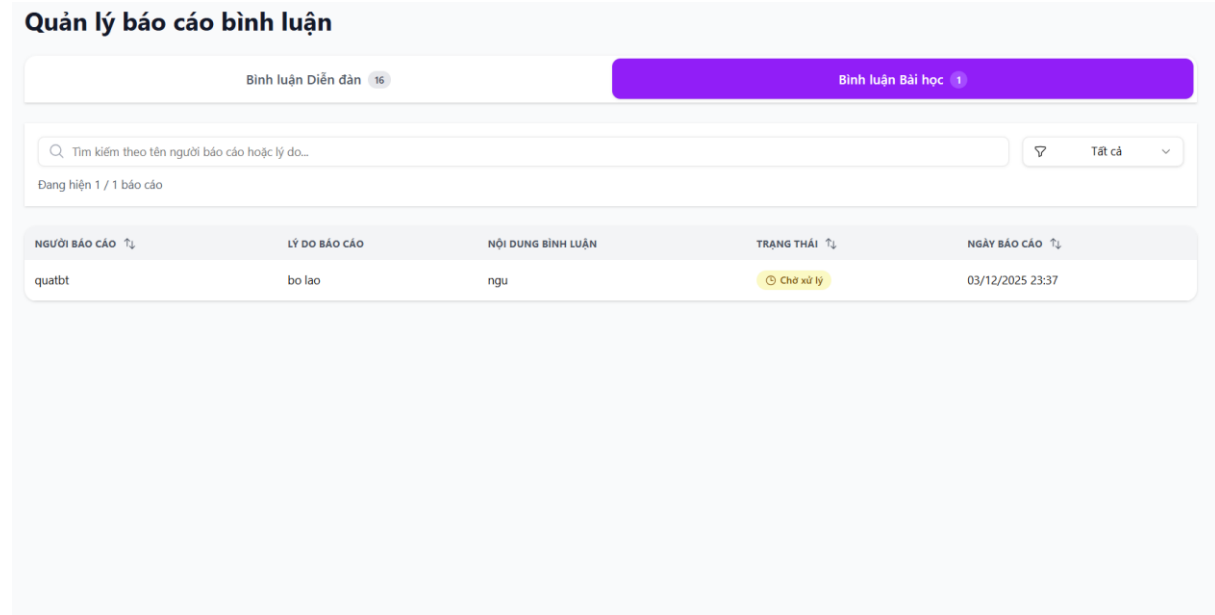


Figure 3.5.2 - lesson comment report list screen interface

This screen allows the Content Moderator to:

- View all lesson comment report: View all comment reports made in the lesson, submitted by the students.
- Filter and Search: Filter reports by its status and search for matching report submitter or reason.

Field Description

Field Name	Description
Bình luận diễn đàn	Type: Button, display all of the comment reports in the Forum.
Bình luận bài học	Type: Button, display all of the comment reports in the Course's Lessons.
Tìm kiếm	Data type: string, max length of 20 characters.
Filter Select Box	Initial values: Pending, Accepted, Rejected.

3.5.3 View Lesson Comment Report Details

Chi tiết báo cáo

Chờ xử lý

Người báo cáo:

quatbt

Lý do báo cáo:

bo lao

Ngày báo cáo:

03/12/2025 23:37:56

Tác giả bình luận:

Student

Nội dung bình luận:

ngu

✓ Chấp nhận & Xóa bình luận

✗ Từ chối báo cáo

Figure 3.5.3 - lesson comment report details screen interface

This screen allows the Content Moderator to:

- View the report details: View the report details and choose to accept and delete the comment or delete the report.

Field Description

Field Name	Description
Chấp nhận & Xóa bình luận	Type: Button, mark the report as Accepted and delete the violating comment.
Từ chối báo cáo	Type: Button, mark the report as Rejected.

3.6 Quiz Feature

3.6.1 External Interfaces

3.6.1.1 User Interfaces

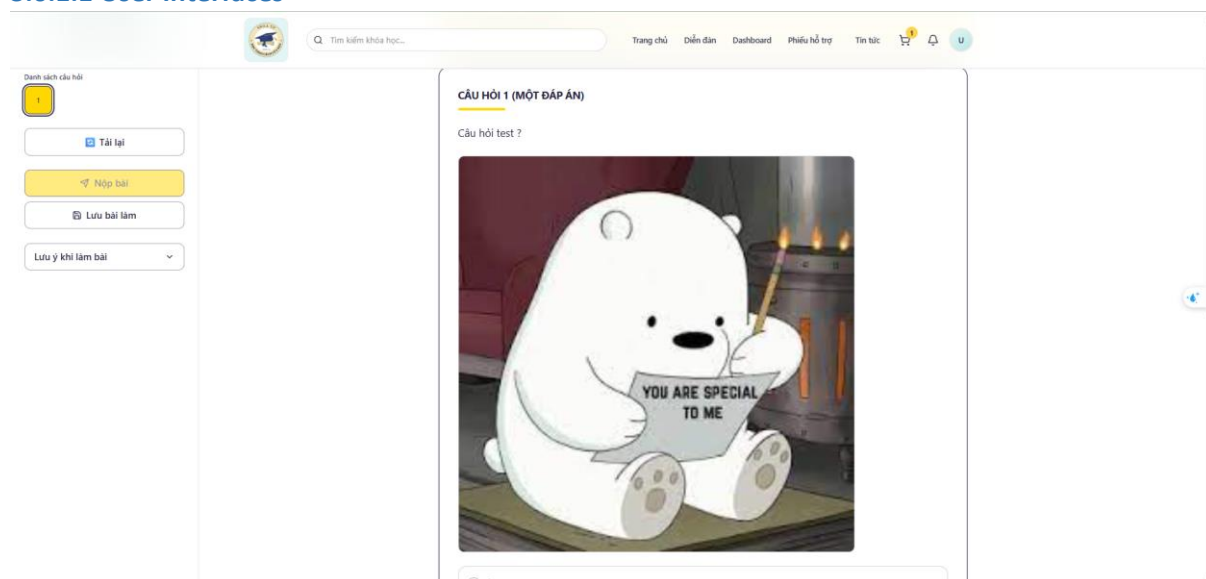


Figure 3.6.1.1 -user screen interface

This screen is the active environment for students to take a quiz. It allows the Student to:

- **View Question:** read the question content, including text and media (images), and identify the question type (e.g., Single Choice, Multiple Choice).
- **Navigate Questions:** use the sidebar question list to jump between questions or track which questions have been answered.
- **Select Answers:** choose options corresponding to the question.
- **Save Progress:** manually save their work to prevent data loss before submission.
- **Submit Quiz:** finalize the attempt and submit answers for grading.

Field Name	Description
Question List Grid	Button/Grid. Action: Clicking a number allows the user to jump directly to that specific question.
"Tải lại" Button	Button. Label: "Tải lại". Action: Refreshes the quiz interface to reload content or images if errors occur.
"Nộp bài" Button	Button. Label: "Nộp bài". Action: Triggers the submission process. Displays a confirmation popup before finalizing the grade.
"Lưu bài làm" Button	Button. Label: "Lưu bài làm". Action: Saves the currently selected answers to the server without submitting the quiz.
"Lưu ý khi làm bài" Dropdown	Button/Accordion. Label: "Lưu ý khi làm bài". Action: Expands or collapses the section to show/hide quiz rules and instructions.
Tùy chọn trả lời	Radio Button / Checkbox. Action: Users click to select one (Radio) or multiple (Checkbox) answers for the current question.

3.6.2 View Detail Result

3.6.2.1 Quiz Result Page

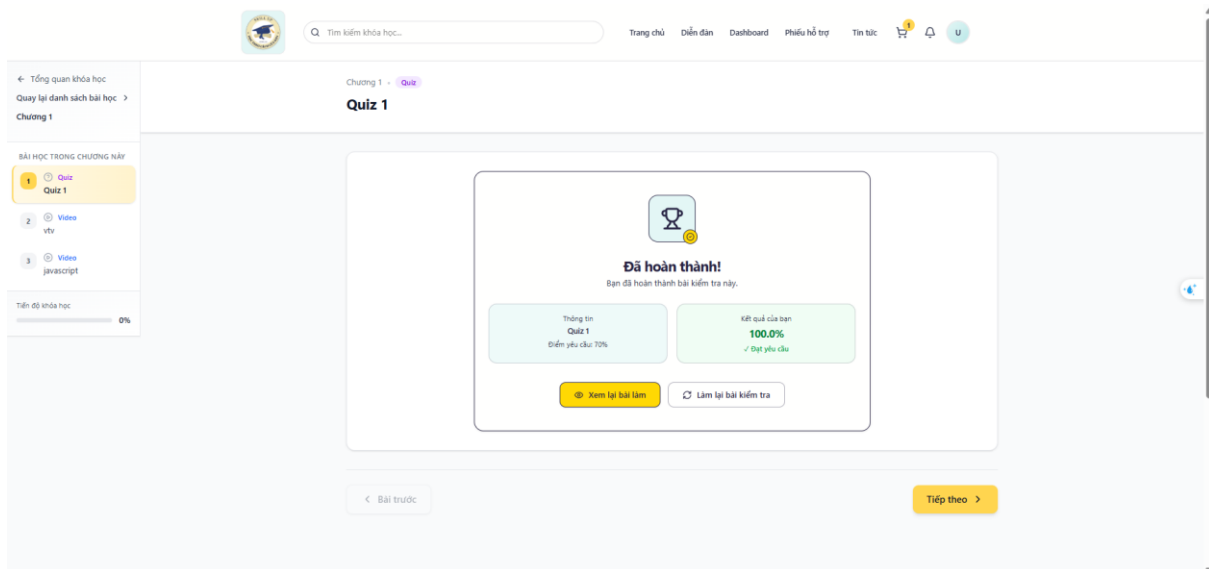


Figure 3.6.2.1 -quiz result screen interface

This screen displays the outcome after a student submits a quiz. It allows the Student to:

- View Performance: instantly see the completion status, achieved score (percentage), and pass/fail result.
- Review Attempt: click "Review" to see detailed feedback on which questions were answered correctly or incorrectly.
- Retry Quiz: check if retakes are allowed and click "Retry" to start a new attempt to improve the score.
- Navigate Forward: proceed to the next lesson in the course syllabus.

Primary Actors	System Moderator	Secondary Actors	none
Description	The "Manage Lecturer Application" use case allows a System Moderator to review the details of lecturer applications. They can check the current approval status (Pending, Approved, Rejected), review the submitted experience text and credentials, and choose to reject or accept the application.		
Preconditions	● The moderator logged into the system with an account. ● The account Role must be "System Moderator". ● The account status must be "Active". ● The Moderator clicked on the "Quản lý ứng tuyển" tab on the side menu and navigate to the "Quản lý đơn ứng tuyển của giảng viên" page.		

Postconditions	● View Application List: The system successfully displays the list of all Lecturer's applications. ● Approve Application: The application status is updated successfully to "Đã chấp nhận", and an email is sent afterward. ● Reject Application: The application status is updated successfully to "Đã từ chối", and an email is sent afterward.
Normal Sequence/Flow	View Application List: 1. The moderator accesses the "Quản lý đơn ứng tuyển của giảng viên" page. 2. The system successfully display the list of all Lecturer's applications, consist of following informations: ○ Họ tên (Full name) ○ Email ○ Tiêu đề (Title) ○ Chức danh (Profession) ○ Mô tả (Short Description) ○ Trạng thái (Status) ○ Ngày tạo (Date of Create) ○ CVs and Degrees buttons. ○ Thao tác (Action: Accept or Reject)
Alternative Sequences/Flows	A1: Approve Application: ○ The moderator accesses the "Quản lý đơn ứng tuyển của giảng viên" page. ○ The system successfully display the list of all Lecturer's applications, consist of following informations: ■ Họ tên (Full name) ■ Email ■ Tiêu đề (Title) ■ Chức danh (Profession)

	■ Mô tả (Short Description) ■ Trạng thái (Status) ■ Ngày tạo (Date of Create) ■ CVs and Degrees buttons. ■ Thao tác (Action: Accept or Reject) ○ The moderator clicks the “Duyệt” button on the Action column to accept the application. ○ The system updates the application status to “Đã chấp nhận”, and sends an email to notify the lecturer of approval. A2: Reject Application: ○ The moderator accesses the “Quản lý đơn ứng tuyển của giảng viên” page. ○ The system successfully display the list of all Lecturer’s applications, consist of following informations: ■ Họ tên (Full name) ■ Email ■ Tiêu đề (Title) ■ Chức danh (Profession) ■ Mô tả (Short Description) ■ Trạng thái (Status) ■ Ngày tạo (Date of Create) ■ CVs and Degrees buttons. ■ Thao tác (Action: Accept or Reject) ○ The moderator clicks the “Từ chối” button on the Action column to reject the application. ○ The system updates the application status to “Đã từ chối”, and sends an email to notify the lecturer of disapproval.
Exception	E1_No Application Found ● Trigger: Occurs at Normal Flow Step 2 (System retrieves latest application record).

	● Sequence: 1. Lecturer navigates to "My Application" section in Dashboard. 2. System attempts to retrieve application record for logged-in user. 3. System detects that no application exists (user never submitted application). 4. System displays empty state screen: a. Empty state illustration (e.g., document icon) b. Message: "Bạn chưa nộp đơn ứng tuyển nào." (MSG-NO-APPLICATION) c. Subtitle: "Hãy nộp đơn ứng tuyển để trở thành giảng viên chính thức." d. Call-to-action button: "Nộp đơn ứng tuyển" (Submit Application) 5. Application detail view not displayed. Postcondition: Empty state displayed; User informed no application exists; Clear guidance to submit new application; Good onboarding experience.
--	--

Field Name	Description
"Xem lại bài làm" Button	Button. Label: "Xem lại bài làm". Action: Redirects the user to the review mode to see correct/incorrect answers and explanations.
"Làm lại bài kiểm tra" Button	Button. Label: "Làm lại bài kiểm tra". Action: Resets the current answers and starts a new quiz session (if retakes are allowed).
"Bài tiếp theo" Button	Button. Label: "Bài tiếp theo" (Next Lesson). Action: Navigates immediately to the next lesson or video in the course syllabus.

3.7 Cart Feature

3.7.1. Add Cart Item

3.7.1.1 Course Detail Page

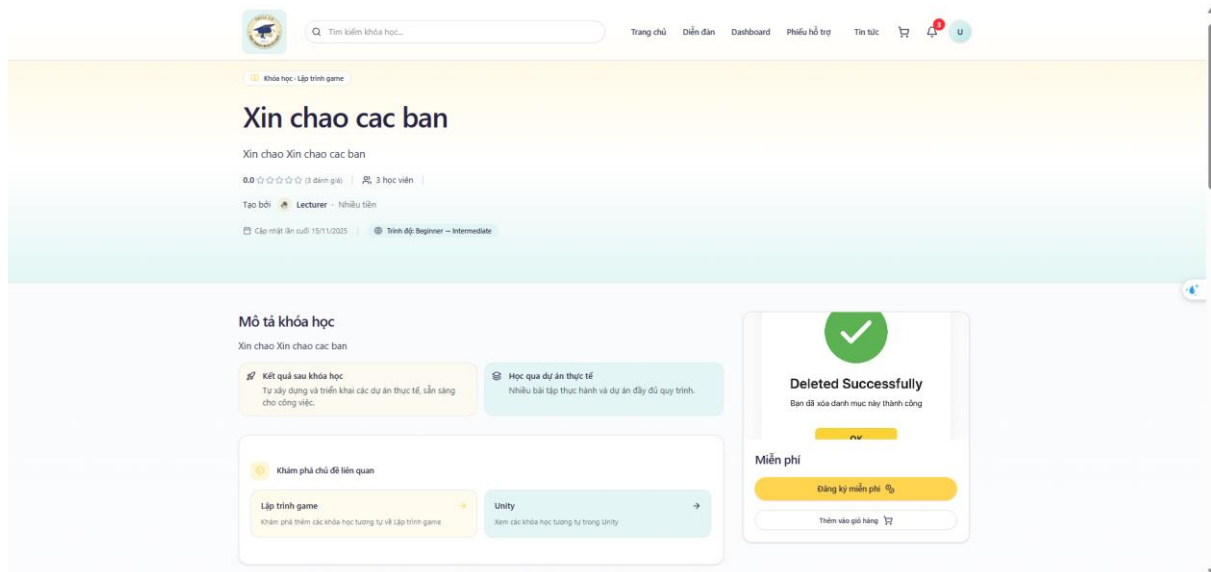


Figure 3.7.1.1 -course detail page screen interface

This function allows the Student to add a course to their shopping cart without immediately checking out.

- **Action:** Click the "Add to Cart" (Thêm vào giỏ hàng) button.
- **System Behavior:** The system validates the request, adds the course to the cart session, updates the cart icon badge number, and displays a success notification (Toast message). The user remains on the current page to continue browsing.

Field Name	Description
"Thêm vào giỏ hàng" Button	Button. Action: Adds item to cart. Visible if the user has not purchased the course yet.
Cart Icon Badge	Icon/Label. Header component. Updates immediately (e.g., from 0 to 1) upon successful addition.

3.7.2. Add Cart Item

3.7.2.1 Shopping Cart Page

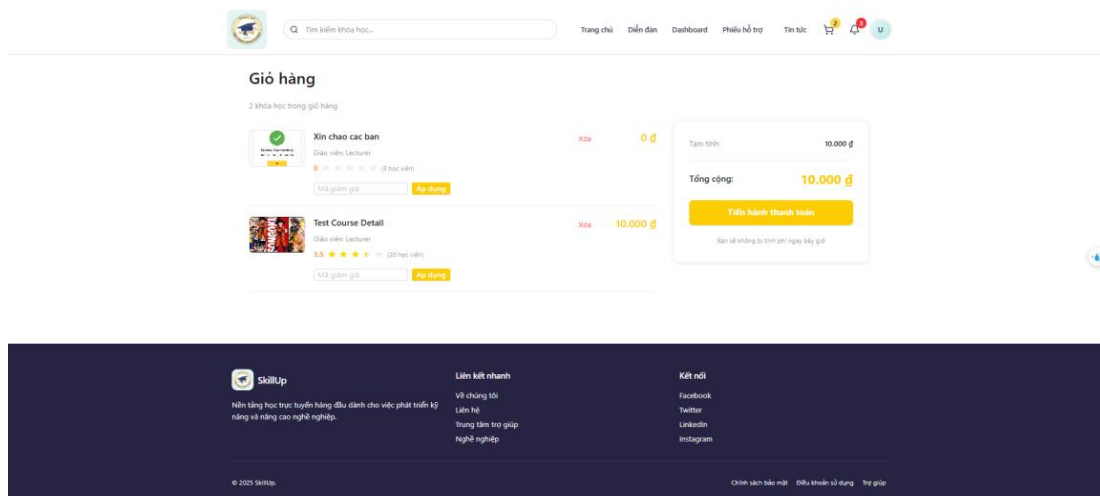


Figure 3.7.2.1 -shopping cart page screen interface

This function allows the Student to remove an unwanted course from the shopping list.

- **Action:** Click the "Remove" ("Xóa") text/button next to a course item.
- **System Behavior:** The system removes the item from the list, recalculates the "Total" price immediately, and displays a confirmation message (e.g., "Deleted Successfully").

Field Name	Description
"Xóa" Button/Link	Button/Link. Label: "Xóa" (Delete/Remove). Action: Triggers immediate removal of the row from the data list. No confirmation popup.
Cart List	List. Automatically refreshes instantly to hide the removed item row.
Order Summary	Label. Automatically recalculates the sum immediately after the item is removed.

3.7.3. Remove Cart Item

3.7.3.1 Shopping Cart Page

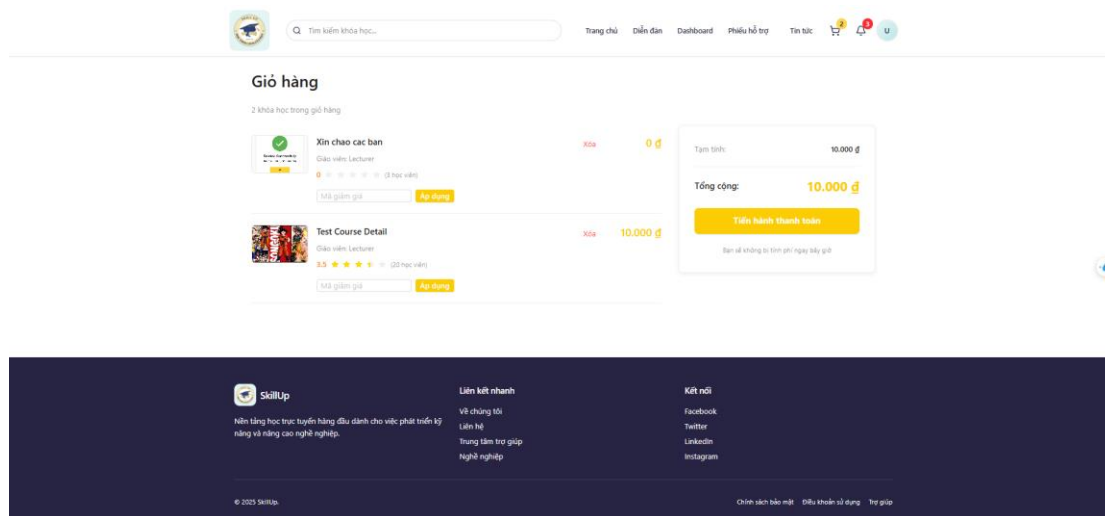


Figure 3.7.3.1 -shopping cart page screen interface

This function allows the Student to enter a promotional code to receive a discount on specific courses.

- **Action:** Enter a code into the "Coupon Code" field and click "Apply" (Áp dụng).
- **System Behavior:** The system checks the validity of the code.
 - If valid: Updates the price (showing the discounted amount) and notifies success.
 - If invalid: Displays an error message (e.g., "Code expired" or "Invalid code").

Field Name	Description
Voucher Input	Textbox. Placeholder: "Mã giảm giá" (Coupon code).
"Áp dụng" Button	Button. Label: "Áp dụng". Trigger validation logic.
Discounted Price	Label. If applied successfully, the old price is struck through, and the new lower price is shown.
Tổng cộng	Label. The "Total" (Tổng cộng) at the bottom right reflects the new sum after discount.

3.8 Shopping cart & payment Feature

3.8.1 Buy course

3.8.1.1 Buy Now

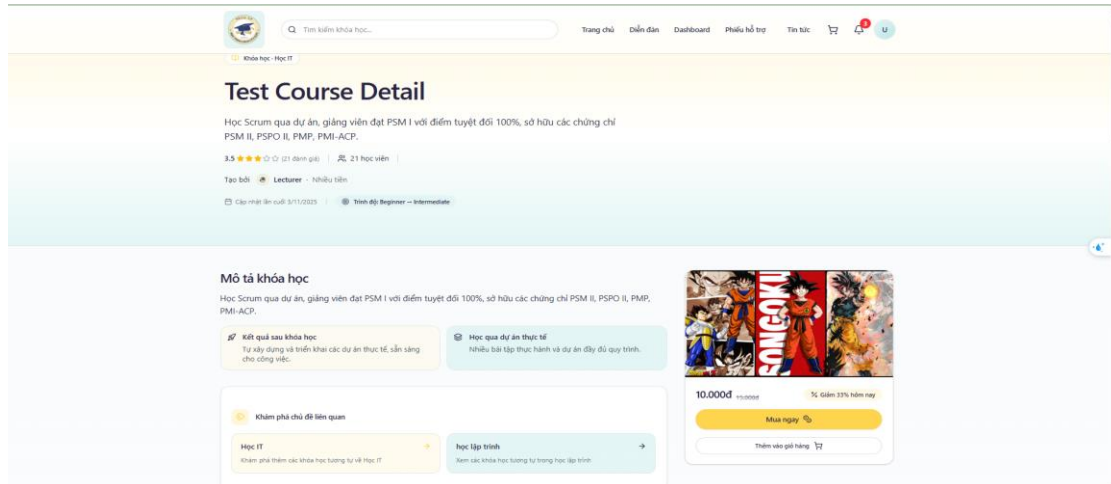


Figure 3.8.1.1 -buy course screen interface

This function allows the Student to purchase a specific course immediately without adding it to the cart.

- Trigger: Click the "Mua ngay" (Buy Now) button on the sticky purchase card.
- System Behavior:
 1. Authentication Check: Checks if the user is logged in. If not, redirects to Login.
 2. Order Creation: Creates a temporary order containing only this single course.
 3. Redirection: Redirects the user directly to the Payment Gateway interface to complete the transaction.

Field Name	Description
"Mua ngay" Button	Button. Label: "Mua ngay". Action: Triggers the immediate purchase flow for the single item.
Price Display	Label. The specific price used for the order calculation.

3.8.2 Checkout from Cart

3.8.2.1 Shopping Cart Page

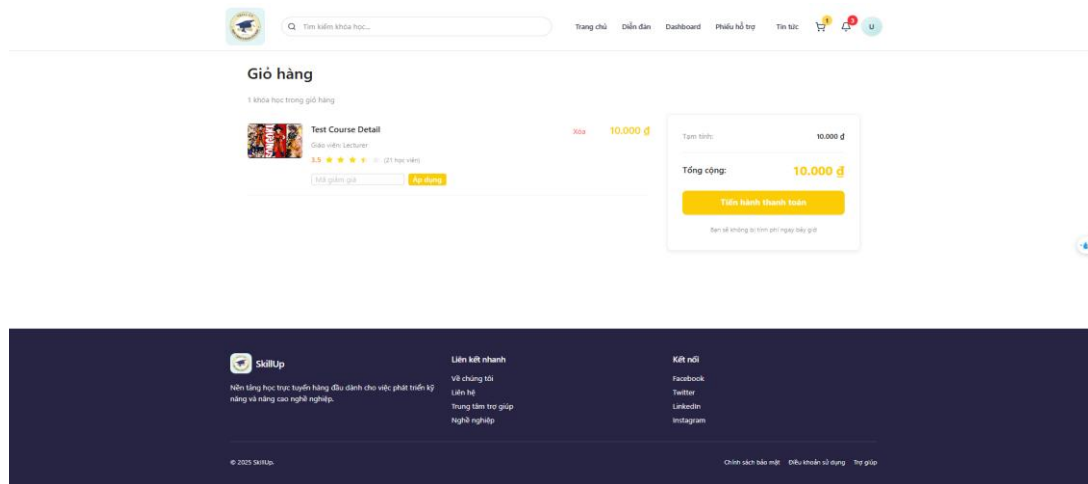


Figure 3.8.2.1 -cart page screen interface

This function allows the Student to finalize the purchase for all items currently in the shopping cart.

- Trigger: Click the "Tiến hành thanh toán" (Proceed to Checkout) button.
- System Behavior:
 1. Validation: Checks if the cart is not empty and validates item availability.
 2. Calculation: Finalizes the total amount after applying any vouchers.
 3. Redirection: Redirects the user to the Payment Gateway interface to pay for the entire cart bundle.

Field Name	Description
"Tiến hành thanh toán" Button	Button. Label: "Tiến hành thanh toán". Action: Initiates the checkout process for the entire cart.
Tổng số tiền	Label. The final sum calculated from all items minus vouchers. This is the amount passed to the payment gateway.

3.8.3 Transaction Management

3.8.3.1 View history transaction

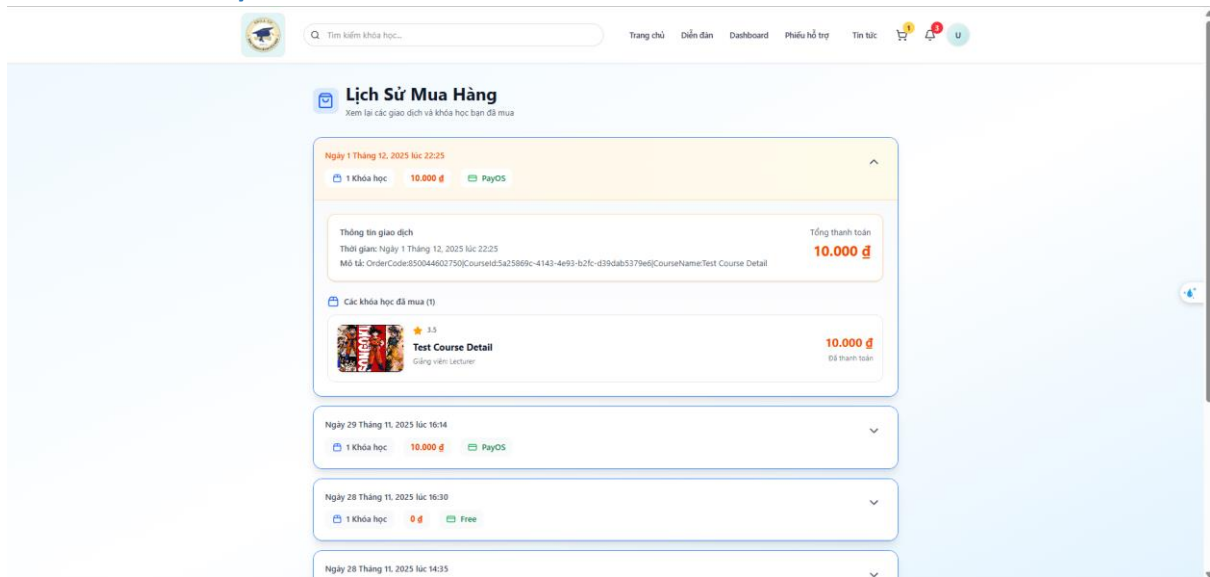


Figure 3.8.3.1 -view history transaction screen interface

This screen allows the Student to review their purchase history in chronological order.

- View Transaction List: browse a list of transaction records sorted by date (newest first).
- Expand Details: click on a transaction row (Accordion style) to reveal detailed information about the order.
- Check Payment Status: view the payment method (e.g., PayOS, Free) and the total amount paid.
- View Purchased Items: see the specific courses included in each transaction, along with their individual prices and payment status.

Field Name	Description
1. Page Header	General Page Info
Lịch Sử Mua Hàng	Label. "Lịch Sử Mua Hàng" (Purchase History).
Xem lại các giao dịch và khóa học bạn đã mua	Label. "Xem lại các giao dịch và khóa học bạn đã mua".

3.8.4 Feedback course

3.8.4.1 Feedback & Rating

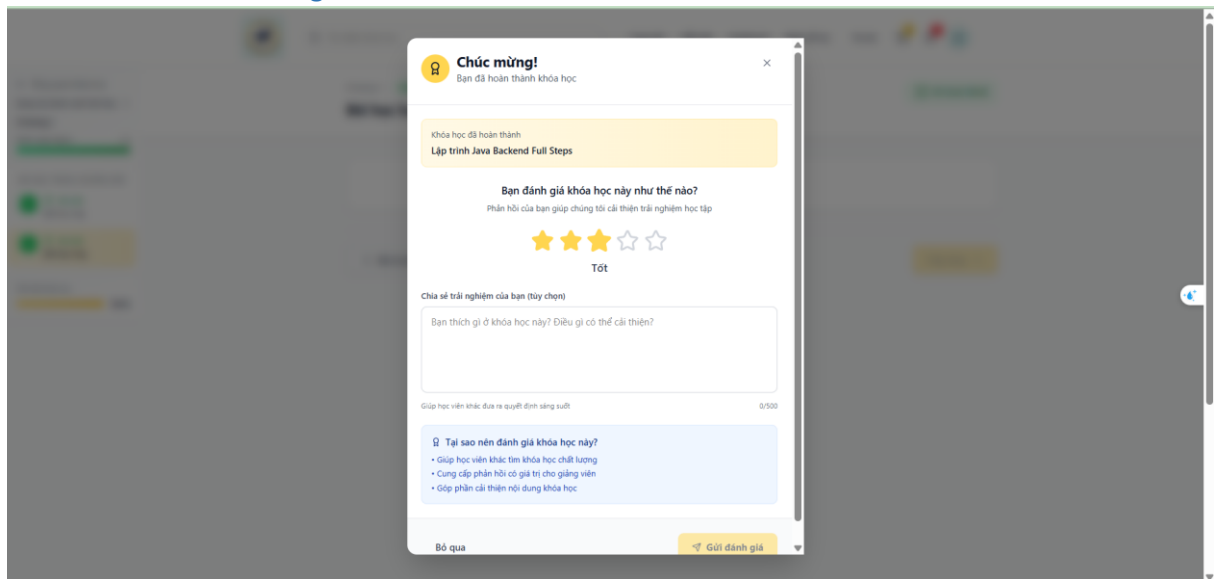


Figure 3.8.4.1 -feedback & rating screen interface

This modal appears automatically when a Student completes 100% of a course. It allows the Student to:

- Rate Quality: provide a star rating (1 to 5 stars) representing their satisfaction.
- Write Review: input a text comment sharing their specific experience or suggestions (optional).
- Submit Feedback: send the review to the system to be displayed on the public Course Detail page.
- Skip: close the popup without leaving a review (can review later manually).

Field Name	Description
Close Icon (X)	Button. Action: Closes the popup modal immediately.
Star Rating Input	Interactive Icon. Users click on the stars to select a rating level (1 to 5). Default is unselected.
Comment Textbox	Text Area. Input field for users to type their feedback or review content. Optional.
"Bỏ qua" Button	Button. Label: "Bỏ qua". Action: Closes the modal without saving any data.
"Gửi đánh giá" Button	Button. Label: "Gửi đánh giá". Action: Validates that a star rating is selected, then submits the feedback to the system.

3.9 Ticket Feature

3.9.1 View Ticket list

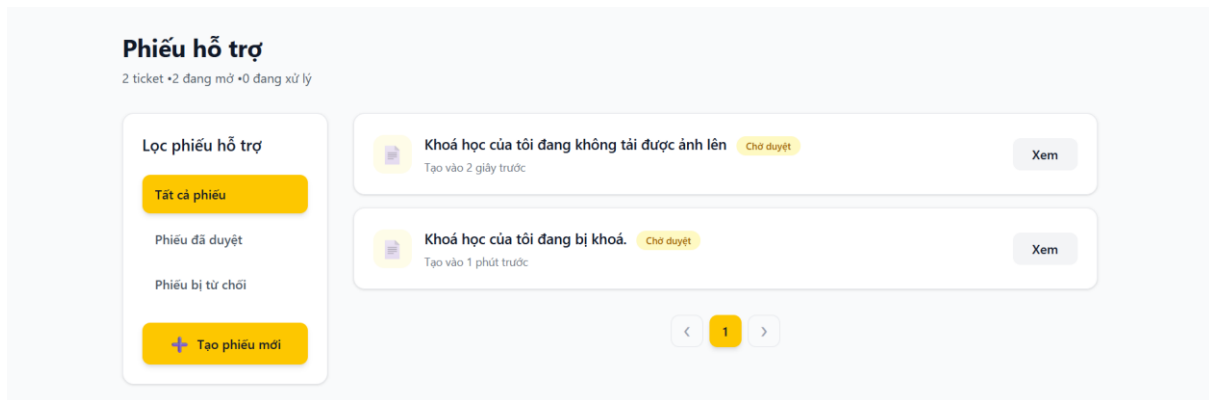


Figure 3.9.1 -ticket list screen interface

This screen allows the student to:

- View their submitted tickets.
- Filter their submitted tickets by status.
- Open the create new Ticket popup window.
- Navigate to the view Ticket details page.

Field Description

Field Name	Description
Tất cả phiếu	Type: Button, Display all user's submitted tickets.
Phiếu đã duyệt	Type: Button, Display all user's accepted tickets.
Phiếu bị từ chối	Type: Button, Display all user's rejected tickets.
Tạo phiếu mới	Type: Button, Open popup window to create a new ticket.
Xem	Type: Button, Display the user's ticket details.

3.9.2 View Ticket Details

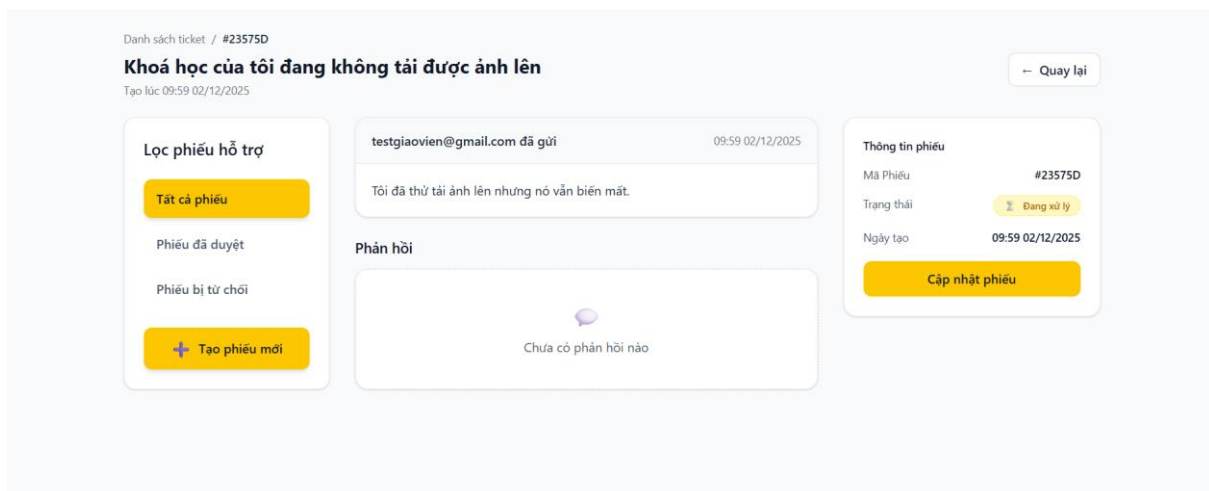


Figure 3.9.2 -ticket details screen interface

This screen allows the student to:

- View their Ticket details.
- Open the Update Ticket window to update the details of the ticket.
- Navigate back to the Ticket list.
- Open the create new ticket popup window.

Field Description

Field Name	Description
Tất cả phiếu	Type: Button, Display all user's submitted tickets.
Phiếu đã duyệt	Type: Button, Display all user's accepted tickets.
Phiếu bị từ chối	Type: Button, Display all user's rejected tickets.
Tạo phiếu mới	Type: Button, Open popup window to create a new ticket.
Cập nhật phiếu	Type: Button, Open popup window to update the selected ticket.
Quay lại	Type: Button, return to the ticket list.

3.12.3 Update Ticket

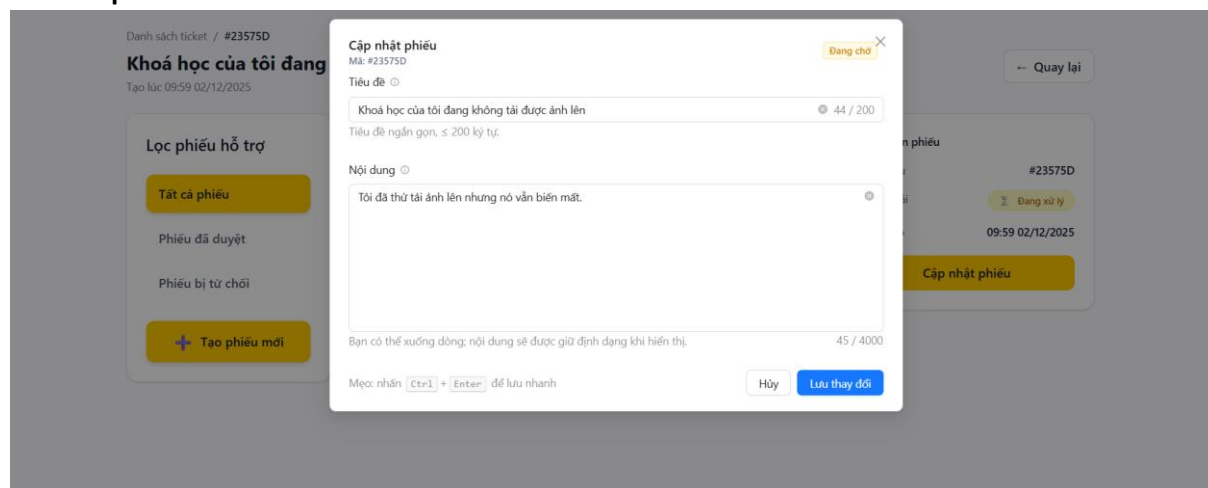


Figure 3.12.3 -update ticket screen interface

This screen allows the student to:

- Update the Ticket details as long as it is not Accepted or Rejected.
- Save the changes to the database.
- Discard the changes made to the ticket as long as it is not saved to the database.

Field Description

Field Name	Description
Tiêu đề	Type: string, max length of 200 characters.
Nội dung	Type: string, max length of 4000 characters.
Hủy	Type: Button, cancel Ticket update.
Lưu thay đổi	Type: Button, save the Ticket changes to the database.

3.9.4 Create Ticket

The screenshot displays the 'Tạo phiếu' (Create Ticket) modal window. It features two main text input areas: 'Tiêu đề' (Title) and 'Nội dung' (Content). The 'Tiêu đề' field includes a placeholder example 'VD: Không đăng nhập được' and a character limit note 'Tiêu đề ngắn gọn, < 200 ký tự.' The 'Nội dung' field has a placeholder 'Mô tả chi tiết vấn đề bạn gặp phải...' and a note about line wrapping: 'Bạn có thể xuống dòng; nội dung sẽ giữ format khi hiển thị.' At the bottom of the modal, there are 'Hủy' (Cancel) and 'Tạo phiếu' (Create Ticket) buttons. The background interface shows a sidebar with filters like 'Tất cả phiếu', 'Phiếu đã duyệt', and 'Phiếu bị từ chối', along with a 'Tạo phiếu mới' (Create new ticket) button. A list of tickets is visible on the right, showing details like '#23575D' and '09:59 02/12/2025'.

Figure 3.9.4 -create ticket screen interface

This screen allows the student to:

- Create a new Ticket to demand support from the Moderators.
- Cancel creating a new ticket.

Field Description

Field Name	Description
Tiêu đề	Type: string, max length of 200 characters.
Nội dung	Type: string, max length of 4000 characters.
Hủy	Type: Button, cancel Ticket update.
Tạo phiếu	Type: Button, save the Ticket to the database.

3.9.5 View All Tickets

Quản lý phiếu hỗ trợ

Tất cả

Chưa xử lý

Đã xử lý

Tìm theo mã, tiêu đề, người gửi...

Sắp xếp mới nhất

Xoá bộ lọc

Xoá tất cả

Tải lại

Mã ticket	Tiêu đề	Người gửi	Trạng thái	Ngày tạo	Hành động
#05EDA7	Yêu cầu khôi phục tài khoản	Lecturer	Đã chấp nhận	11:37 28/10/2025	
#382DBD	Khoá học đang bị khoá	Lecturer	Từ chối	11:37 28/10/2025	
#2F9479	Hỗ trợ hệ thống	Lecturer	Đã chấp nhận	11:37 28/10/2025	
#6A3FFB	Hình ảnh khoá học biến mất	Lecturer	Từ chối	02:32 01/12/2025	

4 bản ghi

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Figure 3.9.5 -Tickets List screen interface

This screen allows the Moderator to:

- View all of the user's(Student and Lecturer) submitted tickets.
- Filter by Ticket's status, date.
- Search for tickets.

Field Description

Field Name	Description
Tất cả	Type: Button, display all user's Tickets.
Chưa xử lý	Type: Button, display all user's pending Tickets.
Đã xử lý	Type: Button, display all user's solved Tickets.
Tìm kiếm	Type: string, max 100 characters.
Sắp xếp mới nhất	Type: Button, sort ticket by date created, from newest to oldest.
Xoá bộ lọc	Type: Button, remove filters.
Xoá tất cả	
Tải lại	
👁 icon	Type: Button, open Ticket Details popup window.

3.9.6 View Ticket Details for moderator

Figure 3.9.6 -ticket details for moderator screen interface

This screen allows the Moderator to:

- View the Ticket details.
- Accept or Reject the Ticket.
- Inputting the reason why.
- Save the response to the database.
- Close the View Details popup window.

Field Description

Field Name	Description
Chấp nhận	Type: Button, Mark the Ticket as accepted.
Từ chối	Type: Button, Mark the Ticket as rejected.
Đóng	Type: Button, close the popup window.
Lưu	Type: Button, Save the response to the database.

3.10 Lecturer Application Feature

3.10.1 Create Lecturer Application

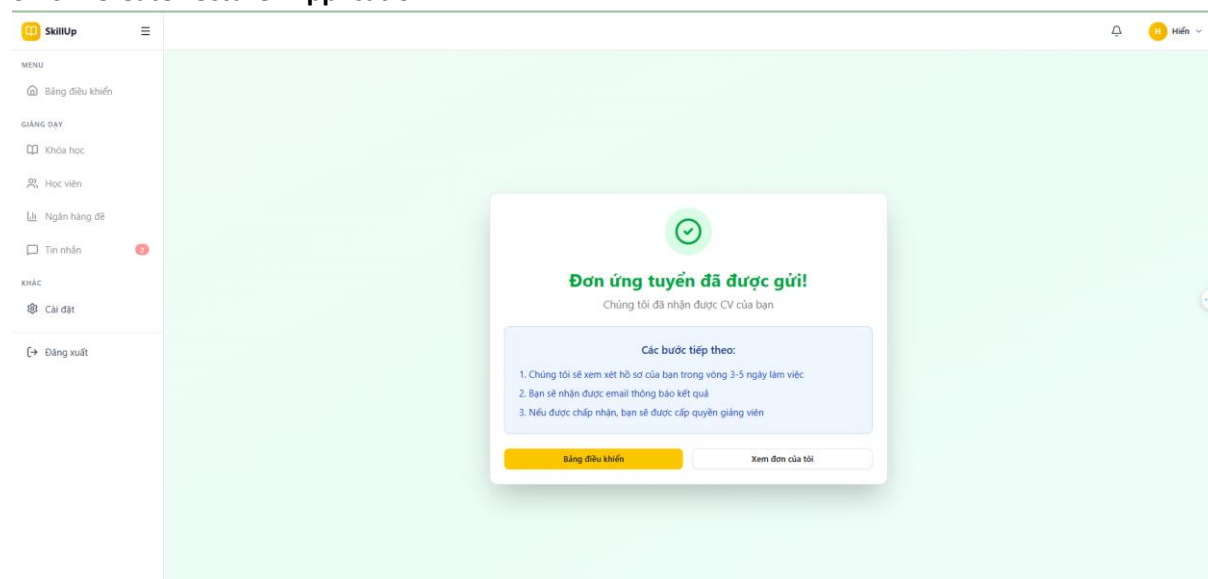


Figure 3.10.1 - create lecturer application screen interface

This modal appears automatically after a user successfully submits their lecturer application form. It serves to:

- **Confirm Submission:** notify the user that their application and CV have been received by the system.
- **Inform Next Steps:** outline the review process (timeline of 3-5 days, email notification, permission granting).
- **Navigate:** provide options to return to the main dashboard or review the specific application details just submitted.

Field Description

Field Name	Description
"Bảng điều khiển" Button	Button. Label: "Bảng điều khiển" (Dashboard). Action: Closes the modal and redirects the user to the main User Dashboard .
"Xem đơn của tôi" Button	Button. Label: "Xem đơn của tôi" (View My Application). Action: Redirects the user to the Application Detail/Status page to track progress.

3.10.2 View detail Lecturer Application

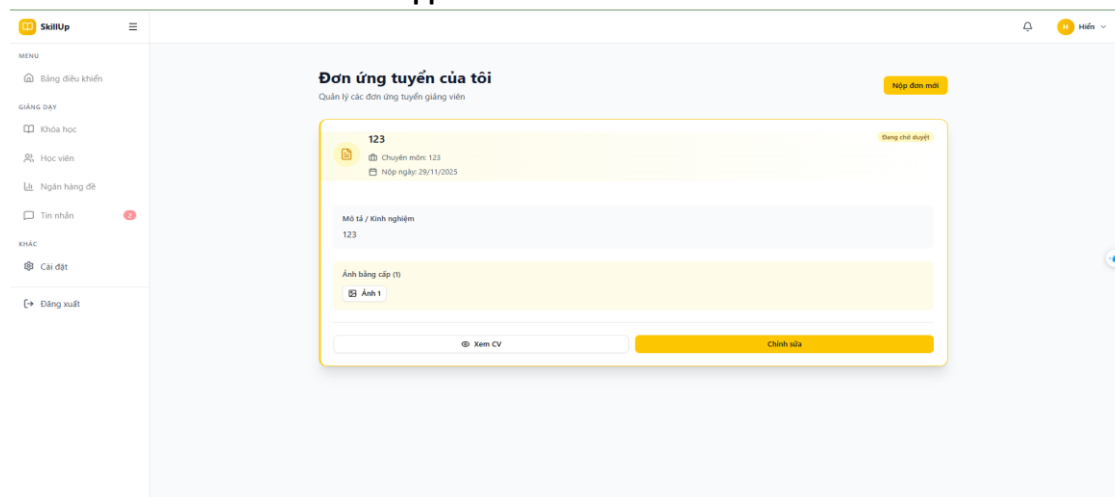


Figure 3.10.2 -Detail lecturer application screen interface

This screen allows the User to track the progress of their lecturer application. It allows the User to:

- **Check Status:** view the current status of the application (e.g., Pending, Approved, Rejected).
- **Review Information:** review the submitted details, including personal info and the attached CV/Portfolio.
- **View Feedback:** read any notes or reasons for rejection provided by the Admin (if applicable).
- **Cancel Application:** retract the application if it is still in the "Pending" state (if the system permits).
- **Navigate Back:** return to the dashboard or home page.

Field Description

Field Name	Description
"Back" Button	Button. Action: Navigates back to the User Dashboard or previous page.
CV/Portfolio Link	Link/Button. Action: Downloads or opens the uploaded CV file (PDF/Doc) for review.
"Cancel Application" Button	Button. Label: "Hủy đơn" (Cancel). Action: Withdraws the application. Typically enabled only when status is "Pending". Triggers a confirmation popup.
"Re-apply" Button	Button. Label: "Gửi lại đơn" (Re-apply). Action: Navigates to the Registration Form to create a new application (visible only if the previous application was Rejected).

3.10.3 View Lecturer Application List

Quản lý đơn ứng tuyển giảng viên

🔍 Tìm theo tiêu đề, trạng thái... Xoá bộ lọc C Tải lại

Tiêu đề	Mô tả	Chức danh	Email	Trạng thái	Ngày tạo	CV & Bằng cấp	Thao tác
Giảng viên	Giảng viên	Web Developer	hocfptkk@gmail.com111	Đã chấp nhận	09:24 02/12/2025		Duyệt Từ chối
123	123	123	collinsbest30026@gmail.com	Chờ xử lý	10:40 29/11/2025		Duyệt Từ chối
Pro top 1	Pro top 1	Pro top 1	quatbthe176382@fpt.edu.vn11...	Từ chối	23:46 28/11/2025		Duyệt Từ chối
hi	hi	hi	holastaynow@gmail.com	Đã chấp nhận	23:13 17/11/2025		Duyệt Từ chối
Giaos vien	thủ việc	Web dev	hhhhhh@gmail.com	Đã chấp nhận	14:52 11/11/2025		Duyệt Từ chối
Senior	hhhhhhafaf	Web dev	hhhy@gmail.com	Đã chấp nhận	14:38 11/11/2025		Duyệt Từ chối
Học sinh	Học sinh	Học sinh	buitienquat2k3@gmail.com	Từ chối	20:57 09/11/2025		Duyệt Từ chối
Top 1	Top 1	Top 1	hhhh@gmail.com	Đã chấp nhận	00:32 07/11/2025		Duyệt Từ chối
senior	sss	Web	testgiaovien@gmail.com	Đã chấp nhận	21:15 05/11/2025		Duyệt Từ chối
Fresher	Fresher	Web	holastaynow11@gmail.com	Đã chấp nhận	22:31 30/10/2025		Duyệt Từ chối

11 bản ghi < 1 2 >

Figure 3.10.3 -Lecturer application list screen interface

This screen allows the System Moderator to view the list of all lecturer applications. It allows the System Moderator to:

- **Check Status:** view the current status of the application (e.g., Pending, Approved, Rejected).
- **Review Information:** review the submitted details, including personal info and the attached CV/Portfolio.
- **Handle Application:** accept or reject the application if it is still in the "Pending" state.

Field Description

Field Name	Description
Tìm kiếm	Type: string . Action: filter and display all applications with matching information.
Xoá bộ lọc	Type: Button. Action: Reset all applying filters.
Tải lại	Type: Button. Action: Refresh the table data.
File icon	Type: Button. Action: Open the application's CVs popup window.
Trophy icon	Type: Button. Action: Open the application's Degrees popup window.
Duyệt	Type: Button. Action: Update the application status to Accepted.
Từ chối	Type: Button. Action: Update the application status to Rejected.

3.11 Question & Answer Feature

3.11.1 Manage Question & Answer

3.11.1.1 Create question bank

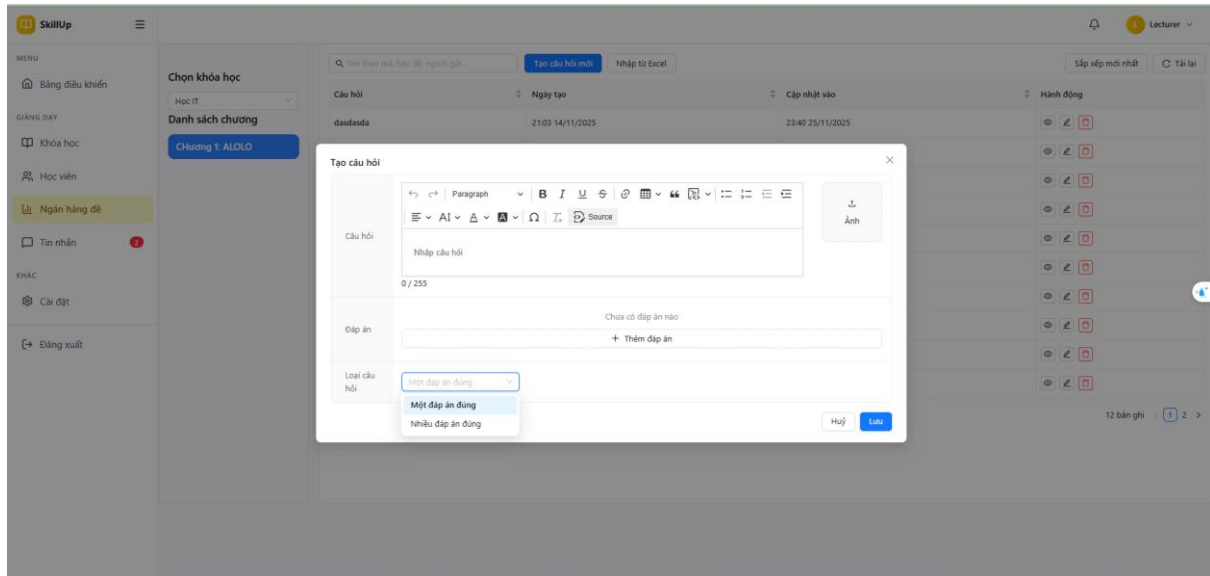


Figure 3.11.1.1 -Create question bank screen interface

This screen allows the Instructor to create a new question and manage its answer options simultaneously.

- **Question Content:** Enter the question text and media.
- **Manage Answers:** Create answer options, delete unnecessary options, and define the correct result.
- **Save:** Store the question and its answers to the database.

Field Description

Field Name	Description
Quiz Title Input	Textbox. Action: Allows the instructor to type the name of the quiz (e.g., "Final Exam").
"Add Question" Button	Button. Action: Adds a new empty question block to the list.
Question Type Dropdown	Dropdown. Action: Selects the format of the question (e.g., Single Choice, Multiple Choice).
Question Text Area	Text Area. Action: Input field for the main content of the question.
"Upload Image" Button	Button. Action: Uploads an image to attach to the specific question (optional).
Answer Option Input	Textbox. Action: Input fields to type the possible answers (Option A, B, C...).
Correct Answer Checkbox/Radio	Checkbox/Radio. Action: Marks which option is the correct answer. Essential for auto-grading.
"Delete Question" Icon	Button. Action: Removes the specific question from the quiz.
"Save Quiz" Button	Button. Action: Validates all fields (must have title, questions, and correct answers) and saves the quiz data.

3.11.1.2 Update question bank

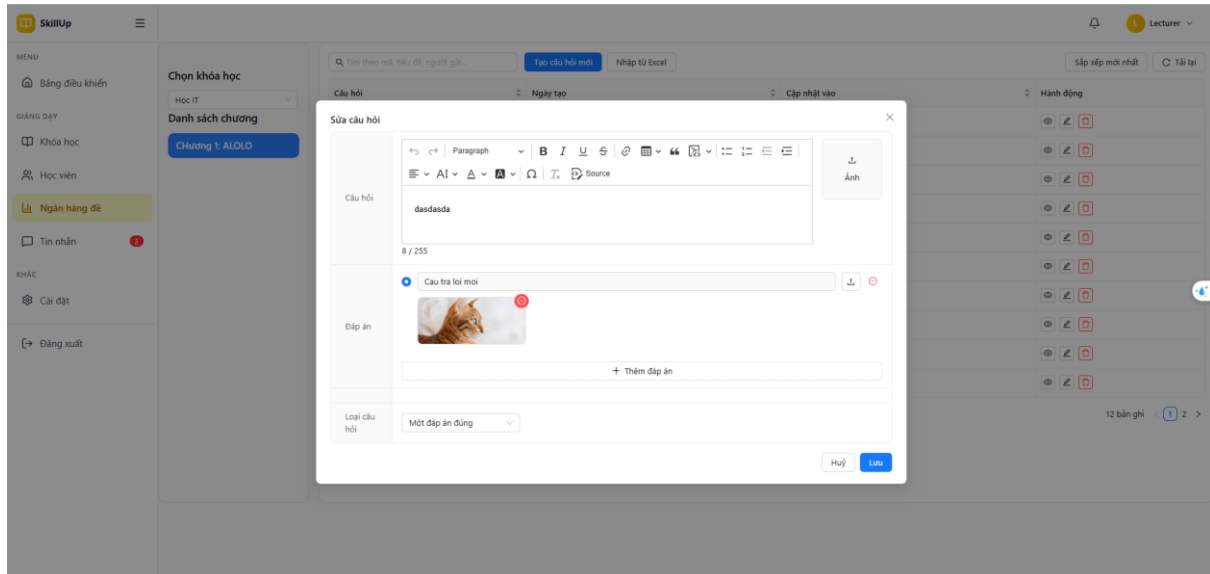


Figure 3.11.1.2 -Update question bank screen interface

This screen allows the Instructor to modify an existing question and its answers.

- **Modify Question:** Edit the text or media.
- **Modify Answers:** Rename answer options (Update), add new options (Create), or remove options (Delete).
- **Update:** Save changes to the database.

Field Description

Field Name	Description
"Update" Button	Button. Action: Validates the modified data and overrides the existing quiz record in the database. Displays a "Updated Successfully" toast.
"Delete Answer" Icon	Button/Icon. Action: Removes a specific answer option (e.g., Option D) from a question.
"Cancel" Button	Button. Action: Discards any unsaved changes and returns to the Course Curriculum page.

3.11.1.3 View question bank

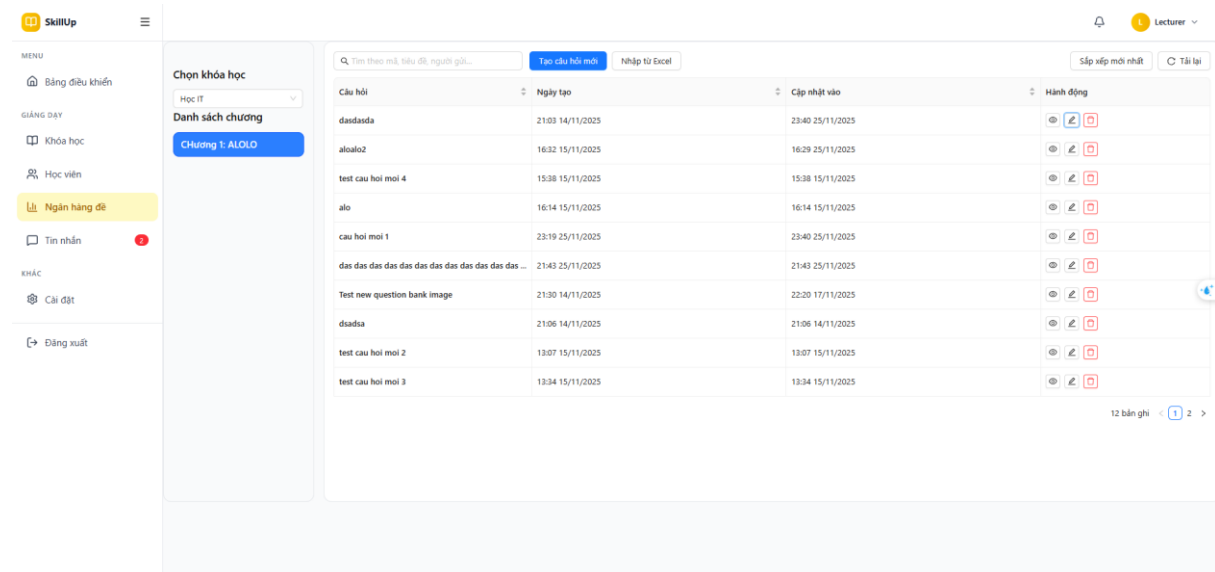


Figure 3.11.1.3 - question bank screen interface

This screen allows the Instructor to preview how a specific question appears to students.

- **Review:** Check question text, image, and options.
- **Actions:** Quick access to Edit or Delete this question.

Field Description

Field Name	Description
"Edit" Button	Button. Action: Navigates to the Quiz Editor screen to modify settings or questions.
"Preview" Button	Button. Action: Opens a simulation of the quiz (Student View) so the instructor can test the flow.
"Delete" Button	Button. Action: Permanently removes the quiz from the course. Triggers a confirmation popup.
"Back" Button	Button. Action: Returns to the Course Curriculum/Section list.
Question Accordion/Toggle	Button/Icon. Action: Expands a specific question row to reveal its answer options and the correct answer key.

3.11.1.4 Delete question bank

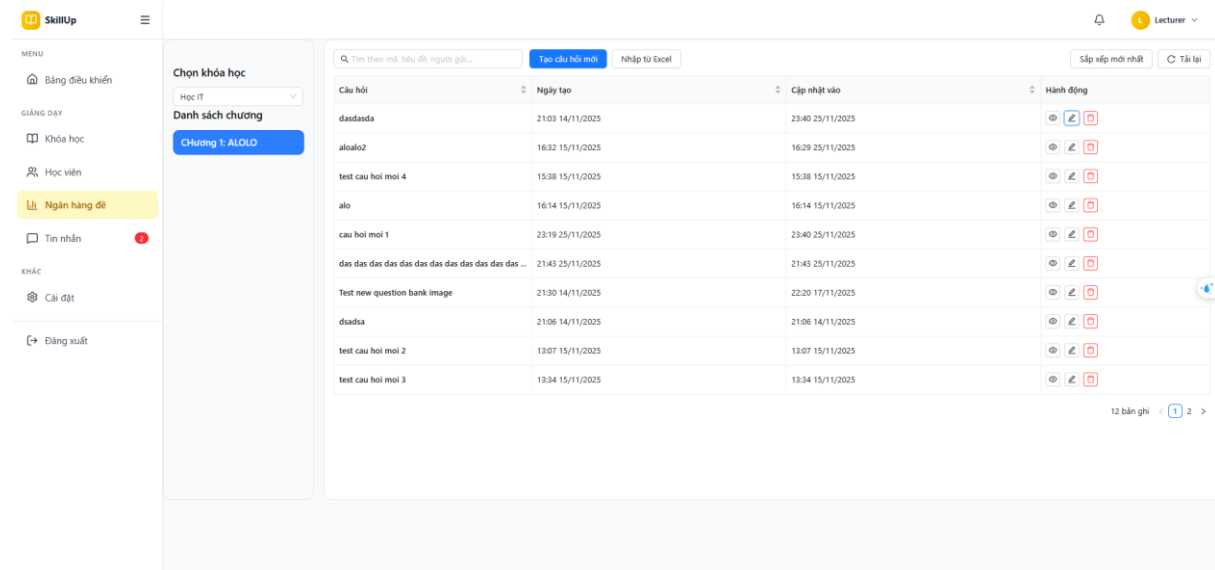


Figure 3.11.1.4 -delete question bank screen interface

This function removes a question permanently.

- **Confirm:** User must confirm the action in a popup.

Field Description

Field Name	Description
"Delete" Icon	Button/Icon. Action: Triggers the delete confirmation popup for the selected question.
"Confirm Delete" Button	Button. (Inside Modal). Action: Permanently removes the question from the quiz data and closes the modal.
"Cancel" Button	Button. (Inside Modal). Action: Closes the confirmation popup without deleting the question.

3.11.1.5 Add question from excel

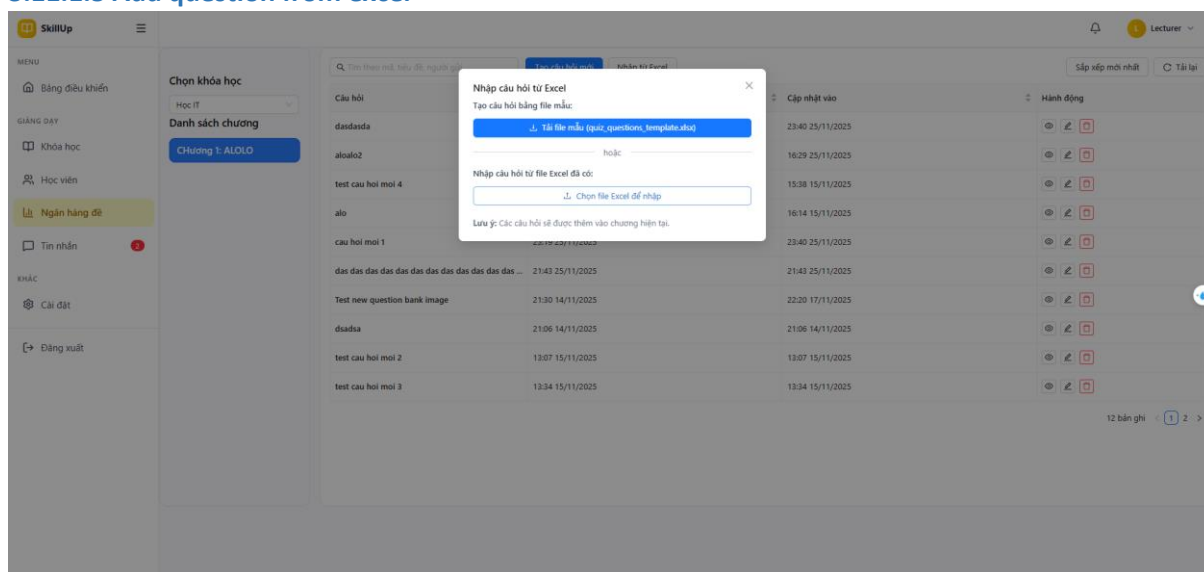


Figure 3.11.1.5 -add question from excel screen interface

This function allows the Instructor to bulk upload questions using a pre-formatted Excel or CSV file.

- **Download Template:** Get the standard file format required by the system.
- **Upload File:** Select and upload the filled data file.
- **Process:** The system validates the file and creates questions automatically.

Field Description

Field Name	Description
"Download Template" Link	Link/Button. Action: Downloads the sample Excel/CSV file containing the required column headers (Question, Option A, Option B, Correct Answer, etc.).
"Select File" Button	Button/Input. Action: Opens the operating system's file picker to select a local file (usually restricted to .xlsx, .xls, .csv).
"Upload" Button	Button. Action: Submits the selected file to the server for processing and data import.
"Cancel" Button	Button. Action: Closes the import modal without performing any action.

3.12 Quiz Feature

3.12.1 Manage quiz

3.12.1.1 Create quiz

QUIZ Cuối cùng

aa

Điểm đạt (%) 80 Thời gian (phút) 30

✓ Tạo quiz × Hủy

+ Thêm nội dung

Figure 3.12.1.1 -create quiz screen interface

This function allows the Instructor to initialize a new quiz.

- Input Data: The instructor fills in the quiz title, description, time limit, and passing score into empty fields.
- Action: Clicking "Create" saves the new quiz to the database.

Field Description

Field Name	Description
Quiz Title Input	Textbox. Action: Input field for the new quiz name.
Description Area	Text Area. Action: Input field for quiz instructions.
Time Limit Input	Number Input. Action: Sets the duration (in minutes).
Passing Score Input	Number Input. Action: Sets the minimum percentage to pass.
"Hủy" Button	Button. Action: Validates that required fields are filled and creates the new quiz record.
"Hủy" Button	Button. Action: Discards input and returns to the quiz list.

3.12.1.2 Update quiz

Quiz Cuối cùng

aa

Điểm đạt (%) 80 Thời gian (phút) 30

✓ Tạo quiz X Hủy

+ Thêm nội dung

Figure 3.12.1.2 -update quiz screen interface

This function allows the Instructor to modify an existing quiz's settings.

- Load Data: The system pre-fills the form with the current quiz details.
- Modify: The instructor edits the information (e.g., changing the time limit).
- Action: Clicking "Update" overrides the old settings.

Field Description

Field Name	Description
Quiz Title Input	Textbox. Action: Edit the existing quiz name.
Description Area	Text Area. Action: Edit existing instructions.
Time Limit Input	Number Input. Action: Modify the duration value.
Passing Score Input	Number Input. Action: Modify the passing percentage.
"Tạo quiz" Button	Button. Action: Saves the changes to the system. Displays a success message.
"Hủy" Button	Button. Action: Discards changes and returns to the quiz list without saving.

3.12.1.3 View quiz

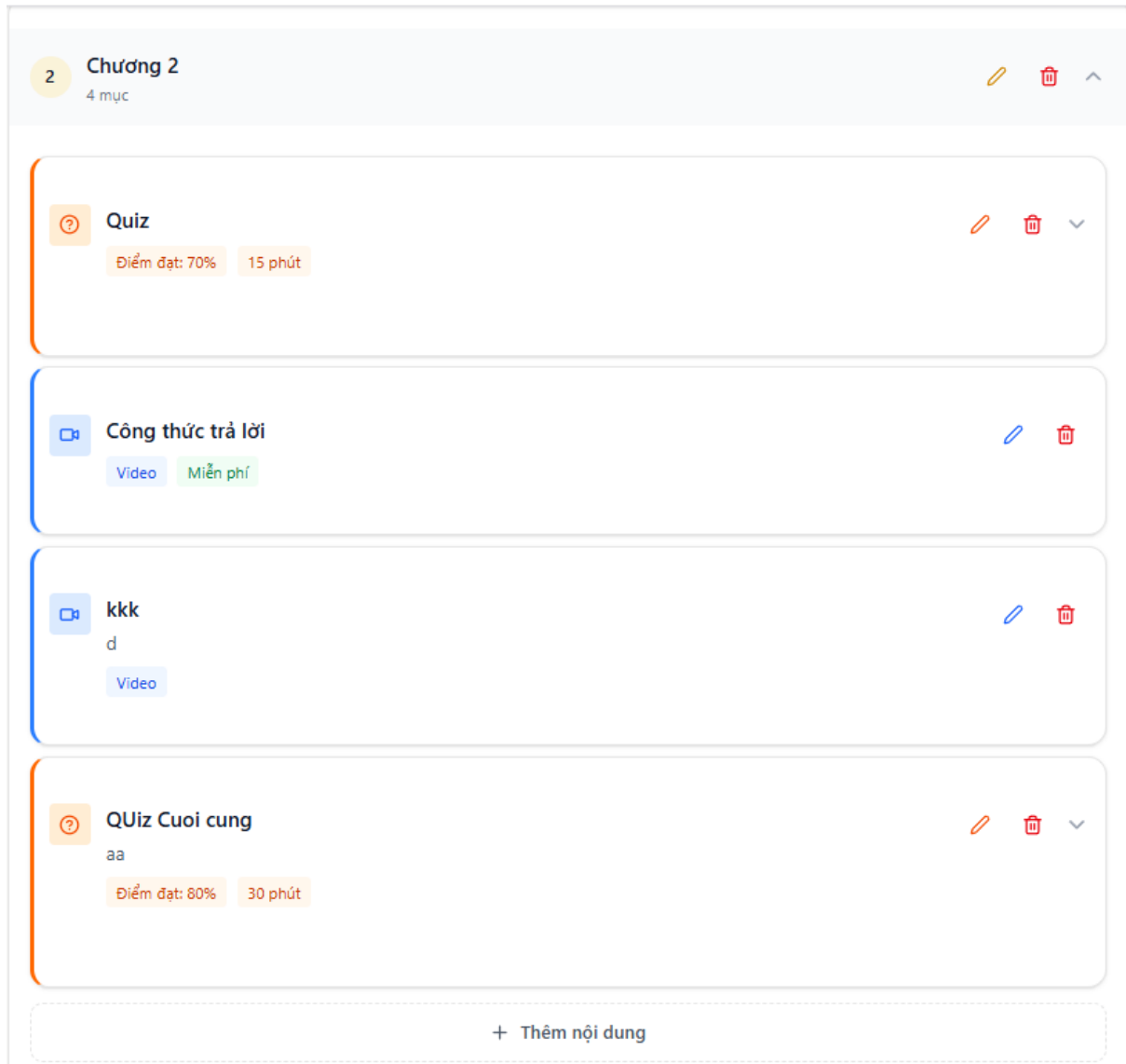


Figure 3.12.1.3 - quiz screen interface

This screen displays all quizzes created within the course. Instructors can track status, search, and access actions like Edit or Delete.

Field Description

Field Name	Description
"Chỉnh sửa" Icon	Button. Action: Navigates to the Quiz Editor to modify settings (Time, Title, etc.).
"xóa" Icon	Button. Action: Triggers the delete confirmation popup for a specific quiz.
Status Toggle	Switch. Action: Toggles the quiz availability (Draft/Published).

3.12.1.4 Delete quiz

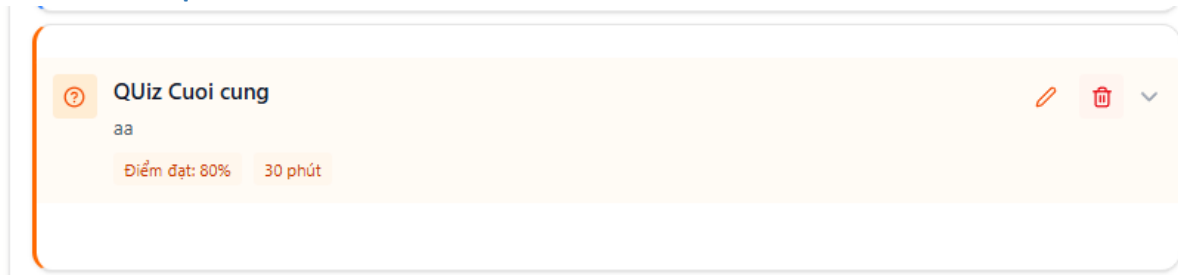


Figure 3.12.1.4 -delete quiz screen interface

This function removes a quiz from the course. Requires strict confirmation to prevent data loss

Field Description

Field Name	Description
"Confirm Delete" Button	Button. Action: Permanently deletes the quiz and all associated questions/results.
"Cancel" Button	Button. Action: Closes the popup without deleting.

3.12.1.5 Add question to quiz



Figure 3.12.1.3 - add question to quiz screen interface

This function allows the Instructor to populate the quiz with questions.

- Methods: Instructors can either create a new question from scratch or select existing questions from the Question Bank.
- Selection: Search and select multiple questions to add at once.
- Action: Confirm to link selected questions to the current quiz.

Field Description

Field Name	Description
"Thêm câu hỏi mới" Button	Button. Action: Opens the Question Creator form to make a brand new question and add it directly to this quiz.
"Thêm từ ngân hàng câu hỏi" Button	Button. Action: Opens a modal displaying the Question Bank to select existing questions.
Ô tìm kiếm	Textbox. Action: Filters questions in the bank by content or tags.
Question Checkbox	Checkbox. Action: Selects specific questions from the list to add.
"Thêm câu hỏi đã chọn" Button	Button. Action: Adds all checked questions to the quiz list.

3.13 Voucher Feature

3.13.1 View list voucher

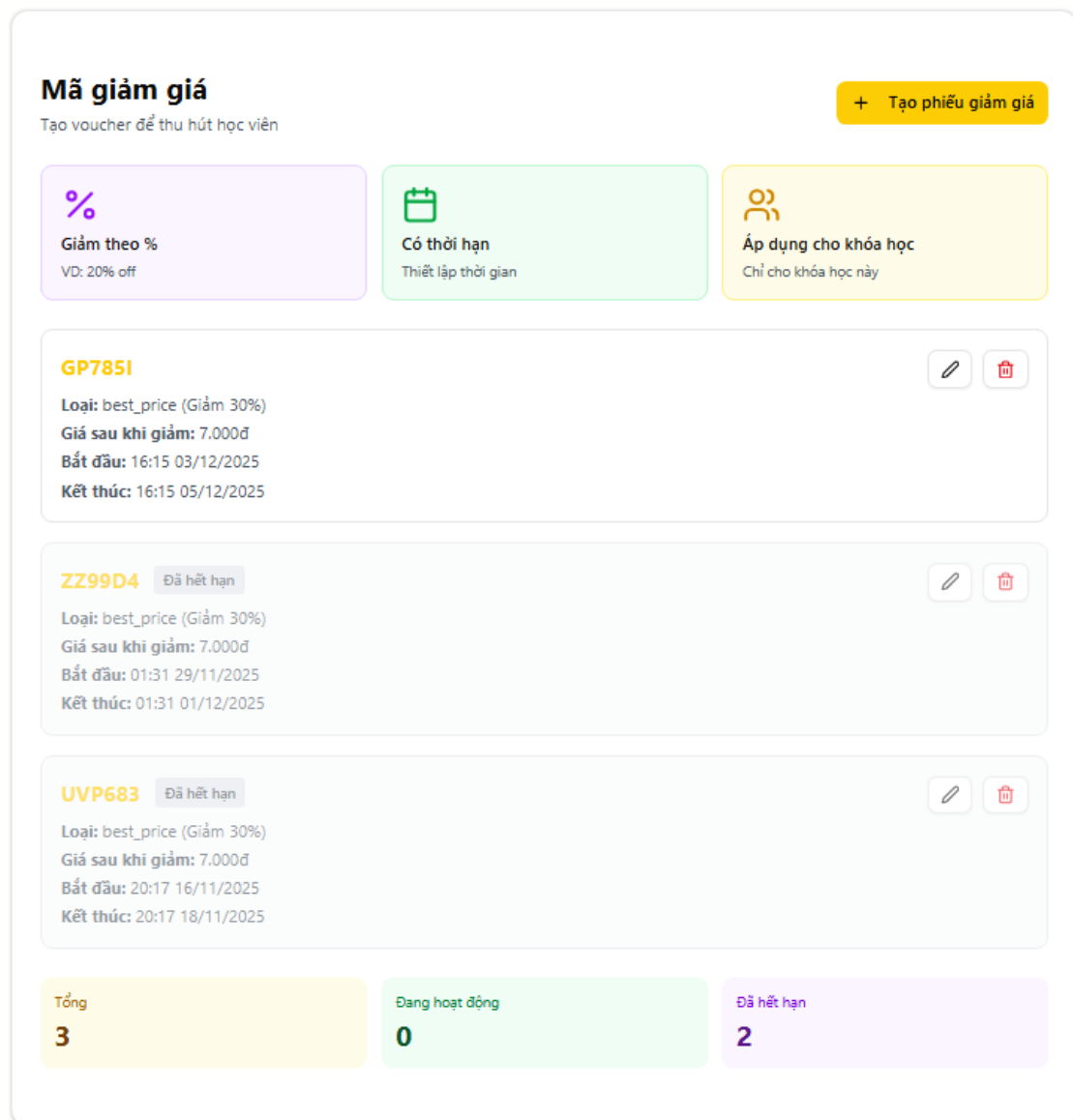


Figure 3.13.1 - list voucher screen interface

This screen allows a Lecturer to view the list of all vouchers configured for the current course, including their basic properties (code, type, discounted price, validity period, status) and quick actions to edit or delete each voucher.

Field Description

Field Name	Description
Voucher Code	Label (e.g., GP78SI, ZZ99D4). Data type: string 6 alphanumeric characters. Unique per course.
Loại	Label. Indicates discount value. Data type: int.
Giá sau khi giảm	Label. Text showing effective price after discount. Data type: decimal.
Bắt đầu	Label. Text showing voucher start date and time. Data type: DateTime; must be $\leq$ End DateTime.
Kết thúc	Label. Text showing voucher end date and time. Data type: DateTime; must be $\geq$ Start DateTime.
Icon Button: chỉnh sửa	Small button with pencil icon. On click, opens Edit Voucher form for this voucher.
Icon Button: xóa	Small button with trash icon. On click, shows confirmation dialog and, if confirmed, deletes this voucher.

3.13.2 Create voucher

Tạo voucher mới ×

Mã voucher *
VD: SALE20, SUMMER2024 ↻

Loại giảm giá *
best_price (30%) ▼

Giá gốc khóa học
10.000đ

Phần trăm giảm (%)
30%

Phần trăm giảm được quy định bởi loại voucher đã chọn

Giá sau khi giảm (VNĐ) *
7.000

Giá được tính tự động: $10.000đ - 30\% = 7.000đ$

Thời gian bắt đầu *
03/12/2025 16:16 📅

Thời gian kết thúc *
05/12/2025 16:16 📅

Tạo phiếu giảm giá Hủy

Figure 3.13.2 - create voucher screen interface

This modal lets a Lecturer define a new voucher for the current course: assign code, choose discount rule, confirm calculated price, and set validity period. It extends the Manage Course Vouchers use case (steps 7-8)

Field Description

Field Name	Description
Close Icon (X)	Dismisses modal without saving; equivalent to “Hủy”
Text Field: Mã voucher	Required string input. Data type: 6 uppercase alphanumeric characters. Must be unique per course. The refresh icon generates a random valid code.
Dropdown: Loại giảm giá	Required. Options correspond to pricing strategies. with display labels including human-readable %
Info Box: Giá gốc khóa học	Read-only text showing current course list price. Updates automatically if underlying course price changes.
Text Field: Phần trăm giảm (%)	Shows discount percentage allowed by selected type
Text Field: Giá sau khi giảm (VND)	Read-only text showing auto-calculated (Giá gốc - %)
DateTime Picker: Thời gian bắt đầu	Required DateTime (format dd/MM/yyyy HH:mm). Must be $\geq$ current time. Includes calendar icon trigger.
DateTime Picker: Thời gian kết thúc	Required DateTime. Must be $>$ Start time.
Button: “Tạo phiếu giảm giá	Submitted form. Disabled until required fields are valid. On success closes modal, refreshes list.
Button: “Hủy”	Submitted form. On success closes modal, refreshes list.

3.13.3 Update voucher

The image shows a modal window titled "Chỉnh sửa voucher" (Edit Voucher) with a close button (X) in the top right corner. The modal contains several form fields and a summary section:

- Mã voucher ***: A text input field containing "GP785I" and a refresh icon.
- Loại giảm giá ***: A dropdown menu showing "best_price (30%)".
- Giá gốc khóa học**: A blue box displaying "10.000đ".
- Phần trăm giảm (%)**: A slider input field set to "30%". Below it, a note states: "Phần trăm giảm được quy định bởi loại voucher đã chọn".
- Giá sau khi giảm (VNĐ) ***: A text input field displaying "7.000". Below it, a calculation is shown: "Giá được tính tự động: 10.000đ - 30% = 7.000đ".
- Thời gian bắt đầu ***: A date and time picker showing "03/12/2025 16:15".
- Thời gian kết thúc ***: A date and time picker showing "05/12/2025 16:15".
- Buttons**: A large yellow "Cập nhật" (Update) button and a smaller "Hủy" (Cancel) button.

Figure 3.13.3 - update voucher screen interface

This modal lets a Lecturer edit an existing voucher's code, discount rule, calculated price, or validity period. Supports the Manage Vouchers use case (Alternative Flow A3 – Edit voucher). Changes persist back to the voucher list when saved.

Field Description

Field Name	Description
Close Icon (X)	Dismisses modal without saving; equivalent to “Hủy”
Text Field: Mã voucher	Required string input. Data type: 6 uppercase alphanumeric characters. Must be unique per course. The refresh icon generates a random valid code.
Dropdown: Loại giảm giá	Required. Options correspond to pricing strategies. with display labels including human-readable %
Info Box: Giá gốc khóa học	Read-only text showing current course list price. Updates automatically if underlying course price changes.
Text Field: Phần trăm giảm (%)	Shows discount percentage allowed by selected type
Text Field: Giá sau khi giảm (VND)	Read-only text showing auto-calculated (Giá gốc - %)
DateTime Picker: Thời gian bắt đầu	Required DateTime (format dd/MM/yyyy HH:mm). Must be $\geq$ current time. Includes calendar icon trigger.
DateTime Picker: Thời gian kết thúc	Required DateTime. Must be $>$ Start time.
Button: “Tạo phiếu giảm giá	Submitted form. Disabled until required fields are valid. On success closes modal, refreshes list.
Button: “Hủy”	Submitted form. On success closes modal, refreshes list.

3.13.4 Delete voucher

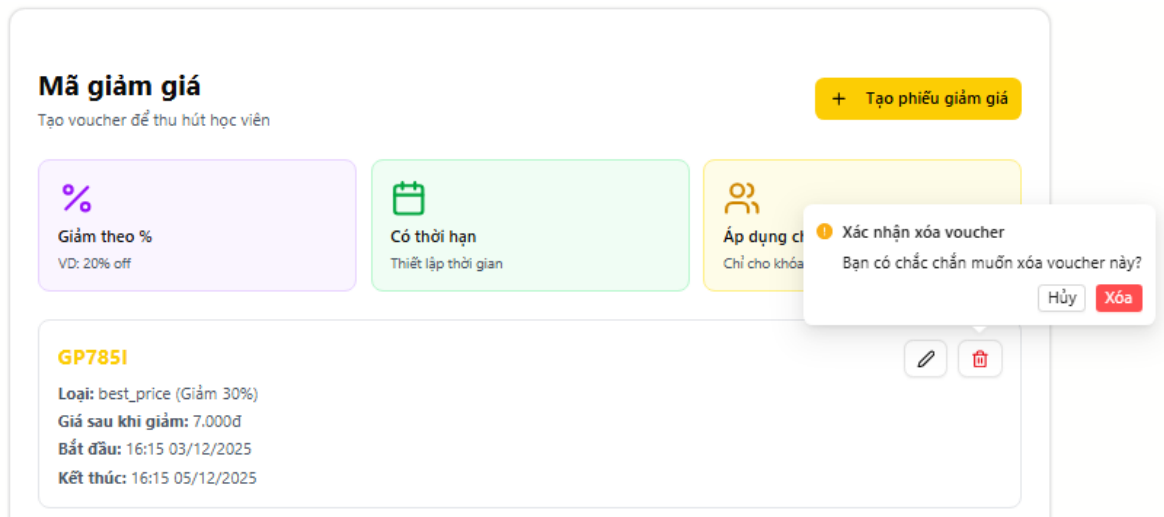


Figure 3.13.4 - delete voucher screen interface

When the Lecturer clicks the trash icon on a voucher, the system asks for confirmation before permanently removing the voucher from the current course. This supports the Manage Course Vouchers use case (Alternative Flow A4 – Delete voucher).

Field Description

Field Name	Description
Button: “Hủy”	Cancels deletion and closes popover. No data change
Button: “Xóa”	On click, calls delete API for the selected voucher. After success, remove the voucher from the list.

3.14 User Manage Feature

3.14.1 View user list

Quản lý tài khoản

Search by name or email

Ảnh đại diện	Họ Tên	Email	Vai trò	Trạng thái	Hành động
	1111	buitienquat2k@gmail.com	Học sinh	Đang hoạt động	Cấm
	badao Binh	fanjfla1989@gmail.com	Học sinh	Đã bị cấm	Bỏ cấm
	binh tran	taolabinhdayk3@gmail.com	Giảng viên	Đang hoạt động	Cấm
	Bùi Tiến Quát	skillup.fpt@gmail.com1	Học sinh	Đang hoạt động	Cấm
	Bùi Tiến Quát	quatbthe17638112@fpt.edu.vn1	Giảng viên	Chờ duyệt	Không khả dụng

< 1 2 3 4 5 ... 8 >

Figure 3.14.1 - user list screen interface

This screen allows a System Moderator to:

- **View User List:** view the list of all Student and Lecturer information.
- **Ban User:** Ban or Unban a Student or Lecturer's account.
- **Search:** Search for the accounts that have matching names or email.

Field Description

Field Name	Description
Tìm kiếm	Type: string, max 20 characters.
Cấm	Type: Button, Action: set an account status to Banned.
Bỏ cấm	Type: Button, Action: set an account status to Active.

3.15 Manage question Feature

3.15.1 Create question

The screenshot shows the 'Create question' interface. At the top, there's a search bar 'Tìm kiếm câu hỏi...'. Below it, a yellow banner says 'Sử dụng công cụ chọn nhanh bên dưới hoặc chọn thủ công từng câu hỏi từ các chương.' (Use the quick selection tool below or manually select each question from the chapters). The main form is titled 'Công cụ chọn nhanh' (Quick selection tool). It has three main sections: 1. 'Chọn cơ bản' (Basic selection) with a 'Tất cả (1)' button and a 'Bỏ chọn' button. 2. 'Chọn nhanh theo số lượng' (Quick selection by quantity) with buttons for 5, 10, 15, 20, and 25, and a 'Chọn đều' button. 3. 'Chọn theo từng chương' (Select by chapter) with a 'Chương 1' dropdown and a '1 câu' input field. There are also buttons for 'Chọn đều' and 'Ngẫu nhiên'. At the bottom, there is a 'Thêm câu hỏi' button and a 'Chưa chọn câu hỏi nào' message.

Figure 3.15.1 - create question screen interface

This function allows the Instructor to input content for a new question.

- **Input Data:** Enter the question text, upload optional media (images), and define answer options.
- **Logic:** Select the correct answer(s) to enable auto-grading.
- **Action:** Save the question to the Question Bank or Quiz.

Field Description

Field Name	Description
Question Content Area	Text Area. Action: Input field for the question description/text. Supports rich text (bold, italic).
"Upload Image" Button	Button. Action: Opens file picker to attach an illustration image to the question.
Answer Option Input	Textbox. Action: Input fields for answer choices (e.g., Option A, Option B).
Correct Answer Selector	Radio/Checkbox. Action: Marks the specific option as the correct answer. - Radio: For Single Choice questions.- Checkbox: For Multiple Choice questions.
"Add Option" Button	Button. Action: Adds a new answer row (e.g., adds Option E) to the list.
"Delete Option" Icon	Button. Action: Removes an answer row (e.g., remove Option D).
"Save" Button	Button. Action: Validates inputs (must have text & correct answer) and saves the question.
"Cancel" Button	Button. Action: Discards inputs and closes the form.

3.15.2 Update question

1

Chương 1

1 mục

?

Chương 1

1

Điểm đạt: 70%

15 phút

+ Thêm câu hỏi

Câu hỏi (1)

Chỉnh sửa câu hỏi

Câu hỏi *

↶ ↷ Paragraph

B *I* U ~~ABC~~

1

Mô tả / Gợi ý (tùy chọn)

1

Loại câu hỏi *

☒ Một đáp án đúng

☐ Nhiều đáp án đúng

Hình ảnh câu hỏi (tùy chọn)

Tải lên hình ảnh

Đáp án *

☒ 1

✓

Figure 3.15.2 - update question screen interface

This function allows the Instructor to modify an existing question.

- **Load Data:** The system automatically pre-fills the form with the current question text, attached image, answer options, and the correct answer key.
- **Modify:** Instructors can edit the text, replace/remove images, change answer choices, or select a different correct answer.
- **Action:** Clicking "Update" saves the changes to the database.

Field Description

Field Name	Description
"Update" Button	Button. Action: Validates the modified inputs and overrides the existing question record. Displays a success toast.
Question Content Area	Text Area. Action: Edit the existing question description/text.
"Change/Remove Image" Button	Button. Action: Replaces the current image with a new one or removes it entirely.
Answer Option Input	Textbox. Action: Edit the text of existing answer options.
Correct Answer Selector	Radio/Checkbox. Action: Change which option is marked as the correct answer.
"Delete Option" Icon	Button. Action: Removes a specific answer row from the list.
"Add Option" Button	Button. Action: Adds a new answer row if more options are needed.
"Cancel" Button	Button. Action: Discards all unsaved changes and returns to the question list.

3.15.3 Delete question

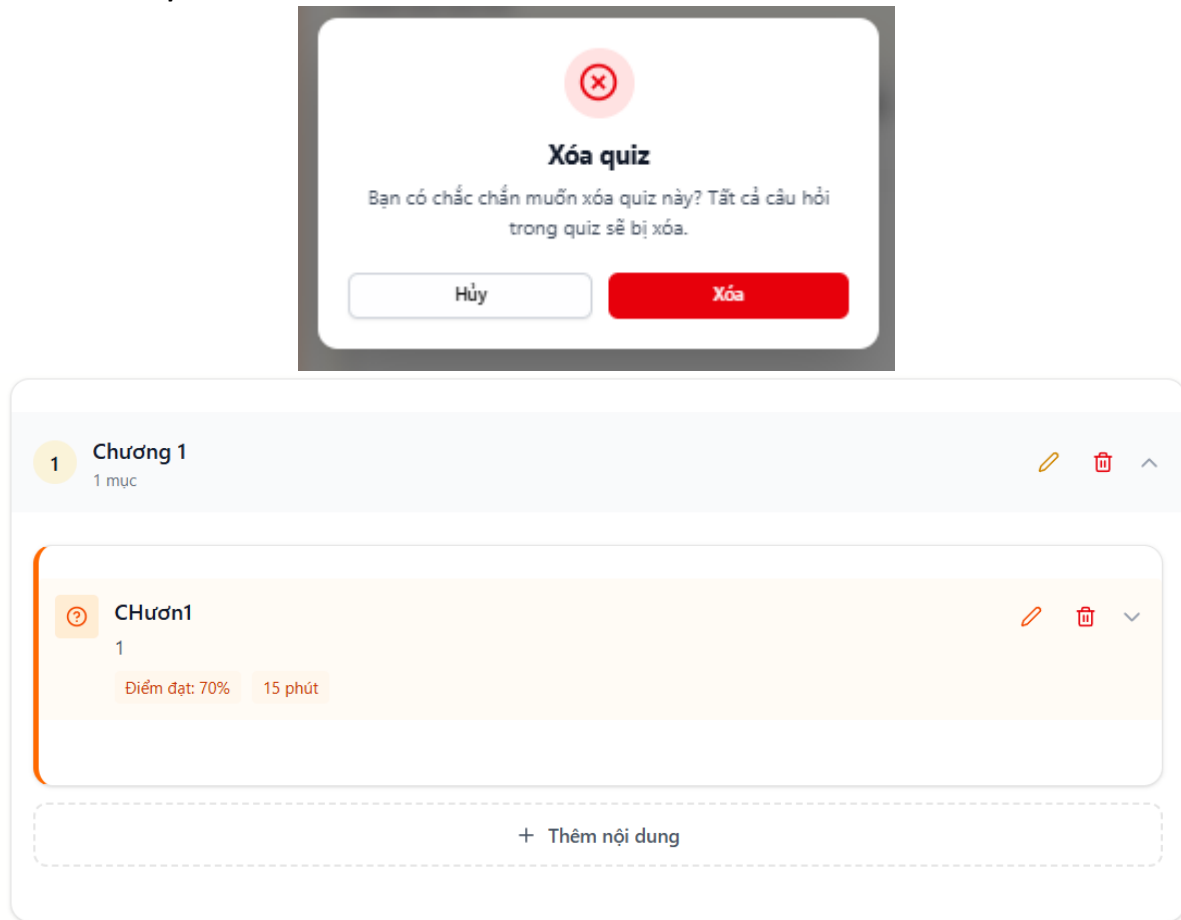


Figure 3.15.3 - delete question screen interface

This function allows the Instructor to permanently remove a question from the bank or quiz.

- **Trigger:** Click the "Delete" (Trash bin) icon on a question row.
- **System Behaviour:** Displays a confirmation modal. If confirmed, the question is removed immediately.

Field Description

Field Name	Description
"Delete" Icon	Button/Icon. Action: Triggers the delete confirmation popup for the specific question.
"Confirm Delete" Button	Button. (Inside Modal). Action: Permanently deletes the question record from the database.
"Cancel" Button	Button. (Inside Modal). Action: Closes the popup without performing the deletion.

3.15.4 View question

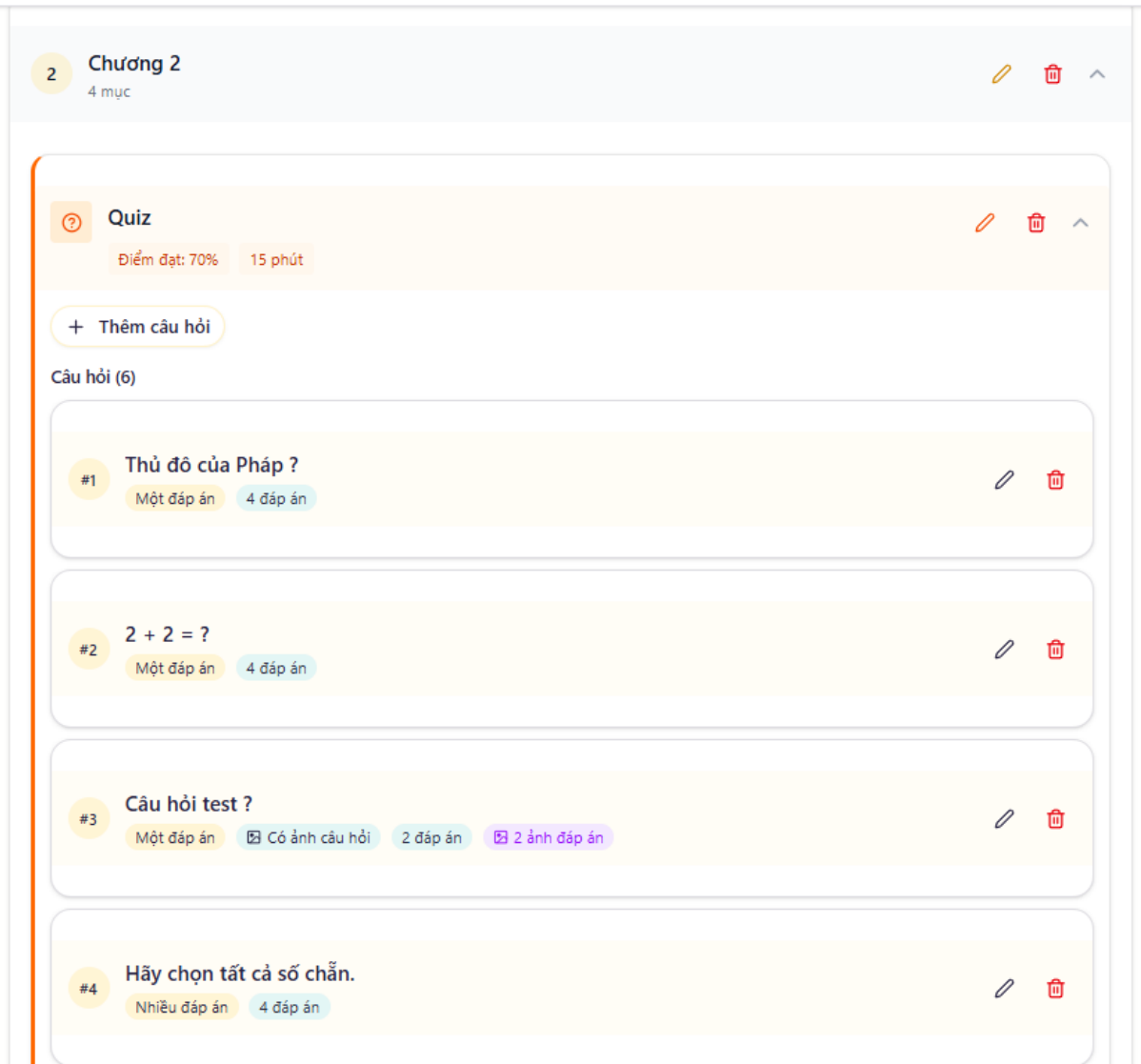


Figure 3.15.4 - list question screen interface

This function allows the Instructor to review the full details of a specific question.

- **View Content:** Displays the question text, attached images, answer options, and highlights the correct answer.
- **Purpose:** Ensures accuracy before adding the question to a quiz.

Field Description

Field Name	Description
"Edit" Button	Button. Action: Navigates to the Question Editor screen to modify the content.
"Delete" Button/Icon	Button. Action: Triggers the delete confirmation popup to remove this question.
"Close" or "Back" Button	Button. Action: Closes the preview modal or returns to the question bank list.

4. Non-Functional Requirements

4.1 External Interfaces

4.1.1 User Interfaces

UI-1: All screens shall have clear information display and easy to use layout.

UI-2: All designs have been implemented by the front-end team based on available resources.

4.1.2 Software Interfaces

SI-1: OLP-SkillUp Interface sends access requests via OAuth2; OLP-SkillUp Back-end verifies authentication tokens.

4.1.3 Hardware Interfaces

No hardware interfaces have been identified

4.1.4 Communication Interfaces

No communication interfaces have been identified.

4.2 Quality Attributes

4.2.1 Usability

APIs:

- APIs are designed following RESTful standards to facilitate seamless integration with client systems, ensuring clear and easy-to-use CRUD (Create, Read, Update, Delete) operations. The APIs support search, filtering, and reporting features through well-structured and extensible endpoints.

React User Interface:

- The UI is optimized for users, ensuring a friendly, intuitive, and easy-to-navigate experience. It integrates multilingual support (English and Vietnamese), allowing users to switch languages during their session. The interface ensures fast search, filtering, and report display, with modularized and reusable components.

4.2.2 Performance

The React frontend is optimized with lazy loading, memorization, and minimizing unnecessary rendering.

5. Requirement Appendix

5.1 Business Rules

ID	Rule Definition
BR-AUTH-01	Inactive or Banned accounts cannot login
BR-AUTH-02	Only roles Student and Lecturer can register via public registration
BR-AUTH-03	OTP must not be used before and must be valid
BR-AUTH-04	Account must be InActive status to verify OTP
BR-AUTH-05	Old password must be correct to change password
BR-AUTH-06	Only Active or Pending accounts can request password reset
User & Role Management Rules	
BR-USER-01	Avatar must be image type (image/jpeg, image/png, image/gif)
BR-USER-02	Only account owner can update their profile
BR-USER-03	Date of birth must be valid date if provided
BR-USER-04	Lecturers cannot edit or delete a voucher once created but can deactivate it; deactivated vouchers do not count towards creating a new one.
BR-USER-05	Courses must undergo a quality review process by moderator before publication, ensuring content meets minimum standards (e.g., HD video, engaging quizzes).
BR-USER-06	Students can rate and review purchased courses only after completing at least 50% of the lectures.
Course Management Rules	
BR-COURSE-01	Only lecturer can create/delete/update their courses
BR-COURSE-02	Course must have valid SubCategory
BR-COURSE-03	Only course owner can publish/unpublish courses
BR-COURSE-04	Course image required and uploaded to Cloudinary
BR-COURSE-05	New courses default to "Draft" status
Enrollment Rules	
BR-ENROLL-01	Student cannot enroll in same course twice
BR-ENROLL-02	Course must be active to allow enrollment
BR-ENROLL-03	Free courses (Price = 0) auto-enroll without payment
BR-ENROLL-04	Only students can enroll in courses
Payment & Transaction Rules	
BR-PAY-01	Course must be active for payment
BR-PAY-02	Cannot pay for already enrolled courses
BR-PAY-03	Transaction status must be "Pending" to cancel
BR-PAY-04	Successful payment auto-enrolls student

BR-PAY-05	Transaction detail created for each course purchase
BR-PAY-06	Student cannot add enrolled courses to cart
Cart Management Rules	
BR-CART-01	Duplicate courses in cart prevented
BR-CART-02	Course must be active to be added to cart
BR-CART-03	Cart items cleared after successful purchase
BR-CART-04	Cannot checkout with empty cart
BR-CART-05	All cart courses must be active at checkout
Voucher Management Rules	
BR-VOUCH-01	Lecturer can only create/update/delete vouchers for their own courses
BR-VOUCH-01	Start time must be before end time
BR-VOUCH-01	Start time must be in future for new vouchers
BR-VOUCH-01	Voucher must be within valid date range to apply
BR-VOUCH-01	Discount amount cannot exceed total price
Voucher Type Rules	
BR-VTYPE-01	Percentage must be between 0 and 100
BR-VTYPE-02	Only Content Moderator can manage voucher types
Section & Lesson Rules	
BR-SECT-01	Only course owner can create, update, delete, reorder sections
BR-SECT-02	Lessons have Type: Text, Video or Quiz
BR-SECT-03	Only enrolled students can access paid lessons
Question Bank Rules	
BR-QB-01	Only lecturer can create question banks
BR-QB-02	Question must belong to lecturer's course
BR-QB-03	At least one correct answer required
BR-QB-04	Question Type: SingleChoice or MultiChoice
BR-QB-05	MultiChoice if >1 correct answer, else SingleChoice
BR-QB-06	Excel import validates question format (7 answer columns A-G)
BR-QB-07	Correct answer column (column 9) format: "A,C"
BR-QB-08	Excel import skips rows with empty question or correct answer
Quiz Rules	
BR-QUIZ-01	Only course owner can create quizzes
BR-QUIZ-02	Quiz questions from QuestionBank
BR-QUIZ-03	Only enrolled students can submit quiz
BR-QUIZ-04	Quiz submission records score and answers

Forum & Post Rules	
BR-FORUM-01	Only authenticated students can create/update/delete posts
BR-FORUM-02	Only Content Moderator can hide/unhide posts
BR-FORUM-03	Post must belong to valid Forum Category
BR-FORUM-04	Authenticated users can comment on posts
Comment Rules	
BR-COMM-01	Only comment owner can edit/delete comments
BR-COMM-02	Users can like/unlike comments
BR-COMM-03	Comments can be reported
BR-COMM-04	Only Content Moderator can delete reported comments
BR-COMM-05	Only enrolled students can comment on lessons
Report Rules	
BR-REP-01	Students can report courses, posts and comments.
BR-REP-02	Only Content Moderator can review reports
BR-REP-03	The report requires reason/description.
Rating Rules	
BR-RATE-01	Only enrolled students can rate courses. Rating value between 1-5.
BR-RATE-02	One rating per student per course, update the rating if exists
Lecturer Application Rules	
BR-LEC-01	Only System Moderator can approve/reject applications
BR-LEC-02	One pending application per user at a time
BR-LEC-03	Application requires additional Title, Profession, CV and Degrees.
Ticket Support Rules	
BR-TICK-01	Ticket owner can only view their tickets
BR-TICK-02	Only System Moderator can respond to tickets
BR-TICK-03	Ticket status: Pending, Resolved, Closed
Banner & News Rules	
BR-NEWS-01	Only Content Moderator can create/edit/delete news
BR-NEWS-02	News can be published/unpublished
BR-BAN-01	Only Content Moderator can manage banners
BR-BAN-02	Banner requires image and link
Upload & Media Rules	
BR-UP-01	All media uploaded to Cloudinary
BR-UP-02	Upload returns public URL
Student Progress Rules	
BR-PROG-01	Progress tracked per student per lesson

BR-PROG-02	Quiz passed updates progress
BR-PROG-03	Course completion when all lessons completed
Chat & Notification Rules	
BR-CHAT-01	Only enrolled students can chat in lesson
BR-CHAT-02	Notifications sent for enrollment, comments, replies
BR-CHAT-03	Notifications can be marked as read by user
Category Management Rules	
BR-CAT-01	Category must have unique name
BR-CAT-02	SubCategory must belong to valid Category
BR-CAT-03	Cannot delete Category with existing courses
BR-CAT-04	Only System Moderator can manage categories

5.2 System Messages

#	Message code	Message Type	Context	Content
1	MSG01	Toast (Success, Green)	User successfully logs into the system	<i>Đăng nhập thành công</i>
2	MSG02	Toast (Success, Green)	User logs in using Google OAuth	<i>Đăng nhập Google thành công</i>
3	MSG03	Toast (Error, Red)	User login fails due to invalid credentials	Đăng nhập thất bại. Vui lòng thử lại!
4	MSG04	Toast (Error, Red)	User logs in but email not verified	<i>Tài khoản chưa được kích hoạt</i>
5	MSG05	Toast (Success, Green)	User successfully registers an account	<i>Đăng ký thành công! Vui lòng kiểm tra email để xác thực tài khoản.</i>
6	MSG06	Toast (Success, Green)	User successfully logs out	<i>Đăng xuất thành công!</i>

7	MSG07	Toast (Success, Green)	User successfully changes password	<i>Đổi mật khẩu thành công!</i>
8	MSG08	Toast (Error, Red)	User enters mismatched passwords	<i>Mật khẩu mới và mật khẩu xác nhận không khớp</i>
9	MSG09	Toast (Error, Red)	Password change fails	<i>Đổi mật khẩu thất bại</i>
10	MSG10	Toast (Success, Green)	User successfully updates profile	<i>Cập nhật hồ sơ thành công!</i>
11	MSG11	Toast (Error, Red)	Profile update fails	<i>Cập nhật thất bại</i>
12	MSG12	Toast (Success, Green)	User successfully uploads avatar	<i>Cập nhật ảnh đại diện thành công!</i>
13	MSG13	Toast (Error, Red)	User selects invalid image file for avatar	<i>Vui lòng chọn file ảnh hợp lệ</i>
14	MSG14	Toast (Error, Red)	Uploaded file exceeds size limit	<i>Kích thước file không được vượt quá 5MB</i>
15	MSG15	Toast (Error, Red)	Avatar upload fails	<i>Upload ảnh đại diện thất bại</i>
16	MSG16	Toast (Success, Green)	Forgot password email sent successfully	<i>Email khôi phục mật khẩu đã được gửi! Vui lòng kiểm tra hộp thư.</i>
17	MSG17	Toast (Error, Red)	User submits empty email in forgot password	<i>Vui lòng nhập email!</i>
18	MSG18	Toast (Error, Red)	User enters invalid email format	<i>Email không đúng định dạng!</i>

19	MSG19	Toast (Error, Red)	Email not found or not verified for password reset	<i>Email không tồn tại hoặc chưa được xác thực</i>
28	MSG28	Toast (Success, Green)	Free course enrollment successful	<i>Đăng ký khóa học miễn phí thành công!</i>
30	MSG30	Toast (Error, Red)	Payment processing error	<i>Có lỗi xảy ra khi thanh toán</i>
31	MSG31	Toast (Info, Blue)	User must login to purchase course	<i>Vui lòng đăng nhập để mua khóa học</i>
32	MSG32	Toast (Success, Green)	Lecturer application submitted	<i>Nộp CV thành công! Chúng tôi sẽ liên hệ với bạn sớm.</i>
33	MSG33	Toast (Error, Red)	Incomplete lecturer application	<i>Vui lòng điền đầy đủ thông tin bắt buộc!</i>
34	MSG34	Toast (Error, Red)	CV must be PDF format	<i>CV chỉ chấp nhận file PDF!</i>
35	MSG35	Toast (Error, Red)	File size exceeds limit (generic)	<i>File không được vượt quá 5MB!</i>
36	MSG36	Toast (Error, Red)	Invalid image file format for degree	<i>File {name} không hợp lệ! Chỉ chấp nhận ảnh JPG, PNG.</i>
37	MSG37	Toast (Success, Green)	Degree images uploaded successfully	<i>Đã thêm {count} ảnh bằng cấp</i>
38	MSG38	Toast (Info, Blue)	Degree image removed	<i>Đã xóa ảnh</i>
39	MSG39	Toast (Error, Red)	Lecturer application submission fails	<i>Không thể nộp CV. Vui lòng thử lại!</i>

40	MSG40	Toast (Success, Green)	Course list loaded successfully	<i>Đã tải danh sách khóa học</i>
41	MSG41	Toast (Error, Red)	Failed to load courses	<i>Lỗi khi tải khóa học</i>
42	MSG42	Toast (Info, Blue)	No courses created yet	<i>Bạn chưa có khóa học nào</i>
43	MSG43	Toast (Error, Red)	Failed to load course list	<i>Lỗi khi tải danh sách khóa học</i>
44	MSG44	Toast (Success, Green)	Quiz created successfully	<i>Tạo quiz thành công!</i>
45	MSG45	Toast (Error, Red)	Failed to load quiz	<i>Không thể tải quiz!</i>
46	MSG46	Toast (Error, Red)	Failed to create quiz	<i>Không thể tạo quiz!</i>
47	MSG47	Toast (Success, Green)	Question created in bank	<i>Tạo câu hỏi trong ngân hàng thành công!</i>
48	MSG48	Toast (Success, Green)	Question updated successfully	<i>Cập nhật câu hỏi thành công!</i>
49	MSG49	Toast (Success, Green)	Question deleted from bank	<i>Xóa câu hỏi khỏi ngân hàng thành công!</i>
50	MSG50	Toast (Error, Red)	Failed to create question	<i>Không thể tạo câu hỏi!</i>
51	MSG51	Toast (Error, Red)	Failed to update question	<i>Không thể cập nhật câu hỏi!</i>

52	MSG52	Toast (Error, Red)	Failed to delete question	<i>Không thể xóa câu hỏi!</i>
53	MSG53	Toast (Success, Green)	Image uploaded successfully	<i>Ảnh đã được tải lên!</i>
54	MSG54	Toast (Error, Red)	Failed to get image URL	<i>Không thể lấy URL ảnh</i>
55	MSG55	Toast(Error, Red)	Failed to add quiz	<i>Dữ Liệu Không Hợp Lệ</i>
56	MSG56	Toast(Error, Red)	Can't find lecturer	<i>Không tìm thấy giảng viên tương ứng</i>